extension architecture

BUILDING CONTROL SUBMISSION

STRUCTURAL CALCULATIONS

FOR

LOFT CONVERSION, INTERNAL RECONFIGURATION & SINGLE STOREY REAR EXTENSION

AT

1 LAMBOURNE DRIVE

COBHAM KT11 3LB

JULY 2024

EXTENSION ARCHITECTURE & INTERIORS LTD 231 Cobden house, Roehampton Lane, London, SW15 4LB Tel (0208) 288 8950 Fax (0871) 918 2798 www.extensionarchitecture.co.uk	extension architecture	Sheet No:
		Prepared by: ARS
PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB		Date: JUL 2024

Contents

- *General Notes*
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- *Loading/ Key Plans*
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ELEMENT: **GENERAL NOTES**

1. These calculations have been produced to comply with the relevant parts of the current Building Regulations and British Standards.

2. In the preparation of these calculations a number of assumptions may have been made based on the design brief supplied. These will be clearly identified with the calculations. The calculations relate to the elements described only and no responsibility is accepted in respect to other elements of the building.

3. Unless otherwise stated, no site visit has been made and no part of the building has been opened up to confirm any assumptions made. The Client should ensure that the relevant areas are opened up or exposed before building materials are ordered and any discrepancies found should be reported to this office. The responsibility for materials purchased remains with the Client.

4. Dimensions have been obtained from information provided by the Architect. Where no figured dimensions have been provided, scaling has been used. Sketches within the calculations should not be scaled. Dimensions indicated in the calculations are for design purposes only and should not be used for constructional purposes.

5. It is assumed that experienced and competent personnel will carry out the works in compliance with the relevant Building Regulations, using traditional construction methods. The Builder shall be responsible for the stability of the structure whilst the works are in progress.

6. The contractor shall be responsible for serving upon the Local Authority any required Commencement Notices, on the District Surveyor the requisite Building Notice or on the Sanitary Authority (EHO) drainage work application and approval: and to comply with any other and all statutory obligations and regulations of the Local Authority.

7. Sketches within the calculations are to demonstrate certain features of the design and are not intended as full constructional drawings. The sketches do not carry any validity for non-structural details.

8. All proprietary products to be installed in accordance with the Manufacturers Recommendations.

9. The contractor in preparing an estimate shall visit the site, acquaint him/herself with the problems of the work involved and shall allow in pricing for all details whether shown on the drawings or not, which may be reasonably considered to be necessary to ensure a neat, structurally sound and workman like job.

10. Where Building Control approval is required, it is essential that this be obtained before the works proceed or materials are ordered. The responsibility for any works carried out before such approval is obtained remains with the Client.

11. Timber details:
    a) All timber is to be grade C24 unless stated otherwise.
    b) All timber materials and workmanship to be in accordance with BS 5268: Part 2 - Structural Use of Timber.
    c) Joists to be doubled up under stud partitions. Double joists/rafters to be bolted together at 600 centres with 12mm diameter bolts and 50 x 50 x 3 washer plates.
    d) Multiple joists/trusses to be bolted together at 600 centres with 12mm dia. bolts and 50 x 50 x 3 washer plates.
    e) No notches, holes or rebates etc. to be cut in any member without the written agreement of the Engineers.
    f) All trusses to be connected to timber wall plate by means of approved truss clips.
    g) All joists spanning greater than 2.5m are to have solid noggin pieces, of a depth equivalent to the joists, at mid-span.
    h) All floor joists, rafters and any external stud walls to be adequately restrained/tied to masonry walls using suitable restraint straps, min 5 x 35mm x 1200 long @ 1.25m ctrs, or as specified in Building Regulation Approved Document A.
    i) All timber joists to have full depth blocking pieces at their ends.
    j) Floor joists are to be positioned with their top surfaces a minimum of 12.5mm above the top flange of any supporting steel beams.
    k) All metal fixings used in the timberwork are to be galvanized or otherwise treated against corrosion.
    l) Timber stud loadbearing partitions comprising: 100 x 100 softwood @ 400 vertical/ 1200 horizontal centres on softwood sole plates with 8mm ply sheeting to each face. Treated timber sole plates on DPC if secured to concrete.

12. Masonry details:
    a) Unless specified otherwise the following masonry strengths are to be used
        i) External facing brick - 20N/mm^2.
        ii) Engineering brick - 50N/mm^2.
        iii) Blockwork below DPC - 7.3 N/mm^2.
        iv) Blockwork above DPC - 3.6 N/mm^2.
    b) Mortar mix to be designation M4 (Class (iii)).
    c) Mass concrete padstones should be formed using a minimum Grade C30 concrete.
    d) Blockwork supporting steel beams to be 7.3N/mm^2.

13. Foundations - Strip/Trench Fill:
    a) Concrete to be GEN3 (min. cement content 300 kg/m^3, max. aggregate size 20mm).
    b) The foundations within these calculations have been designed to limit the ground bearing pressure to 100kN/m^2 unless otherwise noted.
    c) If no soil investigation has been instigated then sulphate resisting cement should be used in the ground construction.
    d) The sides of the un-cast foundations are likely to be prone to collapse and allowance should be made for temporary planking and strutting.
    e) Where required compressible material is to be ''Cordek Ltd'' products or specified.
    f) Where no site investigation has taken place and where trees are adjacent to the site in clay soils the Contractor shall inform the Engineer/B.C.O. and allow for the foundations to be designed in accordance with NHBC Practice Note 4.2 recommendations.

g) Where service invert levels are below the formation level the footing must be increased to a minimum of 300mm below the invert of the pipe and 300mm surround to the pipe. The pipe must be sleeved with a minimum of 25mm clearance to any face of the pipework by either low density polystyrene or UPVC sleeve.

14. Steelwork details

a) All bolts to be a minimum Grade 8.8 to sizes shown unless indicated otherwise.
b) One hour fire protection to be provided to all structural steel members.
c) All structural steelwork to be compliant with BS EN 10025 S275 Hot Rolled Steel.
d) All structural hollow section steelwork to be compliant with BS EN 10210 S355 Hot Formed Steel.
e) All welding to be carried out in accordance with BS5135 by a properly certified welder.
f) Site welding is to not be carried without approval from Structural Engineer.
g) The Steelwork Fabricator to check all dimensions on site and report any discrepancies to the engineer.
h) Any beams supporting walls over, shall be provided with a steel plate to suit the width of the wall supported. Thickness of plate to be confirmed with Extension Architect.
i) Welds to be 6mm continuous fillet, unless noted otherwise.
j) Steelwork which is not required to be encased in concrete to be blast cleaned/wire brushed free from mill scale, rust and other contamination and painted with two coats of approved primer as soon as practicable but not more than four hours after cleaning.
k) Uncased stanchions and beams located within an external wall to have a minimum gap of 40mm from face of external brickwork or alternately 25mm min impermeable insulation from face of steel to the external wall, unless galvanised or similar treatment.
l) All pockets formed in brickwork or blockwork for steel beams to be made good in Grade 35 concrete.
m) All concrete encased steelwork to be unpainted.

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Relevant Design Standards

BS 648:	1964	Schedule of Weights of Building Materials
BS 5950	Part 1	Structural use of steelwork in buildings
BS 6399:	Part 1	Loading for Buildings: Code of Practice for Dead and Imposed Loads
BS5268:	Part 2	Structural use of timber

Materials

Steelwork	S355
Timber	C24
Concrete:	
- Padstones	C40/50
- Substructure	C25/30

ELEMENT: **LOADING / KEY PLANS**

1. Loadings

The Loads is based on BS 6399, the design ultimate load is obtained as follows:

a) Dead load + Imposed load
1.4 x Characteristic Dead Loads + 1.6 x Characteristic Imposed Loads

Building Loads

PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB		Date JUL/24	Page A/01	
		Engineer ARS	Checked -	
		DL	LL	UNITS
BUILDING LOADS				
Pitched Roof no lining (Existing or New)				
Roof clay tiles SW		0.70		
Timber rafters		0.25		
Insulation		0.10		
Imposed Load (Snow/Maintenance)			0.75	
	Total	1.05	0.75	kN/m²
Pitch 35°	Total	1.28	0.92	kN/m²
Pitched Roof with lining (Existing or New)				
Roof clay tiles SW		0.70		
Timber rafters		0.25		
Insulation		0.10		
Battens and Felt		0.05		
Ceiling & Services		0.20		
Imposed Load (Snow/Maintenance)			0.75	
	Total	1.30	0.75	kN/m²
Pitch 35°	Total	1.59	0.92	kN/m²
New Flat Roof with lining (Rear / Side Extension)				
Roof finishes		0.45		
Timber joists		0.20		
Insulation		0.10		
Battens and Felt		0.05		
Ceiling & Services		0.20		
Imposed Load (Snow/Maintenance)			0.75	
	Total	1.00	0.75	kN/m²
Typical timber floor				
Floor finishes		0.10		
Timber Joists		0.20		
Insulation		0.05		
Plywood & Ceiling finishes		0.20		
Services		0.05		
Imposed Load (Residential)			1.50	
	Total	0.60	1.50	kN/m²

Building Loads

PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date JUL/24	Page A/01
	Engineer ARS	Checked -

	DL	LL	UNITS
Internal Timber Stud Wall SW			
Timbers SW	0.25		
Battens and Felt	0.05		
Insulation	0.05		
Finishes (2x)	0.35		
Total	0.70	0.00	kN/m²
Existing Solid BrickWall SW (215mm thk.)			
Brickwall (21kN/mc density)	4.30		
Finishes (1x)	0.20		
Total	4.50	0.00	kN/m²
Internal Block / Brick Wall (100mm THK.)			
100mm Blockwork / Brickwork (20 kN/m³)	2.00		
Finishes (x2)	0.30		
Total	2.30	0.00	kN/m²
New Gable End / Dormer Rear Wall SW (Cladding + timber)			
Cladding	0.60		
Insulation	0.10		
Timbers SW	0.30		
Battens and Felt	0.05		
Finishes (1x)	0.15		
Total	1.20	0.00	kN/m²
New Cavity Wall SW (100mm Cavity)			
102.5mm Brickwork (20 kN/m³)	2.05		
Insulation (100mm)	0.10		
100mm Blockwork (19 kN/m³)	1.90		
Plasterboard (x1) + finishes	0.25		
External Render	0.30		
Total	4.60	0.00	kN/m²

* Service load arising from mechanical and electrical installation, plant, freezer ceiling, equipment & fittings

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	Prepared by:	ARS
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ELEMENT: **LOADING & DESIGN OF TIMBER & STEEL WORKS**

HIP RAFTERS

SPAN OF THE JOIST = 4500 mm

LOADING:

I. Dead Load

Pitched roof = $\left[\frac{1.28kn/\mathrm{m}^2 \text{ x } (2.5+1.5)\mathrm{m}}{2}\right]$ = 2.56 Kn/m

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2.56 Kn/m

II. Imposed Load

Pitched roof = $\left[\frac{0.92kn/\mathrm{m}^2 \text{ x } (2.5+1.5)\mathrm{m}}{2}\right]$ = 1.84 Kn/m

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1.84 Kn/m

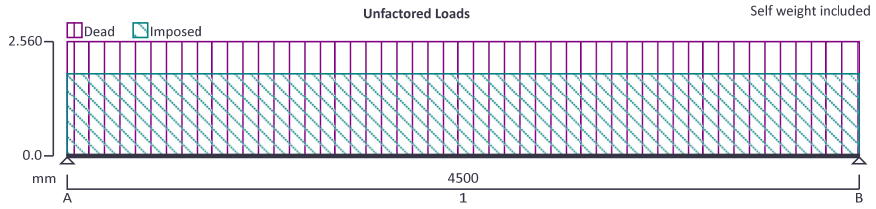
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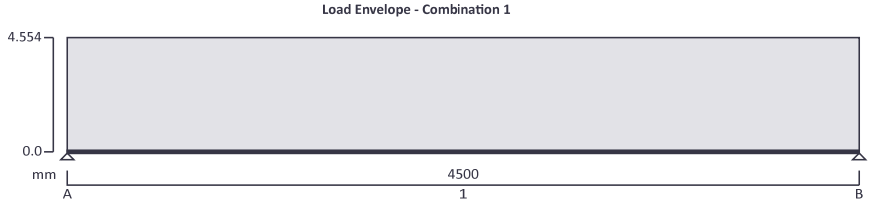
HIP RAFTER DESIGN:

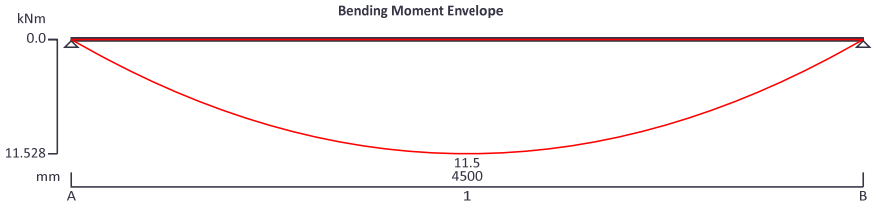
TIMBER BEAM ANALYSIS & DESIGN TO BS5268-2:2002

TEDDS calculation version 1.7.01

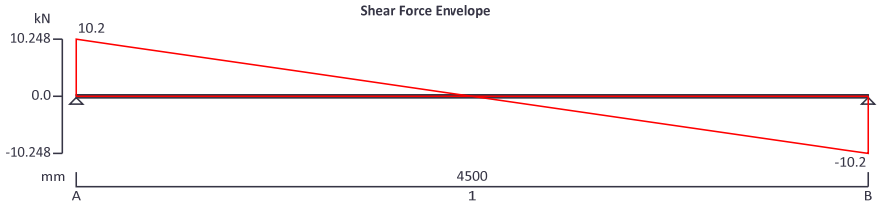








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Applied loading

Beam loads

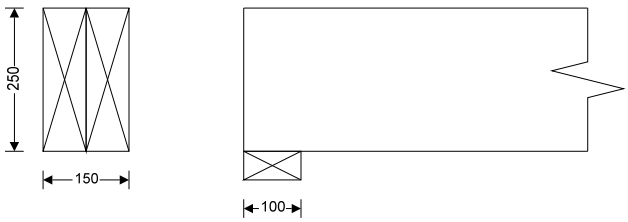
Dead self weight of beam × 1
Dead full UDL 2.560 kN/m
Imposed full UDL 1.840 kN/m

Load combinations

Load combination 1	Support A	Dead × 1.00
		Imposed × 1.00
	Span 1	Dead × 1.00
		Imposed × 1.00
	Support B	Dead × 1.00
		Imposed × 1.00

Analysis results

Maximum moment;	M_{max} = **11.528** kNm;	M_{min} = **0.000** kNm
Design moment;	M = max(abs(M_{max}),abs(M_{min})) = **11.528** kNm	
Maximum shear;	F_{max} = **10.248** kN;	F_{min} = **-10.248** kN
Design shear;	F = max(abs(F_{max}),abs(F_{min})) = **10.248** kN	
Total load on beam;	W_{tot} = **20.495** kN	
Reactions at support A;	R_{A_max} = **10.248** kN;	R_{A_min} = **10.248** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **6.108** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **4.140** kN	
Reactions at support B;	R_{B_max} = **10.248** kN;	R_{B_min} = **10.248** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **6.108** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **4.140** kN	



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Timber section details

Breadth of sections;	b = **75** mm
Depth of sections;	h = **250** mm
Number of sections in member;	N = **2**
Overall breadth of member;	$b_b = N \times b$ = **150** mm
Timber strength class;	**C24**

Member details

Service class of timber;	**1**
Load duration;	**Long term**
Length of span;	L_{s1} = **4500** mm
Length of bearing;	L_b = **100** mm

Section properties

Cross sectional area of member;	$A = N \times b \times h$ = **37500** mm^2
Section modulus;	$Z_x = N \times b \times h^2 / 6$ = **1562500** mm^3
	$Z_y = h \times (N \times b)^2 / 6$ = **937500** mm^3
Second moment of area;	$I_x = N \times b \times h^3 / 12$ = **195312500** mm^4
	$I_y = h \times (N \times b)^3 / 12$ = **70312500** mm^4
Radius of gyration;	$i_x = \sqrt{(I_x / A)}$ = **72.2** mm
	$i_y = \sqrt{(I_y / A)}$ = **43.3** mm

Modification factors

Duration of loading - Table 17;	K_3 = **1.00**
Bearing stress - Table 18;	K_4 = **1.00**
Total depth of member - cl.2.10.6;	$K_7 = (300 \text{ mm} / h)^{0.11}$ = **1.02**
Load sharing - cl.2.10.11;	K_8 = **1.10**
Minimum modulus of elasticity - Table 20;	K_9 = **1.14**

Lateral support - cl.2.10.8

Ends held in position and members held in line, as by purlins or tie rods at centres not more than 30 times the breadth of the member

Permissible depth-to-breadth ratio - Table 19;	**4.00**
Actual depth-to-breadth ratio;	$h / (N \times b)$ = **1.67**

PASS - Lateral support is adequate

Compression perpendicular to grain

Permissible bearing stress (no wane);	$\sigma_{c_adm} = \sigma_{cp1} \times K_3 \times K_4 \times K_8$ = **2.640** N/mm^2
Applied bearing stress;	$\sigma_{c_a} = R_{A_max} / (N \times b \times L_b)$ = **0.683** N/mm^2
	$\sigma_{c_a} / \sigma_{c_adm}$ = **0.259**

PASS - Applied compressive stress is less than permissible compressive stress at bearing

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Bending parallel to grain

Permissible bending stress; $\sigma_{m_adm} = \sigma_m \times K_3 \times K_7 \times K_8 =$ **8.417** N/mm^2

Applied bending stress; $\sigma_{m_a} = M / Z_x =$ **7.378** N/mm^2

$\sigma_{m_a} / \sigma_{m_adm} =$ **0.877**

PASS - Applied bending stress is less than permissible bending stress

Shear parallel to grain

Permissible shear stress; $\tau_{adm} = \tau \times K_3 \times K_8 =$ **0.781** N/mm^2

Applied shear stress; $\tau_a = 3 \times F / (2 \times A) =$ **0.410** N/mm^2

$\tau_a / \tau_{adm} =$ **0.525**

PASS - Applied shear stress is less than permissible shear stress

Deflection

Modulus of elasticity for deflection; $E = E_{min} \times K_9 =$ **10032** N/mm^2

Permissible deflection; $\delta_{adm} = \min(0.551 \text{ in}, 0.003 \times L_{s1}) =$ **13.500** mm

Bending deflection; $\delta_{b_s1} =$ **12.411** mm

Shear deflection; $\delta_{v_s1} =$ **0.588** mm

Total deflection; $\delta_a = \delta_{b_s1} + \delta_{v_s1} =$ **12.999** mm

$\delta_a / \delta_{adm} =$ **0.963**

PASS - Total deflection is less than permissible deflection

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RIDGE BEAM 1

SPAN OF THE BEAM = 7500 mm (6900+600)

LOADING:

I. Dead Load

Pitched roof = $\left\lfloor \frac{1.28kn/m^2 \text{ x } (4.0+4.0)m}{2} \right\rfloor$ = 5.12 Kn/m

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5.12 Kn/m

II. Imposed Load

Pitched roof = $\left\lfloor \frac{0.92kn/m^2 \text{ x } (4.0+4.0)m}{2} \right\rfloor$ = 3.68 Kn/m

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3.68 Kn/m

Point load from Hip Rafters

Pdl = 6.10 kN @ 0.1m

Pil = 4.14 kN @ 0.1m

Point load from Hip Rafters

Pdl = 6.10 kN @ 0.1m

Pil = 4.14 kN @ 0.1m

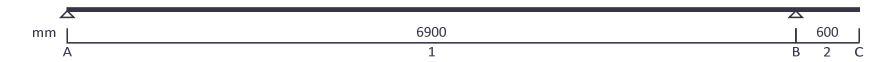
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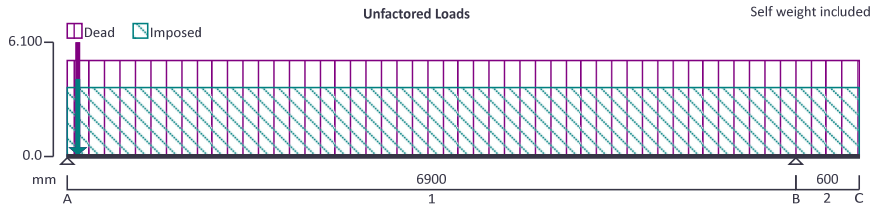
RIDGE BEAM 1 DESIGN:

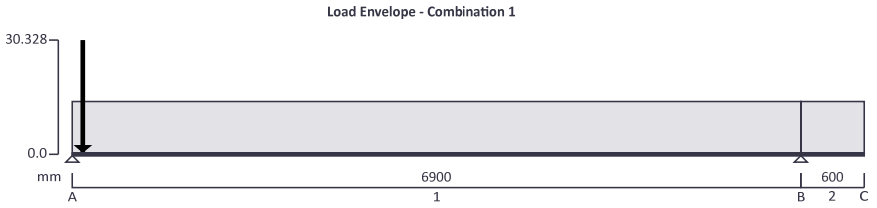
STEEL BEAM ANALYSIS & DESIGN (BS5950)

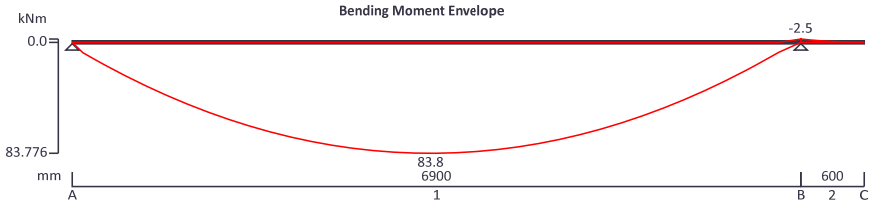
In accordance with BS5950-1:2000 incorporating Corrigendum No.1

TEDDS calculation version 3.0.07









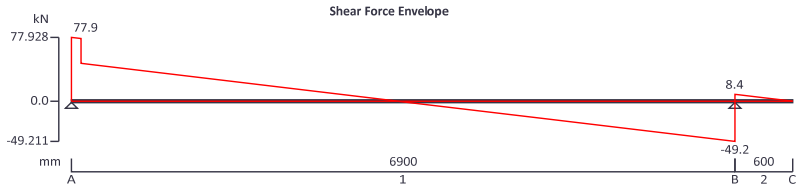
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Support conditions

Support A	Vertically restrained
	Rotationally free
Support B	Vertically restrained
	Rotationally free
Support C	Vertically free
	Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1
	Dead full UDL 5.12 kN/m
	Imposed full UDL 3.68 kN/m
	Dead point load 6.1 kN at 100 mm
	Imposed point load 4.14 kN at 100 mm
	Dead point load 6.1 kN at 100 mm
	Imposed point load 4.14 kN at 100 mm

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support C	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **83.8** kNm;	M_{min} = **-2.5** kNm
Maximum moment span 1;	M_{s1_max} = **83.8** kNm;	M_{s1_min} = **-2.5** kNm
Maximum moment span 2;	M_{s2_max} = **0** kNm;	M_{s2_min} = **-2.5** kNm
Maximum shear;	V_{max} = **77.9** kN;	V_{min} = **-49.2** kN
Maximum shear span 1;	V_{s1_max} = **77.9** kN;	V_{s1_min} = **-49.2** kN
Maximum shear span 2;	V_{s2_max} = **8.4** kN;	V_{s2_min} = **0** kN

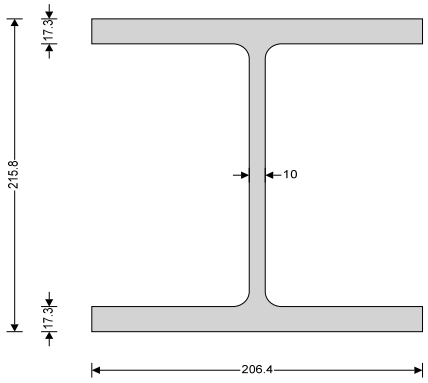
Deflection;	δ_{max} = **18** mm;	δ_{min} = **4.9** mm
Deflection span 1;	δ_{s1_max} = **18** mm;	δ_{s1_min} = **0** mm
Deflection span 2;	δ_{s2_max} = **0** mm;	δ_{s2_min} = **4.9** mm
Maximum reaction at support A;	R_{A_max} = **77.9** kN;	R_{A_min} = **77.9** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **31.9** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **20.8** kN	
Maximum reaction at support B;	R_{B_max} = **57.6** kN;	R_{B_min} = **57.6** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **23.9** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **15.1** kN	
Maximum reaction at support C;	R_{C_max} = **0** kN;	R_{C_min} = **0** kN

Section details

Section type;	**UC 203x203x71 (BS4-1)**
Steel grade;	**S355**

From table 9: Design strength p_y

Thickness of element;	max(T, t) = **17.3** mm
Design strength;	p_y = **345** N/mm^2
Modulus of elasticity;	E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only
Span 2 has lateral restraint at supports only
Cantilever tip is free
Cantilever support is continuous with lateral and torsional restraint

Effective length factors

Effective length factor in major axis;	K_x = **1.00**
Effective length factor in minor axis;	K_y = **1.00**
Effective length factor for lateral-torsional buckling;	$K_{LT.A}$ = **1.20**; + 2 × D
	$K_{LT.B}$ = **1.20**; + 2 × D
	$K_{LT.C}$ = **1.20**; + 2 × D

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB

Date: JUL 2024

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]} =$ **0.89**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

$d / t = 18.0 \times \varepsilon <= 80 \times \varepsilon$; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **103.2** mm

$b / T = 6.7 \times \varepsilon <= 9 \times \varepsilon$; Class 1 plastic

Section is class 1 plastic

Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(abs(V_{max}), abs(V_{min})) =$ **77.9** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **2158** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **446.7** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity at span 1 - Section 4.2.5

Design bending moment; $M = \max(abs(M_{s1_max}), abs(M_{s1_min})) =$ **83.8** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{xx}, 1.5 \times p_y \times Z_{xx}) =$ **275.6** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = 1.2 \times L_{s1} + 2 \times D =$ **8712** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **164.461**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; u = **0.853**

Torsional index; x = **11.926**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.555**

Ratio - cl.4.3.6.9; $\beta_W =$ **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **77.876**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.632**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; α_{LT} = **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0)$ = **0.331**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2$ = **333.6** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2$ = **394.5** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5})$ = **193.2** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; m_{LT} = **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx}$ = **154.3** kNm

M_b / m_{LT} = **154.3** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(20 \text{ mm}, L_{s1} / 250)$ = **20** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min}))$ = **18.012** mm

PASS - Maximum deflection does not exceed deflection limit

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RIDGE BEAM 2 (CRANKED)

SPAN OF THE BEAM = 7000 mm

LOADING:

I. Dead Load

Pitched roof = $\left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (3.0+2.0)\mathrm{m}}{2} \right\rfloor$ = 3.20 Kn/m

---

3.20 Kn/m

II. Imposed Load

Pitched roof = $\left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (3.0+2.0)\mathrm{m}}{2} \right\rfloor$ = 2.30 Kn/m

---

2.30 Kn/m

Point load from Ridge Beam 1

Pdl = 31.90 kN @ 3.5m

Pil = 20.80 kN @ 3.5m

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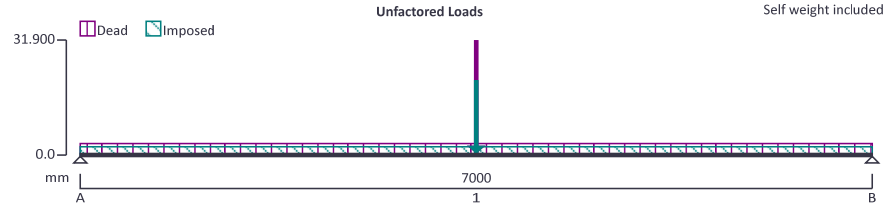
RIDGE BEAM 2 (CRANKED) DESIGN:

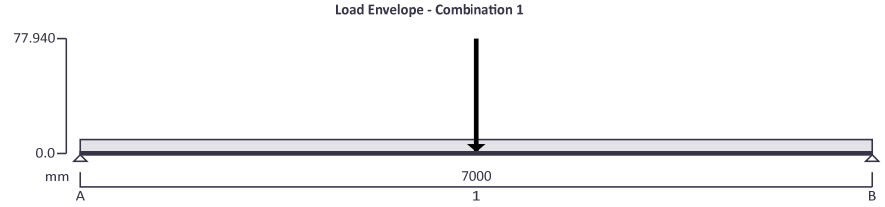
STEEL BEAM ANALYSIS & DESIGN (BS5950)

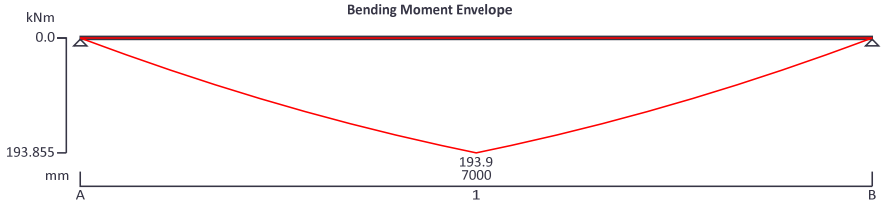
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TEDDS calculation version 3.0.07









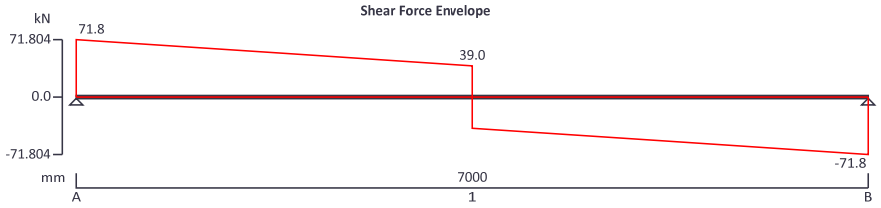
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Support conditions

Support A	Vertically restrained
	Rotationally free
Support B	Vertically restrained
	Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1
	Dead full UDL 3.2 kN/m
	Imposed full UDL 2.3 kN/m
	Dead point load 31.9 kN at 3500 mm
	Imposed point load 20.8 kN at 3500 mm

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **193.9** kNm;	M_{min} = **0** kNm
Maximum shear;	V_{max} = **71.8** kN;	V_{min} = **-71.8** kN
Deflection;	δ_{max} = **19.7** mm;	δ_{min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **71.8** kN;	R_{A_min} = **71.8** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **30.2** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **18.5** kN	
Maximum reaction at support B;	R_{B_max} = **71.8** kN;	R_{B_min} = **71.8** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **30.2** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **18.5** kN	

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Section details

Section type; **UC 254x254x89 (BS4-1)**

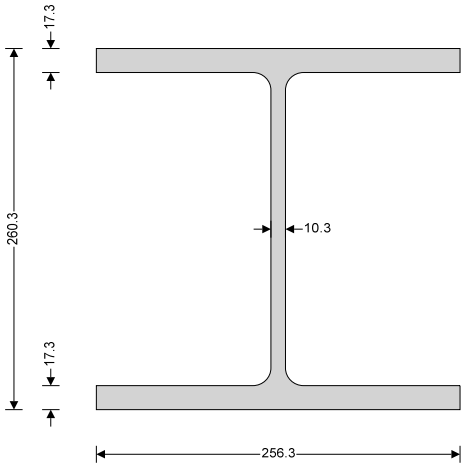
Steel grade; **S355**

From table 9: Design strength p_y

Thickness of element; max(T, t) = **17.3** mm

Design strength; p_y = **345** N/mm^2

Modulus of elasticity; E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Effective length factors

Effective length factor in major axis; K_x = **1.00**

Effective length factor in minor axis; K_y = **1.00**

Effective length factor for lateral-torsional buckling; $K_{LT.A}$ = **1.20**; + 2 × D

$K_{LT.B}$ = **1.20**; + 2 × D

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.89**

Internal compression parts - Table 11

Depth of section; d = **200.3** mm

d / t = 21.8 × ε <= 80 × ε; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **128.2** mm

b / T = 8.3 × ε <= 9 × ε; Class 1 plastic

Section is class 1 plastic

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB

Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(\text{abs}(V_{max}), \text{abs}(V_{min})) =$ **71.8** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **2681** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **555** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity - Section 4.2.5

Design bending moment; $M = \max(\text{abs}(M_{s1_max}), \text{abs}(M_{s1_min})) =$ **193.9** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{xx}, 1.2 \times p_y \times Z_{xx}) =$ **422.2** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = 1.2 \times L_{s1} + 2 \times D =$ **8921** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **136.247**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; $u =$ **0.850**

Torsional index; $x =$ **14.472**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.655**

Ratio - cl.4.3.6.9; $\beta_W =$ **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **75.831**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.632**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; $\alpha_{LT} =$ **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.316**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2 =$ **351.8** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **404.1** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5}) =$ **199.4** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; $m_{LT} =$ **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx} =$ **244** kNm

$M_b / m_{LT} =$ **244** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(20 \text{ mm}, L_{s1} / 250) =$ **20** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min})) =$ **19.686** mm

PASS - Maximum deflection does not exceed deflection limit

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BEAM 1

SPAN OF THE BEAM = 5300 mm

LOADING (Worst Case):

I. Dead Load

Flat Roof (Ext)	= $\left\lfloor \frac{1.0kn/\mathrm{m}^2 \text{ x } (1.5)\mathrm{m}}{2} \right\rfloor$	= 0.75 Kn/m
Pitch Roof	= $\left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 1.92 Kn/m
Floor (Concrete)	= $\left\lfloor \frac{3.6kn/\mathrm{m}^2 \text{ x } (5.3)\mathrm{m}}{2} \right\rfloor$	= 9.54 Kn/m
Ceiling (Timber)	= $\left\lfloor \frac{0.3kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 0.45 Kn/m
Brick work	= 20 Kn/m³ x 0.100 x 2.5	= 5.00 Kn/m
Block work	= 19 Kn/m³ x 0.100 x 2.5	= 4.75 Kn/m
		22.41 Kn/m

II. Imposed Load

Flat Roof (Ext)	= $\left\lfloor \frac{1.0kn/\mathrm{m}^2 \text{ x } (1.5)\mathrm{m}}{2} \right\rfloor$	= 0.56 Kn/m
Pitch Roof	= $\left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 1.38 Kn/m
Floor (Concrete)	= $\left\lfloor \frac{1.5kn/\mathrm{m}^2 \text{ x } (5.3)\mathrm{m}}{2} \right\rfloor$	= 3.98 Kn/m
Ceiling	= $\left\lfloor \frac{0.5kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 0.75 Kn/m
		6.67 Kn/m

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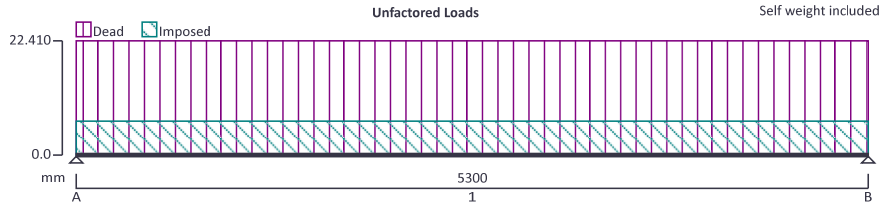
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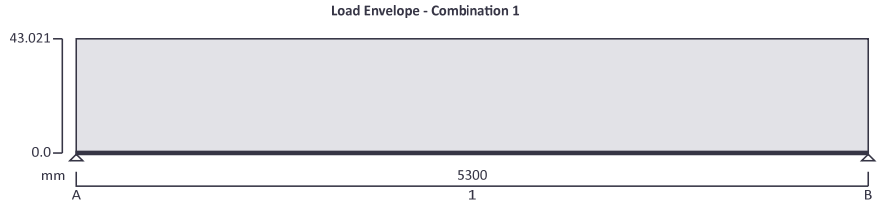
STEEL BEAM ANALYSIS & DESIGN (BS5950)

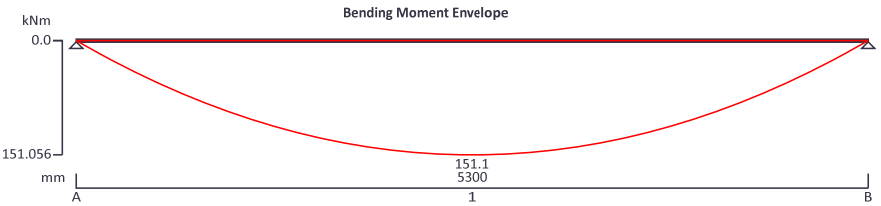
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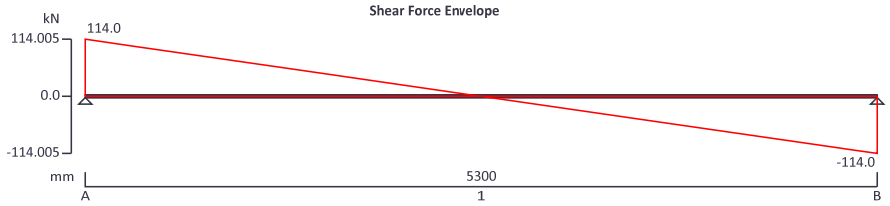








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Support conditions

Support A	Vertically restrained
	Rotationally free
Support B	Vertically restrained
	Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1
	Dead full UDL 22.41 kN/m
	Imposed full UDL 6.67 kN/m

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **151.1** kNm;	M_{min} = **0** kNm
Maximum shear;	V_{max} = **114** kN;	V_{min} = **-114** kN
Deflection;	δ_{max} = **19.6** mm;	δ_{min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **114** kN;	R_{A_min} = **114** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **61.2** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **17.7** kN	
Maximum reaction at support B;	R_{B_max} = **114** kN;	R_{B_min} = **114** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **61.2** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **17.7** kN	

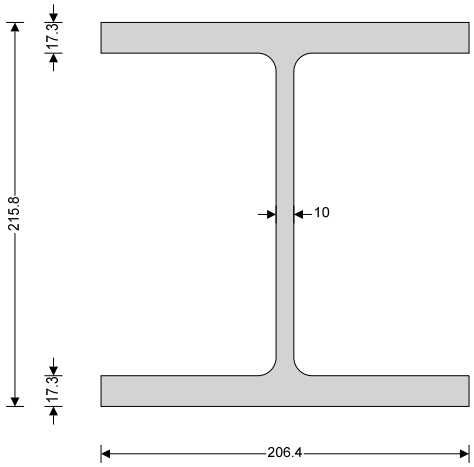
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Section details

Section type;	**UC 203x203x71 (BS4-1)**
Steel grade;	**S355**
From table 9: Design strength p_y	
Thickness of element;	max(T, t) = **17.3** mm
Design strength;	p_y = **345** N/mm^2
Modulus of elasticity;	E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Effective length factors

Effective length factor in major axis;	K_x = **1.00**
Effective length factor in minor axis;	K_y = **1.00**
Effective length factor for lateral-torsional buckling;	$K_{LT.A}$ = **1.20**; + 2 × D
	$K_{LT.B}$ = **1.00**;

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.89**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

d / t = 18.0 × ε <= 80 × ε; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **103.2** mm

b / T = 6.7 × ε <= 9 × ε; Class 1 plastic

Section is class 1 plastic

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Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(\text{abs}(V_{max}), \text{abs}(V_{min})) =$ **114** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **2158** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **446.7** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity - Section 4.2.5

Design bending moment; $M = \max(\text{abs}(M_{s1_max}), \text{abs}(M_{s1_min})) =$ **151.1** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{xx}, 1.2 \times p_y \times Z_{xx}) =$ **275.6** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = ((1.2 + 1.2) \times L_{s1} + 2 \times D) / 2 =$ **6576** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **124.141**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; $u =$ **0.853**

Torsional index; $x =$ **11.926**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.628**

Ratio - cl.4.3.6.9; $\beta_W =$ **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **66.495**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.632**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; $\alpha_{LT} =$ **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.251**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2 =$ **457.6** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **458.7** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5}) =$ **229.5** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; $m_{LT} =$ **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx} =$ **183.3** kNm

$M_b / m_{LT} =$ **183.3** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(20 \text{ mm}, L_{s1} / 250) =$ **20** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min})) =$ **19.589** mm

PASS - Maximum deflection does not exceed deflection limit

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BEAM 2

SPAN OF THE BEAM = 4900 mm

LOADING (Worst Case):

I. Dead Load

Floor (Concrete) = $\left\lfloor \frac{3.6kn/\mathrm{m}^2 \text{ x } (5.3+3.2)\mathrm{m}}{2} \right\rfloor$ = 15.30 Kn/m

Ceiling (Timber) = $\left\lfloor \frac{0.3kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$ = 0.12 Kn/m

15.42 Kn/m

II. Imposed Load

Floor (Concrete) = $\left\lfloor \frac{1.5kn/\mathrm{m}^2 \text{ x } (5.3+3.2)\mathrm{m}}{2} \right\rfloor$ = 6.38 Kn/m

Ceiling = $\left\lfloor \frac{0.5kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$ = 0.20 Kn/m

6.58 Kn/m

Partial UDL of brick & Block wall from 3.1m - 4.9m

Pitch Roof = $\left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (2.0)\mathrm{m}}{2} \right\rfloor$ = 1.28 Kn/m

Pitch Roof = $\left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (2.0)\mathrm{m}}{2} \right\rfloor$ = 0.92 Kn/m

Partial UDL of brick & Block wall from 3.1m - 4.9m

Brick work = 20 Kn/m³ x 0.100 x 2.5 = 5.00 Kn/m

Block work = 19 Kn/m³ x 0.100 x 2.5 = 4.75 Kn/m

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Point load from Beam 1

Pdl = 61.20 Kn @ 3.1m

Pil = 17.70 kN @3.1m

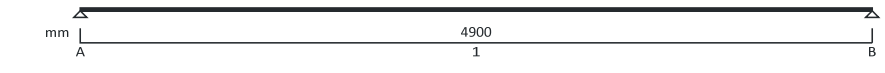
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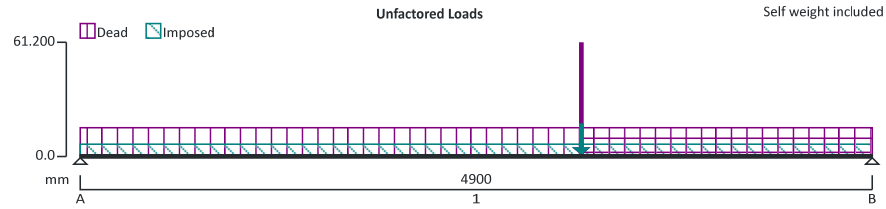
BEAM 2 DESIGN:

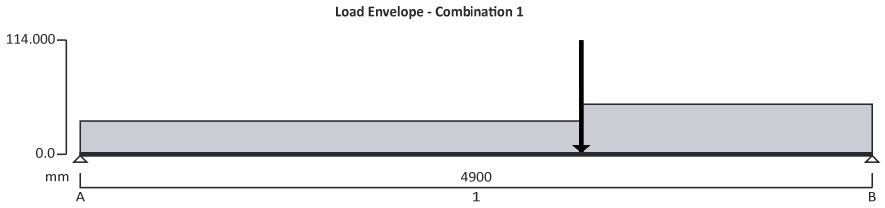
STEEL BEAM ANALYSIS & DESIGN (BS5950)

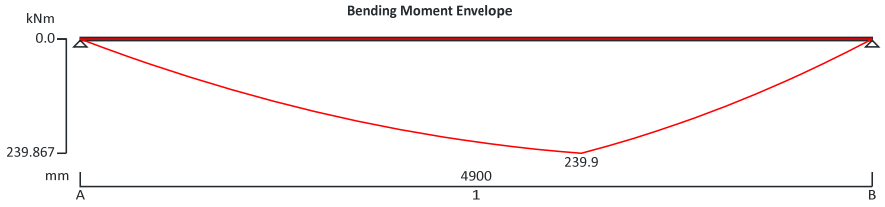
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TEDDS calculation version 3.0.08

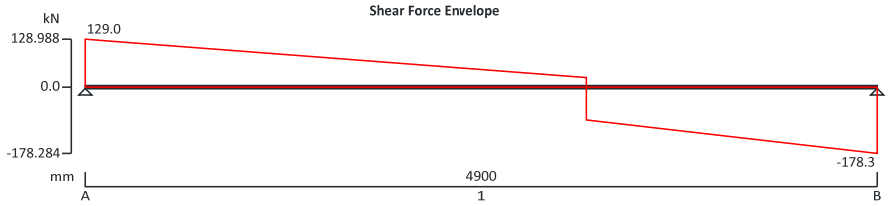








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Support conditions

Support A	Vertically restrained
	Rotationally free
Support B	Vertically restrained
	Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1
	Dead full UDL 15.42 kN/m
	Imposed full UDL 6.58 kN/m
	Dead partial UDL 2.2 kN/m from 3100 mm to 4900 mm
	Dead partial UDL 9.75 kN/m from 3100 mm to 4900 mm
	Dead point load 61.2 kN at 3100 mm
	Imposed point load 17.7 kN at 3100 mm

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **239.9** kNm;	M_{min} = **0** kNm
Maximum shear;	V_{max} = **129** kN;	V_{min} = **-178.3** kN
Deflection;	δ_{max} = **19.3** mm;	δ_{min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **129** kN;	R_{A_min} = **129** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **66.3** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **22.6** kN	
Maximum reaction at support B;	R_{B_max} = **178.3** kN;	R_{B_min} = **178.3** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **96.1** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **27.3** kN	

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Section details

Section type; **UC 203x203x86 (BS4-1)**

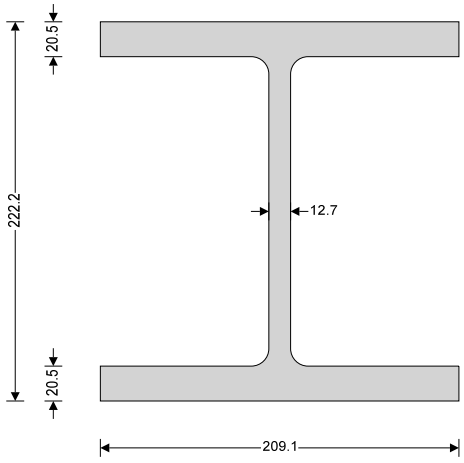
Steel grade; **S355**

From table 9: Design strength p_y

Thickness of element; max(T, t) = **20.5** mm

Design strength; p_y = **345** N/mm^2

Modulus of elasticity; E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Effective length factors

Effective length factor in major axis; K_x = **1.00**

Effective length factor in minor axis; K_y = **1.00**

Effective length factor for lateral-torsional buckling; $K_{LT.A}$ = **1.20**; + 2 × D

$K_{LT.B}$ = **1.00**;

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.89**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

d / t = 14.2 × ε <= 80 × ε; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **104.6** mm

b / T = 5.7 × ε <= 9 × ε; Class 1 plastic

Section is class 1 plastic

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Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(\text{abs}(V_{max}), \text{abs}(V_{min})) =$ **178.3** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **2822** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **584.1** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity - Section 4.2.5

Design bending moment; $M = \max(\text{abs}(M_{s1_max}), \text{abs}(M_{s1_min})) =$ **239.9** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{xx}, 1.2 \times p_y \times Z_{xx}) =$ **337** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = ((1.2 + 1.2) \times L_{s1} + 2 \times D) / 2 =$ **6102** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **114.253**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; $u =$ **0.850**

Torsional index; $x =$ **10.220**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.609**

Ratio - cl.4.3.6.9; $\beta_W =$ **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **59.187**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.632**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; $\alpha_{LT} =$ **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.200**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2 =$ **577.6** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **519** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5}) =$ **254.2** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; $m_{LT} =$ **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx} =$ **248.3** kNm

$M_b / m_{LT} =$ **248.3** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(20 \text{ mm}, L_{s1} / 250) =$ **19.6** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min})) =$ **19.286** mm

PASS - Maximum deflection does not exceed deflection limit

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BEAM 3

SPAN OF THE BEAM = 7000 mm (3600+3400)

LOADING (Worst Case):

I. Dead Load

Pitch Roof	$= \left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 1.92 Kn/m
Pitch Roof (Ext)	$= \left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (4.0)\mathrm{m}}{2} \right\rfloor$	= 2.56 Kn/m
Floor (Concrete)	$= \left\lfloor \frac{3.6kn/\mathrm{m}^2 \text{ x } (5.3)\mathrm{m}}{2} \right\rfloor$	= 9.54 Kn/m
Ceiling (Timber)	$= \left\lfloor \frac{0.3kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$	= 0.12 Kn/m
		14.14 Kn/m

II. Imposed Load

Pitch Roof	$= \left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 1.38 Kn/m
Pitch Roof (Ext)	$= \left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (4.0)\mathrm{m}}{2} \right\rfloor$	= 1.84 Kn/m
Floor (Concrete)	$= \left\lfloor \frac{1.5kn/\mathrm{m}^2 \text{ x } (5.3)\mathrm{m}}{2} \right\rfloor$	= 3.98 Kn/m
Ceiling	$= \left\lfloor \frac{0.5kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$	= 0.20 Kn/m
		7.40 Kn/m

Partial UDL of brick & Block wall from 0.0m – 5.5m

Brick work = 20 Kn/m³ x 0.100 x 2.5 = 5.00 Kn/m

Block work = 19 Kn/m³ x 0.100 x 2.5 = 4.75 Kn/m

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Partial UDL of brick & Block wall from 5.5m – 7.0m

Brick work = 20 Kn/m³ x 0.100 x 1.0 = 2.00 Kn/m

Block work = 19 Kn/m³ x 0.100 x 1.0 = 1.90 Kn/m

Point load from Beam 1

Pdl = 61.20 Kn @ 5.5m

Pil = 17.70 kN @5.5m

Considering Point load from Ridge Beam 2

Pdl = 16.00 Kn @ 0.8m

Pil = 10.00 kN @0.8m

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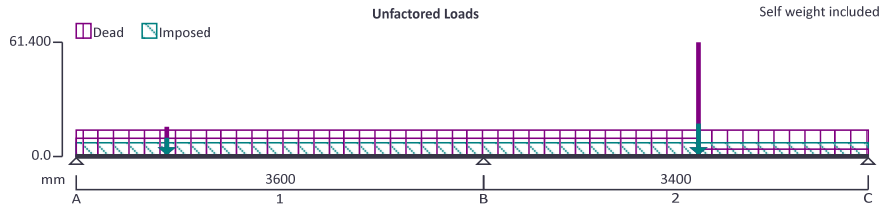
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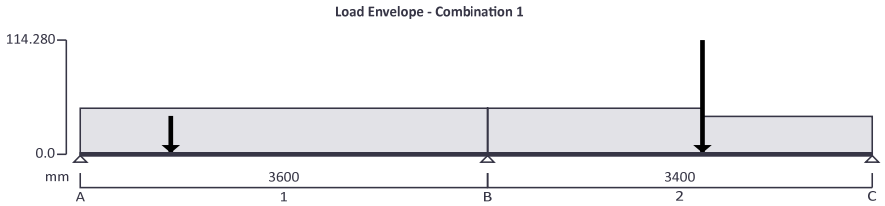
STEEL BEAM ANALYSIS & DESIGN (BS5950)

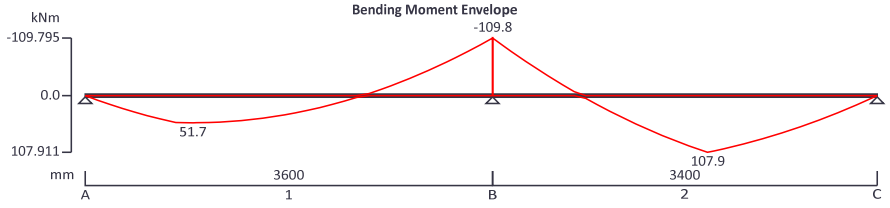
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TEDDS calculation version 3.0.07

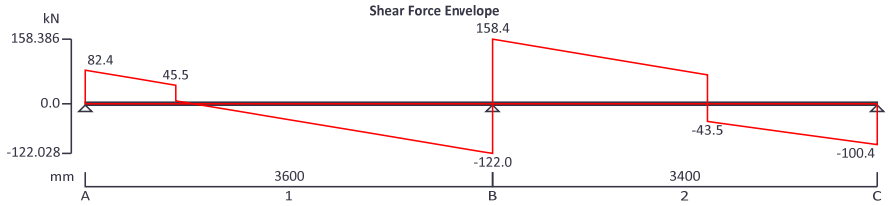








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Support conditions

Support A	Vertically restrained
	Rotationally free
Support B	Vertically restrained
	Rotationally free
Support C	Vertically restrained
	Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1
	Dead full UDL 14.14 kN/m
	Imposed full UDL 7.4 kN/m
	Dead partial UDL 9.75 kN/m from 0 mm to 5500 mm
	Dead partial UDL 3.9 kN/m from 5500 mm to 7000 mm
	Dead point load 16 kN at 800 mm
	Imposed point load 10 kN at 800 mm
	Dead point load 61.4 kN at 5500 mm
	Imposed point load 17.7 kN at 5500 mm

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support C	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **107.9** kNm;	M_{min} = **-109.8** kNm
Maximum moment span 1;	M_{s1_max} = **51.7** kNm;	M_{s1_min} = **-109.8** kNm
Maximum moment span 2;	M_{s2_max} = **107.9** kNm;	M_{s2_min} = **-109.8** kNm

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Maximum shear;	V_{max} = **158.4** kN;	V_{min} = **-122** kN
Maximum shear span 1;	V_{s1_max} = **82.4** kN;	V_{s1_min} = **-122** kN
Maximum shear span 2;	V_{s2_max} = **158.4** kN;	V_{s2_min} = **-100.4** kN
Deflection;	δ_{max} = **4.9** mm;	δ_{min} = **0.2** mm
Deflection span 1;	δ_{s1_max} = **2.2** mm;	δ_{s1_min} = **0.2** mm
Deflection span 2;	δ_{s2_max} = **4.9** mm;	δ_{s2_min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **82.4** kN;	R_{A_min} = **82.4** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **40.6** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **16** kN	
Maximum reaction at support B;	R_{B_max} = **280.4** kN;	R_{B_min} = **280.4** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **147.1** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **46.5** kN	
Maximum reaction at support C;	R_{C_max} = **100.4** kN;	R_{C_min} = **100.4** kN
Unfactored dead load reaction at support C;	R_{C_Dead} = **52.2** kN	
Unfactored imposed load reaction at support C;	$R_{C_Imposed}$ = **17** kN	

Section details

Section type; **UC 203x203x60 (BS4-1)**

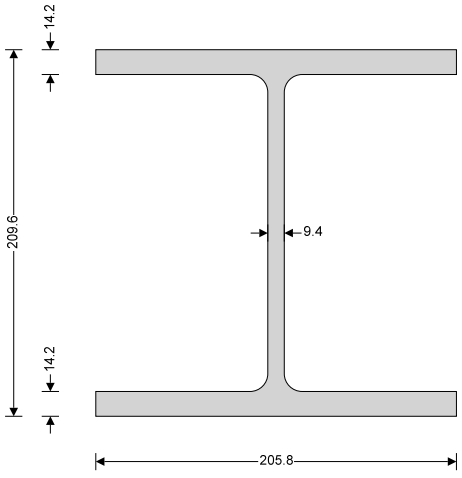
Steel grade; **S355**

From table 9: Design strength p_y

Thickness of element; max(T, t) = **14.2** mm

Design strength; p_y = **355** N/mm^2

Modulus of elasticity; E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Span 2 has full lateral restraint

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Effective length factors

Effective length factor in major axis; K_x = **1.00**

Effective length factor in minor axis; K_y = **1.00**

Effective length factor for lateral-torsional buckling; $K_{LT.A}$ = **1.20**; + 2 × D

$K_{LT.B}$ = **1.00**;

$K_{LT.C}$ = **1.00**;

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.88**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

$d / t = 19.4 \times \varepsilon <= 80 \times \varepsilon$; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **102.9** mm

$b / T = 8.2 \times \varepsilon <= 9 \times \varepsilon$; Class 1 plastic

Section is class 1 plastic

Shear capacity - Section 4.2.3

Design shear force; $F_v = max(abs(V_{max}), abs(V_{min}))$ = **158.4** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D$ = **1970** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v$ = **419.7** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity at span 1 - Section 4.2.5

Design bending moment; $M = max(abs(M_{s1_max}), abs(M_{s1_min}))$ = **109.8** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = min(p_y \times S_{xx}, 1.5 \times p_y \times Z_{xx})$ = **232.9** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = ((1.2 + 1.2) \times L_{s1} + 2 \times D) / 2$ = **4530** mm

Slenderness ratio; $\lambda = L_E / r_{yy}$ = **87.119**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; u = **0.846**

Torsional index; x = **14.064**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25}$ = **0.765**

Ratio - cl.4.3.6.9; β_W = **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]}$ = **56.395**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5}$ = **30.198**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; α_{LT} = **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0)$ = **0.183**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2$ = **636.2** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2$ = **553.9** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5})$ = **269.3** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; m_{LT} = **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx}$ = **176.7** kNm

M_b / m_{LT} = **176.7** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(14 \text{ mm}, L_{s2} / 250)$ = **13.6** mm

Maximum deflection span 2; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min}))$ = **4.897** mm

PASS - Maximum deflection does not exceed deflection limit

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BEAM 4

SPAN OF THE BEAM = 5300 mm

LOADING

I. Dead Load

Flat Roof (Ext)	= $\left\lfloor \frac{1.0Kn/\mathrm{m}^2 \text{ x } (1.5)\mathrm{m}}{2} \right\rfloor$	= 0.75 Kn/m
Bifolding Door	= 1.2 Kn/m^2 x 2.1 m	= 2.52 Kn/m
Brick work	= 20 Kn/m^3 x 0.100 x 1.2	= 2.40 Kn/m
Block work	= 19 Kn/m^3 x 0.100 x 1.2	= 2.28 Kn/m
		7.95 Kn/m

II. Imposed Load

Flat Roof (Ext)	= $\left\lfloor \frac{0.75kn/\mathrm{m}^2 \text{ x } (1.5)\mathrm{m}}{2} \right\rfloor$	= 0.56 Kn/m
		0.56 Kn/m

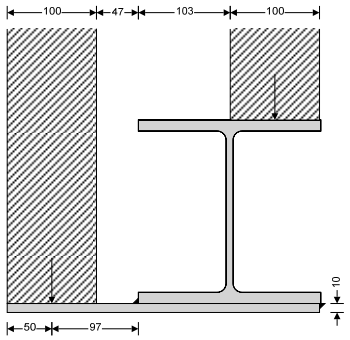
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BEAM 4 DESIGN:

STEEL MASONRY SUPPORT

In accordance with BS5950-1:2000 incorporating Corrigendum No.1

Tedds calculation version 1.0.04



Steel member details

Torsion beam;	**UC 203x203x52**
Masonry support plate;	User
Steel grade of support plate;	**S355**
Design strength of support plate;	p_{ysb} = **355** N/mm^2
Modulus of elasticity;	E = 205000 N/mm^2
Constant;	$\varepsilon = \sqrt{(275N/mm^2 / p_{ysb})}$ = **0.880**
Length of plate beyond beam;	l_h = **147** mm
Total length of plate;	l_{plate} = **350** mm
Thickness of plate;	t_{sb} = **10** mm
Width of main beam;	B_{mb} = **204** mm
Area of plate;	$A_{sbu} = t_{sb} \times l_{plate}$ = **3500.0** mm^2
Distance from weld position to CoG;	$c_{yysb} = l_h / 2 - (l_{plate} - l_h) / 2$ = **-28** mm

Supported materials detail

Density of masonry on main beam;	$\rho_{m,mb}$ = **20.0** kN/m^3
Width of masonry on main beam;	b_{mmb} = **100** mm
Height of masonry on main beam;	h_{mmb} = **1200** mm
Eccentricity of main beam material;	e_{mb} = **103** mm
Add dead force main beam (not from masonry);	P_{Gaddmb} = **8.0** kN/m
Add live force main beam (not from masonry);	P_{Qaddmb} = **0.6** kN/m
Density of masonry on support beam;	$\rho_{m,sb}$ = **19.0** kN/m^3
Width of masonry on support beam;	b_{msb} = **100** mm
Height of masonry on support beam;	h_{msb} = **1200** mm
Add dead force support beam (not from masonry);	P_{Gaddsb} = **0.0** kN/m
Add live force support beam (not from masonry);	P_{Qaddsb} = **0.0** kN/m

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Geometry

Cavity width;	c = **150** mm
Supported width of masonry;	$d_m = l_h + e_{mb} - c$ = **100** mm

Biaxial stress effects in the plate (SCI-P-110)

Maximum overall bending moment;	M_x = **69.1** kNm
Dist to NA combined section (CoG torsion beam);	$y_{e,all} = (D_{mb} + t_{sb}) \times A_{sbu} / (2 \times (A_{mb} + A_{sbu}))$ = **37** mm
Second moment of area of combined section;	$I_{xx,all} = (I_{xxmb} + A_{mb} \times y_{e,all}^2) + A_{sbu} \times (D_{mb} / 2 + t_{sb} / 2 - y_{e,all})^2$ = **7935** cm^4
Elastic section modulus of combined section;	$Z_{xx,all} = I_{xx,all} / (D_{mb} / 2 + t_{sb} - y_{e,all})$ = **1047.66** cm^3
Section modulus of plate;	$Z_{xx,plate} = 1m \times t_{sb}^2 / (6 \times 1m)$ = **16.67** cm^3/m
Eccentricity of support beam masonry;	e_1 = **97** mm
Force of masonry on support plate;	$P_1 = (b_{msb} \times h_{msb} \times \rho_{m,sb} + P_{Gaddsb}) \times \gamma_{fG} + P_{Qaddsb} \times \gamma_{fQ}$ = **3.2** kN/m
Bending at heel;	$M_{x,plate} = P_1 \times e_1$ = **0.3** kNm/m
Moment capacity of plate;	$M_c = 1.2 \times Z_{x,plate} \times p_{ysb}$ = **7.1** kNm/m

PASS - Design strength exceeds stress at heel

Longitudinal stress due to overall bending;	$\sigma_1 = M_x / Z_{xx,all}$ = **66.0** N/mm^2
Constant relating to Von Mises curve;	$c_{fp} = (4 \times p_{ysb}^2 - 3 \times \sigma_1^2)^{0.5}$ = **700.7** N/mm^2
Transverse bending stress ratio limit;	$\alpha_{ts} = (c_{fp}^2 - \sigma_1^2) / (2 \times c_{fp} \times p_{ysb})$ = **0.978**
Transverse bending stress ratio;	$\alpha_{ls} = M_{x,plate} / M_c$ = **0.044**

PASS - Transverse bending stress ratio less than allowable limit

Deflection at toe

Unfactored force on support angle;	$P_{1SLS} = b_{msb} \times h_{msb} \times \rho_{m,sb} + P_{Gaddsb} + P_{Qaddsb}$ = **2.3** kN/m
Distance from weld to load position;	$a_m = e_1$ = **97** mm
Length of load resultant to edge of plate;	$b_m = l_h - e_1$ = **50** mm
Dist from weld to load position as ratio of length;	$a_l = a_m / (a_m + b_m)$ = **0.660**
Effective second moment of inertia;	$I_{eff_def} = t_{sb}^3 / 12$ = **83333** mm^4/m
Deflection at toe;	$\delta = (a_l^2 \times (3 - a_l) / 6) \times (P_{1SLS} \times (a_m + b_m)^3) / (E_{S5950} \times I_{eff_def})$ = **0.07** mm
Deflection limit;	δ_{lim} = **1.67** mm

PASS - Deflection is within specified criteria

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Weld details - assume a full length weld and that the plate acts as a propped cantilever with the prop at the weld position and the fixed end at the centre of the torsion beam

Leg length of weld; s_{weld} = **6** mm

Throat size of weld; $a_{weld} = 1/\sqrt{(2)} \times s_{weld}$ = **4.2** mm

Shear force at weld position; $R_A = P_1 \times maX((1 + (3 \times e_1) / (2 \times B_{mb} / 2)), 1.4)$ = **7.7** kN/m

Maximum possible force in plate; $R_p = (l_h + B_{mb}) \times t_{sb} \times p_{ysb}$ = **1247.1** kN

Longitudinal shear between beam and plate; $R_l = 2 \times R_p / L$ = **470.6** kN/m

Horizontal shear between beam and plate; $R_h = P_1 \times e_1 / (s_{weld} / 2 + t_{sb} / 2)$ = **38.7** kN/m

Resultant weld force; $R_{weld} = (R_A^2 + R_l^2 + R_h^2)^{0.5}$ = **0.472** kN/mm

Strength of weld (Table 37); p_{weld} = **220.0** N/mm^2

Capacity of full length weld; $p_{c,weld} = a_{weld} \times p_{weld}$ = **0.933** kN/mm

PASS - Capacity of weld exceeds resultant force on weld

Torsional loading ULS

Loading of support beam masonry; $w_{1ULS} = (h_{msb} \times b_{msb} \times \rho_{m,sb} + P_{Gaddsb}) \times \gamma_{fG} + P_{Qaddsb} \times \gamma_{fQ}$ = **3.19** kN/m

Loading of main beam masonry; $w_{2ULS} = (h_{mmb} \times b_{mmb} \times \rho_{m,mb} + P_{Gaddmb}) \times \gamma_{fG} + P_{Qaddmb} \times \gamma_{fQ}$ = **15.39** kN/m

Self weight of support beam; $w_{3ULS} = A_{sbu} \times \rho_{sb} \times \gamma_{fG}$ = **0.38** kN/m

Torsional loading SLS

Loading of support beam masonry; $w_{1SLS} = h_{msb} \times b_{msb} \times \rho_{m,sb} + P_{Gaddsb} + P_{Qaddsb}$ = **2.28** kN/m

Loading of main beam masonry; $w_{2SLS} = h_{mmb} \times b_{mmb} \times \rho_{m,mb} + P_{Gaddmb} + P_{Qaddmb}$ = **10.91** kN/m

Self weight of support beam; $w_{3SLS} = A_{sbu} \times \rho_{sb}$ = **0.27** kN/m

Eccentricities

Distance to shear centre of main beam; e_{0mb} = **0** mm

Eccentricity of support beam masonry; $e_{1mb} = (B_{mb} + b_{msb}) / 2 + c - e_{mb}$ = **199** mm

Eccentricity of main beam masonry; $e_{2mb} = (B_{mb} - b_{mmb}) / 2 - e_{mb}$ = **-51** mm

Eccentricity of support beam; $e_{3mb} = B_{mb} / 2 + c_{yysb}$ = **74** mm

Torsional effects

Applied torque (ULS); $T_{qULS} = abs(w_{1ULS} \times e_{1mb} + w_{2ULS} \times e_{2mb} + w_{3ULS} \times e_{3mb})$ = **0.12** kNm/m

Total torque (ULS); $T_q = T_{qULS} \times L$ = **0.63** kNm

Applied torque (SLS); $T_{qSLS} = abs(w_{1SLS} \times e_{1mb} + w_{2SLS} \times e_{2mb} + w_{3SLS} \times e_{3mb})$ = **0.08** kNm/m

Total torque (SLS); $T_{qu} = T_{qSLS} \times L$ = **0.43** kNm

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STEEL BEAM TORSION DESIGN

In accordance with BS5950-1:2000 incorporating Corrigendum No.1

Tedds calculation version 2.0.02

Section details

Section type;	**UC 203x203x52**
Steel grade;	**S355**
Design stength;	$p_{yw} = p_y =$ **355** N/mm^2
Constant;	$\varepsilon = \sqrt{(275\ \text{N/mm}^2 / p_y)} =$ **0.880**

Geometry - Beam unrestrained against lateral-torsional buckling between supports.

Effective span;	L = **5300** mm
Length of segment for LT buckling;	L_{LT} = **5300** mm

Compression flanges laterally unrestrained

Partial torsional restraint against rotation about longitudinal axis provided by connection of bottom flange to supports

Effective length for LT buckling; $L_{E_LT} = L_{LT} \times 1.2 + 2 \times D =$ **6772** mm

Loading - Torsional loading comprises only full-length uniformly distributed load(s)

Internal forces & moments on member under factored loading for uls design

Applied shear force;	F_{vy} = **52.2** kN
Maximum bending moment;	$M_{LT} = M_x =$ **69.14** kNm
Applied torque;	$T_q =$ **0.63** kNm
Minor axis bending moment;	M_y = 0 kNm
Compression force;	F_c = 0 kN

Equivalent uniform moment factors

EUM factor (Cl. 4.3.6.6 and T18); $m_{LT} =$ **1.000**

Torsional deflection parameters

Beam is torsion fixed and warping free at each end. (as defined in SCI-P-057 section 2.1.6) - Appendix B case 4

Dist along the beam for first derivative of twist; z_1 = 0 mm

Dist along the beam for second derivative of twist; $z_2 = L / 2 =$ **2650** mm

First derivative of angle of twist; $\phi'_1 = T_q / (G \times J) \times a / L \times [L^2 / (2 \times a) \times (1 / L - 2 \times z_1 / L^2) + \sinh(z_1 / a) - \tanh(L / (2 \times a)) \times \cosh(z_1 / a)] \times 1$ rads = **7.11×10^{-3}** rads/m

Third derivative of angle of twist; $\phi'''_1 = T_q / (G \times J \times a^2) \times a/L \times [\sinh(z_1 / a) - \tanh(L / (2 \times a)) \times \cosh(z_1 / a)] \times 1$ rads = **-3.95×10^{-3}** rads/m^3

Angle of twist; $\phi_2 = T_q \times a / (G \times J) \times a / L \times [L^2 / (2 \times a^2) \times (z_2 / L - z_2^2 / L^2) + \cosh(z_2 / a) - \tanh(L / (2 \times a)) \times \sinh(z_2 / a) - 1] \times 1$ rads = **0.011** rads

Second derivative of angle of twist; $\phi''_2 = T_q / (G \times J \times a) \times a / L \times [\cosh(z_2 / a) - \tanh(L / (2 \times a)) \times \sinh(z_2 / a) - 1] \times 1$ rads = **-3.75×10^{-3}** rads/m^2

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Design parameters

Total angle of twist; $\phi = abs(\phi_2) =$ **0.011** rads

First derivative of ϕ; $\phi' = abs(\phi'_1) =$ **7.11×10^{-3}** rads/m

Second derivative of ϕ; $\phi'' = abs(\phi''_2) =$ **3.75×10^{-3}** rads/m^2

Third derivative of ϕ; $\phi''' = abs(\phi'''_1) =$ **3.95×10^{-3}** rads/m^3

Section classification

$b / T =$ **8.2**

$d / t =$ **20.4**

$r_{1s} = \min(1.0, \max(-1.0, F_c / (d \times t \times p_{yw}))) =$ **0.000**

$r_{2s} = F_c / (A_g \times p_{yw}) =$ **0.000**

Section classification is compact

Shear capacity (parallel to y-axis)

Design shear force; $F_{vy} =$ **52.2** kN

Design shear resistance (Cl. 4.2.3); $P_{vy} = 0.6 \times p_y \times A_{vy} =$ **347.0** kN

Pass - Shear

Moment capacity (x-axis)

Design bending moment; $M_x =$ **69.1** kNm

Moment capacity; $M_{cxu} = p_y \times S_x =$ **201.4** kNm

Moment capacity low shear (Cl. 4.2.5.1); $M_{cx} = \min(p_y \times S_x, 1.2 \times p_y \times Z_x) =$ **201.4** kNm

Pass - Moment capacity exceeds design bending moment

Lateral torsional buckling

Effective length for lateral torsional buckling; $L_{E_LT} =$ **6772** mm

Slenderness ratio; $\lambda = L_{E_LT} / r_y =$**131**

Buckling parameter; $u =$ **0.848**

Flange ratio; $\eta = 0.5$

Torsional index; $x =$ **15.8**

Slenderness factor; $v = 1 / (1 + 0.05 \times (\lambda / x)^2)^{0.25} =$ **0.69**

Ratio - cl 4.3.6.9; $\beta_w = 1.0 =$ **1.000**

Equvalent slenderness – cl 4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{(\beta_w)} =$ **77**

Limiting slendernes – Annex B2.2; $\lambda_{L0} = 0.4 \times \sqrt{(\pi^2 \times E_{S5950} / p_y)} =$ **30**

Euler stress; $p_E = \pi^2 \times E_{S5950} / \lambda_{LT}^2 =$ **345** N/mm^2

Perry factor; $\eta_{LT} = \max(7.0 \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.324**

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **406232770.716**

Bending strength; $p_b = p_E \times p_y / (\phi_{LT} + \sqrt{(\phi_{LT}^2 - p_E \times p_y)}) =$ **200** N/mm^2

Buckling resistance moment; $M_b = p_b \times S_x =$ **113.7** kNm

Max moment governing buckling resistance; $M_{LT} =$ **69.1** kNm

Equiv uniform moment factor for LTB; $m_{LT} =$ **1.00**

$M_b / m_{LT} =$ **113.7** kNm

Pass - lat. tors. buckling

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Buckling under combined bending & torsion -SCI-P-057 section 2.3

For simplicity, a conservative check is applied using the maximum stresses due to each of the separate load effects, even though these do not necessarily all occur at the same section along the member.

Span factor;	L / a = **4.53**
Angle of twist;	ϕ = **0.011** rads
Second derivative of ϕ;	ϕ'' = **3.75×10^{-3}** rads/m^2
Induced minor axis moment;	$M_{yt} = M_x \times \phi$ / 1 rad = **0.79** kNm
Normal stress at flange tip due to M_{yt};	$\sigma_{byt} = M_{yt} / Z_y$ = **5** N/mm^2
Normal stress at flange tip due to warping;	$\sigma_w = E_{S5950} \times W_{n0} \times \phi''$ / 1 rad = **8** N/mm^2
Interaction index;	$i_b = M_x \times m_{LT} / M_b + (\sigma_{byt} + \sigma_w) / p_y \times (1 + 0.5 \times M_x \times m_{LT} / M_b)$ = **0.65**

Pass - Combined bending and torsion check satisfied

Local capacity under combined bending & torsion

For simplicity, a conservative check is applied using the maximum stresses due to each of the separate load effects, even though these do not necessarily all occur at the same section along the member.

Max. direct stress due to M_x;	$\sigma_{bx} = M_x / Z_x$ = **136** N/mm^2
Combined stress - eqn 2.22;	$\sigma_{bx} + \sigma_{byt} + \sigma_w$ = **148** N/mm^2
Design strength;	p_y = **355** N/mm^2

Pass - Local capacity

Combined shear stresses - SCI-P-057 section 2.3

For simplicity, a conservative check is applied using the maximum shear stresses due to each of the separate load effects, even though these do not necessarily all occur at the same section along the member.

Max shear stresses due to bending in web;	$\tau_{bw} = F_{vy} \times Q_w / (I_x \times t)$ = **36** N/mm^2
Max shear stresses due to bending in flange;	$\tau_{bf} = F_{vy} \times Q_f / (I_x \times T)$ = **10** N/mm^2
Max shear stresses due to torsion in web;	τ_{tw} = abs($G \times t \times \phi'$ / 1rad) = **4** N/mm^2
Max shear stresses due to torsion in flange;	τ_{tf} = abs($G \times T \times \phi'$ / 1 rad) = **7** N/mm^2
Max shear stresses due to warping in flange;	τ_{wf} = abs($-E_{S5950} \times S_{w1} \times \phi'''$ / 1 rad / T) = **0** N/mm^2
Amp shear stress torsion & warping in web;	$\tau_{vtw} = \tau_{tw} \times (1 + 0.5 \times M_x \times m_{LT} / M_b)$ = **6** N/mm^2
Amp shear stress torsion & warping in flange;	$\tau_{vtf} = (\tau_{tf} + \tau_{wf}) \times (1 + 0.5 \times M_x \times m_{LT} / M_b)$ = **10** N/mm^2

Combined shear stresses due to bending, torsion & warping:

Combined shear stresses in web;	$\tau_w = \tau_{bw} + \tau_{vtw}$ = **41** N/mm^2
Combined shear stresses in flange;	$\tau_f = \tau_{bf} + \tau_{vtf}$ = **19** N/mm^2
Shear strength;	$p_v = 0.6 \times p_y$ = **213** N/mm^2

Pass - Combined shear stresses

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Twist check

Total applied torque (unfactored); T_{qu} = **0.43** kNm

Maximum twist under sls loading; $\phi_{sls} = \phi \times T_{qu} / T_q$ = **0.45** deg

Twist limit; ϕ_{lim} = **3.00** deg

Pass - Twist

Deflection

Maximum y-axis deflection; δ_{y_max} = **13.3** mm

Deflection limit - cl. 2.5.2; $\delta_{lim} = \min(L/ k_{\delta}, \delta_{lim_abs})$ = **14.0** mm

Pass - Deflection within specified limit

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BEAM 5

SPAN OF THE BEAM = 5500 mm

LOADING

I. Dead Load

Pitch Roof (Ext)	$= \left\lfloor \frac{1.28Kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 1.98 Kn/m
Floor (Timber)	$= \left\lfloor \frac{0.6kn/\mathrm{m}^2 \text{ x } (2.7)\mathrm{m}}{2} \right\rfloor$	= 0.81 Kn/m
Garage Shutter	= 1.5 Kn/m^2 x 2.6 m	= 3.90 Kn/m
Brick work	= 20 Kn/m³ x 0.200 x 0.3	= 1.20 Kn/m
		7.89 Kn/m

II. Imposed Load

Pitch Roof (Ext)	$= \left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (3.0)\mathrm{m}}{2} \right\rfloor$	= 1.38 Kn/m
Floor (Timber)	$= \left\lfloor \frac{1.5kn/\mathrm{m}^2 \text{ x } (2.7)\mathrm{m}}{2} \right\rfloor$	= 2.03 Kn/m
		3.41 Kn/m

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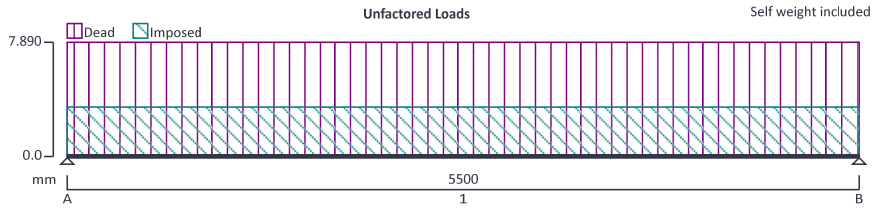
BEAM 5 DESIGN:

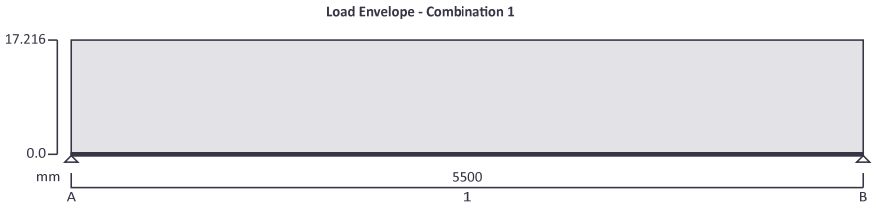
STEEL BEAM ANALYSIS & DESIGN (BS5950)

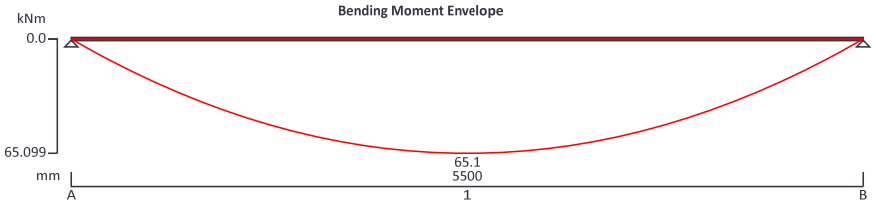
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TEDDS calculation version 3.0.07

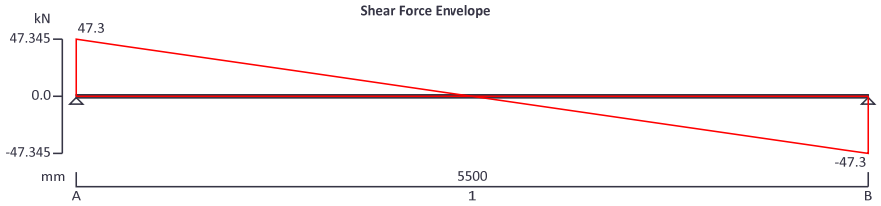








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Support conditions

Support A	Vertically restrained	
	Rotationally free	
Support B	Vertically restrained	
	Rotationally free	

Applied loading

Beam loads	Dead self weight of beam × 1	
	Dead full UDL 7.89 kN/m	
	Imposed full UDL 3.41 kN/m	

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **65.1** kNm;	M_{min} = **0** kNm
Maximum shear;	V_{max} = **47.3** kN;	V_{min} = **-47.3** kN
Deflection;	δ_{max} = **13.1** mm;	δ_{min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **47.3** kN;	R_{A_min} = **47.3** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **23.1** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **9.4** kN	
Maximum reaction at support B;	R_{B_max} = **47.3** kN;	R_{B_min} = **47.3** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **23.1** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **9.4** kN	

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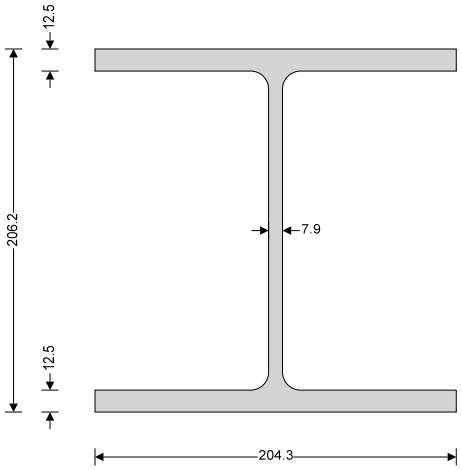
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Section details

Section type;	**UC 203x203x52 (BS4-1)**
Steel grade;	**S355**

From table 9: Design strength p_y

Thickness of element;	max(T, t) = **12.5** mm
Design strength;	p_y = **355** N/mm^2
Modulus of elasticity;	E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Effective length factors

Effective length factor in major axis;	K_x = **1.00**
Effective length factor in minor axis;	K_y = **1.00**
Effective length factor for lateral-torsional buckling;	$K_{LT.A}$ = **1.20**; + 2 × D
	$K_{LT.B}$ = **1.00**;

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.88**

Internal compression parts - Table 11

Depth of section;	d = **160.8** mm	
	d / t = 23.1 × ε <= 80 × ε;	Class 1 plastic

Outstand flanges - Table 11

Width of section;	b = B / 2 = **102.2** mm	
	b / T = 9.3 × ε <= 10 × ε;	Class 2 compact

Section is class 2 compact

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB

Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(\text{abs}(V_{max}), \text{abs}(V_{min})) =$ **47.3** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **1629** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **347** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity - Section 4.2.5

Design bending moment; $M = \max(\text{abs}(M_{s1_max}), \text{abs}(M_{s1_min})) =$ **65.1** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{xx}, 1.2 \times p_y \times Z_{xx}) =$ **201.4** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = ((1.2 + 1.2) \times L_{s1} + 2 \times D) / 2 =$ **6806** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **131.427**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; $u =$ **0.848**

Torsional index; $x =$ **15.838**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.689**

Ratio - cl.4.3.6.9; $\beta_W =$ **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **76.767**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.198**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; $\alpha_{LT} =$ **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.326**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2 =$ **343.3** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **405.1** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5}) =$ **199.6** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; $m_{LT} =$ **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx} =$ **113.2** kNm

$M_b / m_{LT} =$ **113.2** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(14 \text{ mm}, L_{s1} / 250) =$ **14** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min})) =$ **13.053** mm

PASS - Maximum deflection does not exceed deflection limit

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BEAM 6

SPAN OF THE BEAM = 5900 mm

LOADING

I. Dead Load

Floor (Timber) = $\left\lfloor \frac{0.6kn/\mathrm{m}^2 \text{ x } (2.7+2.7)\mathrm{m}}{2} \right\rfloor$ = 1.62 Kn/m

Stud Wall = 0.7 Kn/m^2 x 2.0 m = 1.40 Kn/m

3.02 Kn/m

II. Imposed Load

Floor (Timber) = $\left\lfloor \frac{1.5kn/\mathrm{m}^2 \text{ x } (2.7+2.7)\mathrm{m}}{2} \right\rfloor$ = 4.05 Kn/m

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4.05 Kn/m

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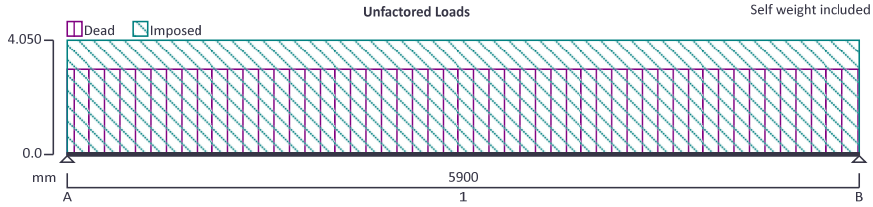
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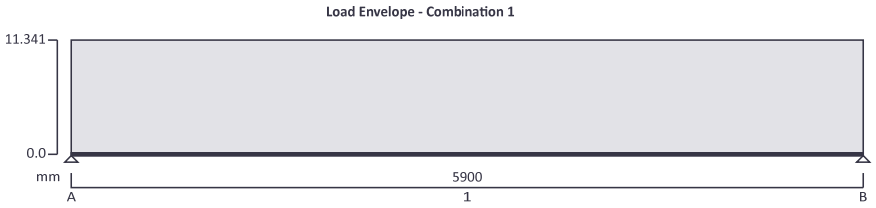
STEEL BEAM ANALYSIS & DESIGN (BS5950)

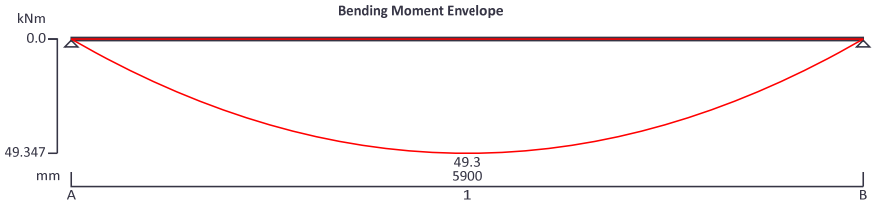
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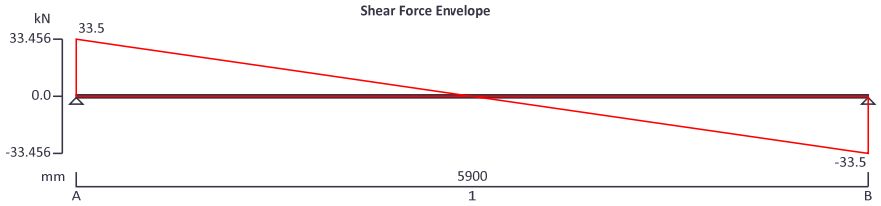








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Support conditions

Support A	Vertically restrained	
	Rotationally free	
Support B	Vertically restrained	
	Rotationally free	

Applied loading

Beam loads	Dead self weight of beam × 1	
	Dead full UDL 3.02 kN/m	
	Imposed full UDL 4.05 kN/m	

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **49.3** kNm;	M_{min} = **0** kNm
Maximum shear;	V_{max} = **33.5** kN;	V_{min} = **-33.5** kN
Deflection;	δ_{max} = **12.7** mm;	δ_{min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **33.5** kN;	R_{A_min} = **33.5** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **10.2** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **11.9** kN	
Maximum reaction at support B;	R_{B_max} = **33.5** kN;	R_{B_min} = **33.5** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **10.2** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **11.9** kN	

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Section details

Section type; **UC 203x203x46 (BS4-1)**

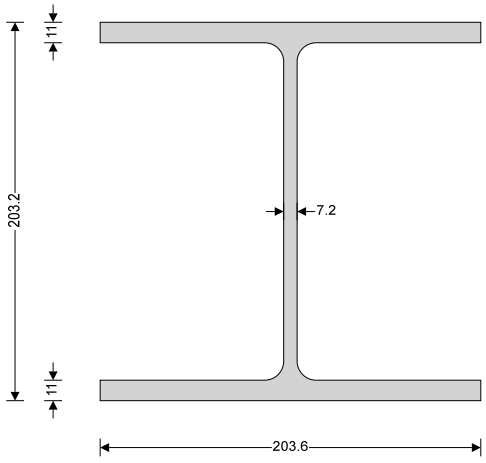
Steel grade; **S355**

From table 9: Design strength p_y

Thickness of element; max(T, t) = **11.0** mm

Design strength; p_y = **355** N/mm^2

Modulus of elasticity; E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Effective length factors

Effective length factor in major axis; K_x = **1.00**

Effective length factor in minor axis; K_y = **1.00**

Effective length factor for lateral-torsional buckling; $K_{LT.A}$ = **1.20**; + 2 × D

$K_{LT.B}$ = **1.00**;

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.88**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

d / t = 25.4 × ε <= 80 × ε; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **101.8** mm

b / T = 10.5 × ε <= 15 × ε; Class 3 semi-compact

Section is class 3 semi-compact

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Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(\text{abs}(V_{max}), \text{abs}(V_{min})) =$ **33.5** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **1463** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **311.6** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity - Section 4.2.5

Design bending moment; $M = \max(\text{abs}(M_{s1_max}), \text{abs}(M_{s1_min})) =$ **49.3** kNm

Effective plastic modulus - Section 3.5.6

Limiting value for class 2 compact flange; $\beta_{2f} = 10 \times \varepsilon =$ **8.801**

Limiting value for class 3 semi-compact flange; $\beta_{3f} = 15 \times \varepsilon =$ **13.202**

Limiting value for class 2 compact web; $\beta_{2w} = 100 \times \varepsilon =$ **88.014**

Limiting value for class 3 semi-compact web; $\beta_{3w} = 120 \times \varepsilon =$ **105.617**

Effective plastic modulus - cl.3.5.6.2

$S_{eff} = \min(Z_{xx} + (S_{xx} - Z_{xx}) \times \min([((\beta_{3w} / (d / t))^2 - 1) / ((\beta_{3w} / \beta_{2w})^2 - 1)], [(\beta_{3f} / (b / T) - 1) / (\beta_{3f} / \beta_{2f} - 1)]), S_{xx}) =$ **490411** mm^3

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{eff}, 1.2 \times p_y \times Z_{xx}) =$ **174.1** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = ((1.2 + 1.2) \times L_{s1} + 2 \times D) / 2 =$ **7283** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **141.855**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; $u =$ **0.847**

Torsional index; $x =$ **17.713**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.698**

Ratio - cl.4.3.6.9; $\beta_W = S_{eff} / S_{xx} =$ **0.986**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **83.252**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.198**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

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Bending strength - Section 4.3.6.5

Robertson constant; α_{LT} = **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0)$ = **0.371**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2$ = **291.9** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2$ = **377.7** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5})$ = **180.2** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; m_{LT} = **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{eff}$ = **88.4** kNm

M_b / m_{LT} = **88.4** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(14 \text{ mm}, L_{s1} / 250)$ = **14** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min}))$ = **12.674** mm

PASS - Maximum deflection does not exceed deflection limit

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BEAM 7

SPAN OF THE BEAM = 5200 mm

LOADING

I. Dead Load

Pitch Roof (Ext)	= $\left\lfloor \frac{1.28Kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$	= 0.51 Kn/m
Glass panel	= 1.0 Kn/m^2 x 1.5 m	= 1.50 Kn/m
Bifolding Door	= 1.2 Kn/m^2 x 2.1 m	= 2.52 Kn/m
Brick work	= 20 Kn/m^3 x 0.100 x 1.0	= 2.00 Kn/m
Block work	= 19 Kn/m^3 x 0.100 x 1.0	= 1.90 Kn/m
		8.43 Kn/m

II. Imposed Load

Pitch Roof (Ext)	= $\left\lfloor \frac{0.75kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$	= 0.37 Kn/m
		0.37 Kn/m

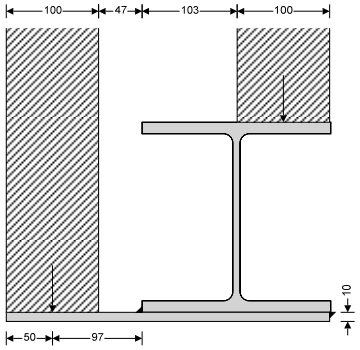
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BEAM 7 DESIGN:

STEEL MASONRY SUPPORT

In accordance with BS5950-1:2000 incorporating Corrigendum No.1

Tedds calculation version 1.0.04



Steel member details

Torsion beam;	**UC 203x203x52**
Masonry support plate;	User
Steel grade of support plate;	**S355**
Design strength of support plate;	p_{ysb} = **355** N/mm^2
Modulus of elasticity;	E = 205000 N/mm^2
Constant;	$\varepsilon = \sqrt{(275N/mm^2 / p_{ysb})}$ = **0.880**
Length of plate beyond beam;	l_h = **147** mm
Total length of plate;	l_{plate} = **350** mm
Thickness of plate;	t_{sb} = **10** mm
Width of main beam;	B_{mb} = **204** mm
Area of plate;	$A_{sbu} = t_{sb} \times l_{plate}$ = **3500.0** mm^2
Distance from weld position to CoG;	$c_{yysb} = l_h / 2 - (l_{plate} - l_h) / 2$ = **-28** mm

Supported materials detail

Density of masonry on main beam;	$\rho_{m,mb}$ = **20.0** kN/m^3
Width of masonry on main beam;	b_{mmb} = **100** mm
Height of masonry on main beam;	h_{mmb} = **1000** mm
Eccentricity of main beam material;	e_{mb} = **103** mm
Add dead force main beam (not from masonry);	P_{Gaddmb} = **8.4** kN/m
Add live force main beam (not from masonry);	P_{Qaddmb} = **0.4** kN/m
Density of masonry on support beam;	$\rho_{m,sb}$ = **19.0** kN/m^3
Width of masonry on support beam;	b_{msb} = **100** mm
Height of masonry on support beam;	h_{msb} = **1000** mm
Add dead force support beam (not from masonry);	P_{Gaddsb} = **0.0** kN/m
Add live force support beam (not from masonry);	P_{Qaddsb} = **0.0** kN/m

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Geometry

Cavity width;	c = **150** mm
Supported width of masonry;	$d_m = l_h + e_{mb} - c$ = **100** mm

Biaxial stress effects in the plate (SCI-P-110)

Maximum overall bending moment;	M_x = **66.6** kNm
Dist to NA combined section (CoG torsion beam);	$y_{e,all} = (D_{mb} + t_{sb}) \times A_{sbu} / (2 \times (A_{mb} + A_{sbu}))$ = **37** mm
Second moment of area of combined section;	$I_{xx,all} = (I_{xxmb} + A_{mb} \times y_{e,all}^2) + A_{sbu} \times (D_{mb} / 2 + t_{sb} / 2 - y_{e,all})^2$ = **7935** cm^4
Elastic section modulus of combined section;	$Z_{xx,all} = I_{xx,all} / (D_{mb} / 2 + t_{sb} - y_{e,all})$ = **1047.66** cm^3
Section modulus of plate;	$Z_{xx,plate} = 1m \times t_{sb}^2 / (6 \times 1m)$ = **16.67** cm^3/m
Eccentricity of support beam masonry;	e_1 = **97** mm
Force of masonry on support plate;	$P_1 = (b_{msb} \times h_{msb} \times \rho_{m,sb} + P_{Gaddsb}) \times \gamma_{fG} + P_{Qaddsb} \times \gamma_{fQ}$ = **2.7** kN/m
Bending at heel;	$M_{x,plate} = P_1 \times e_1$ = **0.3** kNm/m
Moment capacity of plate;	$M_c = 1.2 \times Z_{x,plate} \times p_{ysb}$ = **7.1** kNm/m

PASS - Design strength exceeds stress at heel

Longitudinal stress due to overall bending;	$\sigma_1 = M_x / Z_{xx,all}$ = **63.6** N/mm^2
Constant relating to Von Mises curve;	$c_{fp} = (4 \times p_{ysb}^2 - 3 \times \sigma_1^2)^{0.5}$ = **701.4** N/mm^2
Transverse bending stress ratio limit;	$\alpha_{ts} = (c_{fp}^2 - \sigma_1^2) / (2 \times c_{fp} \times p_{ysb})$ = **0.980**
Transverse bending stress ratio;	$\alpha_{ls} = M_{x,plate} / M_c$ = **0.036**

PASS - Transverse bending stress ratio less than allowable limit

Deflection at toe

Unfactored force on support angle;	$P_{1SLS} = b_{msb} \times h_{msb} \times \rho_{m,sb} + P_{Gaddsb} + P_{Qaddsb}$ = **1.9** kN/m
Distance from weld to load position;	$a_m = e_1$ = **97** mm
Length of load resultant to edge of plate;	$b_m = l_h - e_1$ = **50** mm
Dist from weld to load position as ratio of length;	$a_l = a_m / (a_m + b_m)$ = **0.660**
Effective second moment of inertia;	$I_{eff_def} = t_{sb}^3 / 12$ = **83333** mm^4/m
Deflection at toe;	$\delta = (a_l^2 \times (3 - a_l) / 6) \times (P_{1SLS} \times (a_m + b_m)^3) / (E_{S5950} \times I_{eff_def})$ = **0.06** mm
Deflection limit;	δ_{lim} = **1.67** mm

PASS - Deflection is within specified criteria

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Weld details - assume a full length weld and that the plate acts as a propped cantilever with the prop at the weld position and the fixed end at the centre of the torsion beam

Leg length of weld; s_{weld} = **6** mm

Throat size of weld; $a_{weld} = 1/\sqrt{(2)} \times s_{weld}$ = **4.2** mm

Shear force at weld position; $R_A = P_1 \times maX((1 + (3 \times e_1) / (2 \times B_{mb} / 2)), 1.4)$ = **6.4** kN/m

Maximum possible force in plate; $R_p = (l_h + B_{mb}) \times t_{sb} \times p_{ysb}$ = **1247.1** kN

Longitudinal shear between beam and plate; $R_l = 2 \times R_p / L$ = **470.6** kN/m

Horizontal shear between beam and plate; $R_h = P_1 \times e_1 / (s_{weld} / 2 + t_{sb} / 2)$ = **32.3** kN/m

Resultant weld force; $R_{weld} = (R_A^2 + R_l^2 + R_h^2)^{0.5}$ = **0.472** kN/mm

Strength of weld (Table 37); p_{weld} = **220.0** N/mm^2

Capacity of full length weld; $p_{c,weld} = a_{weld} \times p_{weld}$ = **0.933** kN/mm

PASS - Capacity of weld exceeds resultant force on weld

Torsional loading ULS

Loading of support beam masonry; $w_{1ULS} = (h_{msb} \times b_{msb} \times \rho_{m,sb} + P_{Gaddsb}) \times \gamma_{fG} + P_{Qaddsb} \times \gamma_{fQ}$ = **2.66** kN/m

Loading of main beam masonry; $w_{2ULS} = (h_{mmb} \times b_{mmb} \times \rho_{m,mb} + P_{Gaddmb}) \times \gamma_{fG} + P_{Qaddmb} \times \gamma_{fQ}$ = **15.19** kN/m

Self weight of support beam; $w_{3ULS} = A_{sbu} \times \rho_{sb} \times \gamma_{fG}$ = **0.38** kN/m

Torsional loading SLS

Loading of support beam masonry; $w_{1SLS} = h_{msb} \times b_{msb} \times \rho_{m,sb} + P_{Gaddsb} + P_{Qaddsb}$ = **1.90** kN/m

Loading of main beam masonry; $w_{2SLS} = h_{mmb} \times b_{mmb} \times \rho_{m,mb} + P_{Gaddmb} + P_{Qaddmb}$ = **10.80** kN/m

Self weight of support beam; $w_{3SLS} = A_{sbu} \times \rho_{sb}$ = **0.27** kN/m

Eccentricities

Distance to shear centre of main beam; e_{0mb} = **0** mm

Eccentricity of support beam masonry; $e_{1mb} = (B_{mb} + b_{msb}) / 2 + c - e_{mb}$ = **199** mm

Eccentricity of main beam masonry; $e_{2mb} = (B_{mb} - b_{mmb}) / 2 - e_{mb}$ = **-51** mm

Eccentricity of support beam; $e_{3mb} = B_{mb} / 2 + c_{yysb}$ = **74** mm

Torsional effects

Applied torque (ULS); $T_{qULS} = abs(w_{1ULS} \times e_{1mb} + w_{2ULS} \times e_{2mb} + w_{3ULS} \times e_{3mb})$ = **0.21** kNm/m

Total torque (ULS); $T_q = T_{qULS} \times L$ = **1.14** kNm

Applied torque (SLS); $T_{qSLS} = abs(w_{1SLS} \times e_{1mb} + w_{2SLS} \times e_{2mb} + w_{3SLS} \times e_{3mb})$ = **0.15** kNm/m

Total torque (SLS); $T_{qu} = T_{qSLS} \times L$ = **0.80** kNm

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STEEL BEAM TORSION DESIGN

In accordance with BS5950-1:2000 incorporating Corrigendum No.1

Tedds calculation version 2.0.02

Section details

Section type;	**UC 203x203x52**
Steel grade;	**S355**
Design stength;	$p_{yw} = p_y =$ **355** N/mm^2
Constant;	$\varepsilon = \sqrt{(275\ \text{N/mm}^2 / p_y)} =$ **0.880**

Geometry - Beam unrestrained against lateral-torsional buckling between supports.

Effective span;	L = **5300** mm
Length of segment for LT buckling;	L_{LT} = **5300** mm
Compression flanges laterally unrestrained	
Partial torsional restraint against rotation about longitudinal axis provided by connection of bottom flange to supports	
Effective length for LT buckling;	$L_{E_LT} = L_{LT} \times 1.2 + 2 \times D =$ **6772** mm

Loading - Torsional loading comprises only full-length uniformly distributed load(s)

Internal forces & moments on member under factored loading for uls design

Applied shear force;	F_{vy} = **50.3** kN
Maximum bending moment;	$M_{LT} = M_x =$ **66.60** kNm
Applied torque;	$T_q =$ **1.14** kNm
Minor axis bending moment;	M_y = 0 kNm
Compression force;	F_c = 0 kN

Equivalent uniform moment factors

EUM factor (Cl. 4.3.6.6 and T18);	$m_{LT} =$ **1.000**

Torsional deflection parameters

Beam is torsion fixed and warping free at each end. (as defined in SCI-P-057 section 2.1.6) - Appendix B case 4	
Dist along the beam for first derivative of twist;	z_1 = 0 mm
Dist along the beam for second derivative of twist;	$z_2 = L / 2 =$ **2650** mm
First derivative of angle of twist;	$\phi'_1 = T_q / (G \times J) \times a / L \times [L^2 / (2 \times a) \times (1 / L - 2 \times z_1 / L^2) + \sinh(z_1 / a) - \tanh(L / (2 \times a)) \times \cosh(z_1 / a)] \times 1$ rads = **1.29×10^{-2}** rads/m
Third derivative of angle of twist;	$\phi'''_1 = T_q / (G \times J \times a^2) \times a/L \times [\sinh(z_1 / a) - \tanh(L / (2 \times a)) \times \cosh(z_1 / a)] \times 1$ rads = **-7.17×10^{-3}** rads/m^3
Angle of twist;	$\phi_2 = T_q \times a / (G \times J) \times a / L \times [L^2 / (2 \times a^2) \times (z_2 / L - z_2^2 / L^2) + \cosh(z_2 / a) - \tanh(L / (2 \times a)) \times \sinh(z_2 / a) - 1] \times 1$ rads = **0.021** rads
Second derivative of angle of twist;	$\phi''_2 = T_q / (G \times J \times a) \times a / L \times [\cosh(z_2 / a) - \tanh(L / (2 \times a)) \times \sinh(z_2 / a) - 1] \times 1$ rads = **-6.81×10^{-3}** rads/m^2

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Design parameters

Total angle of twist; $\phi = abs(\phi_2) =$ **0.021** rads

First derivative of ϕ; $\phi' = abs(\phi'_1) =$ **1.29×10^{-2}** rads/m

Second derivative of ϕ; $\phi'' = abs(\phi''_2) =$ **6.81×10^{-3}** rads/m^2

Third derivative of ϕ; $\phi''' = abs(\phi'''_1) =$ **7.17×10^{-3}** rads/m^3

Section classification

$b / T =$ **8.2**

$d / t =$ **20.4**

$r_{1s} = \min(1.0, \max(-1.0, F_c / (d \times t \times p_{yw}))) =$ **0.000**

$r_{2s} = F_c / (A_g \times p_{yw}) =$ **0.000**

Section classification is compact

Shear capacity (parallel to y-axis)

Design shear force; $F_{vy} =$ **50.3** kN

Design shear resistance (Cl. 4.2.3); $P_{vy} = 0.6 \times p_y \times A_{vy} =$ **347.0** kN

Pass - Shear

Moment capacity (x-axis)

Design bending moment; $M_x =$ **66.6** kNm

Moment capacity; $M_{cxu} = p_y \times S_x =$ **201.4** kNm

Moment capacity low shear (Cl. 4.2.5.1); $M_{cx} = \min(p_y \times S_x, 1.2 \times p_y \times Z_x) =$ **201.4** kNm

Pass - Moment capacity exceeds design bending moment

Lateral torsional buckling

Effective length for lateral torsional buckling; $L_{E_LT} =$ **6772** mm

Slenderness ratio; $\lambda = L_{E_LT} / r_y =$**131**

Buckling parameter; $u =$ **0.848**

Flange ratio; $\eta = 0.5$

Torsional index; $x =$ **15.8**

Slenderness factor; $v = 1 / (1 + 0.05 \times (\lambda / x)^2)^{0.25} =$ **0.69**

Ratio - cl 4.3.6.9; $\beta_w = 1.0 =$ **1.000**

Equvalent slenderness – cl 4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{(\beta_w)} =$ **77**

Limiting slendernes – Annex B2.2; $\lambda_{L0} = 0.4 \times \sqrt{(\pi^2 \times E_{S5950} / p_y)} =$ **30**

Euler stress; $p_E = \pi^2 \times E_{S5950} / \lambda_{LT}^2 =$ **345** N/mm^2

Perry factor; $\eta_{LT} = \max(7.0 \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.324**

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **406232770.716**

Bending strength; $p_b = p_E \times p_y / (\phi_{LT} + \sqrt{(\phi_{LT}^2 - p_E \times p_y)}) =$ **200** N/mm^2

Buckling resistance moment; $M_b = p_b \times S_x =$ **113.7** kNm

Max moment governing buckling resistance; $M_{LT} =$ **66.6** kNm

Equiv uniform moment factor for LTB; $m_{LT} =$ **1.00**

$M_b / m_{LT} =$ **113.7** kNm

Pass - lat. tors. buckling

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Buckling under combined bending & torsion -SCI-P-057 section 2.3

For simplicity, a conservative check is applied using the maximum stresses due to each of the separate load effects, even though these do not necessarily all occur at the same section along the member.

Span factor;	L / a = **4.53**
Angle of twist;	ϕ = **0.021** rads
Second derivative of ϕ;	ϕ'' = **6.81×10^{-3}** rads/m^2
Induced minor axis moment;	$M_{yt} = M_x \times \phi$ / 1 rad = **1.38** kNm
Normal stress at flange tip due to M_{yt};	$\sigma_{byt} = M_{yt} / Z_y$ = **8** N/mm^2
Normal stress at flange tip due to warping;	$\sigma_w = E_{S5950} \times W_{n0} \times \phi''$ / 1 rad = **14** N/mm^2
Interaction index;	$i_b = M_x \times m_{LT} / M_b + (\sigma_{byt} + \sigma_w) / p_y \times (1 + 0.5 \times M_x \times m_{LT} / M_b)$ = **0.67**

Pass - Combined bending and torsion check satisfied

Local capacity under combined bending & torsion

For simplicity, a conservative check is applied using the maximum stresses due to each of the separate load effects, even though these do not necessarily all occur at the same section along the member.

Max. direct stress due to M_x;	$\sigma_{bx} = M_x / Z_x$ = **131** N/mm^2
Combined stress - eqn 2.22;	$\sigma_{bx} + \sigma_{byt} + \sigma_w$ = **152** N/mm^2
Design strength;	p_y = **355** N/mm^2

Pass - Local capacity

Combined shear stresses - SCI-P-057 section 2.3

For simplicity, a conservative check is applied using the maximum shear stresses due to each of the separate load effects, even though these do not necessarily all occur at the same section along the member.

Max shear stresses due to bending in web;	$\tau_{bw} = F_{vy} \times Q_w / (I_x \times t)$ = **34** N/mm^2
Max shear stresses due to bending in flange;	$\tau_{bf} = F_{vy} \times Q_f / (I_x \times T)$ = **9** N/mm^2
Max shear stresses due to torsion in web;	τ_{tw} = abs($G \times t \times \phi'$ / 1rad) = **8** N/mm^2
Max shear stresses due to torsion in flange;	τ_{tf} = abs($G \times T \times \phi'$ / 1 rad) = **13** N/mm^2
Max shear stresses due to warping in flange;	τ_{wf} = abs($-E_{S5950} \times S_{w1} \times \phi'''$ / 1 rad / T) = **1** N/mm^2
Amp shear stress torsion & warping in web;	$\tau_{vtw} = \tau_{tw} \times (1 + 0.5 \times M_x \times m_{LT} / M_b)$ = **10** N/mm^2
Amp shear stress torsion & warping in flange;	$\tau_{vtf} = (\tau_{tf} + \tau_{wf}) \times (1 + 0.5 \times M_x \times m_{LT} / M_b)$ = **17** N/mm^2

Combined shear stresses due to bending, torsion & warping:

Combined shear stresses in web;	$\tau_w = \tau_{bw} + \tau_{vtw}$ = **45** N/mm^2
Combined shear stresses in flange;	$\tau_f = \tau_{bf} + \tau_{vtf}$ = **27** N/mm^2
Shear strength;	$p_v = 0.6 \times p_y$ = **213** N/mm^2

Pass - Combined shear stresses

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Twist check

Total applied torque (unfactored); T_{qu} = **0.80** kNm

Maximum twist under sls loading; $\phi_{sls} = \phi \times T_{qu} / T_q$ = **0.84** deg

Twist limit; ϕ_{lim} = **3.00** deg

Pass - Twist

Deflection

Maximum y-axis deflection; δ_{y_max} = **12.9** mm

Deflection limit - cl. 2.5.2; $\delta_{lim} = \min(L/ k_{\delta}, \delta_{lim_abs})$ = **14.0** mm

Pass - Deflection within specified limit

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BEAM 8 (CRANKED)

SPAN OF THE BEAM = 6200 mm

LOADING (Worst Case):

I. Dead Load

Pitch Roof(Ext)	= $\left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$	= 0.51 Kn/m
Window Panel	= 1.0 Kn/m^2 x 1.5 m	= 1.50 Kn/m
Brick work	= 20 Kn/m³ x 0.100 x 0.6	= 1.20 Kn/m
Block work	= 19 Kn/m³ x 0.100 x 0.6	= 1.14 Kn/m
		4.35 Kn/m

II. Imposed Load

Pitch Roof(Ext)	= $\left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$	= 0.37 Kn/m
		0.37 Kn/m

Point load from Ridge Beam 1

Pdl = 24.00 Kn @ 3.1m

Pil = 15.00 kN @3.1m

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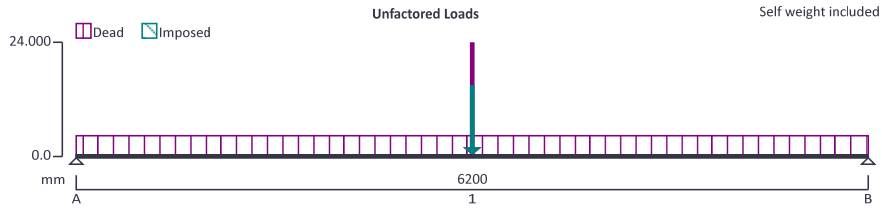
BEAM 8 (CRANKED) DESIGN:

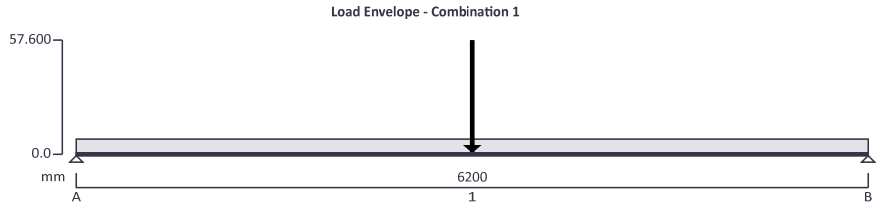
STEEL BEAM ANALYSIS & DESIGN (BS5950)

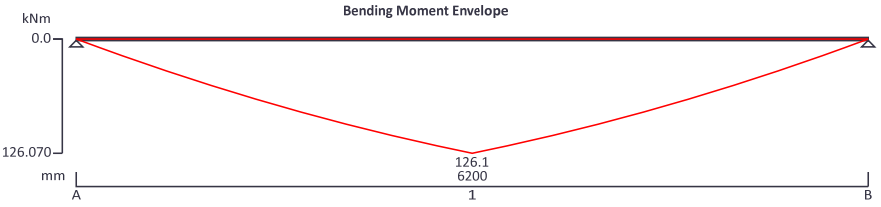
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TEDDS calculation version 3.0.07

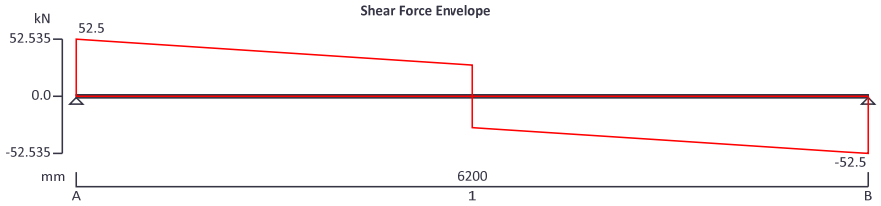








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Support conditions

Support A	Vertically restrained
	Rotationally free
Support B	Vertically restrained
	Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1
	Dead full UDL 4.35 kN/m
	Imposed full UDL 0.37 kN/m
	Dead point load 24 kN at 3100 mm
	Imposed point load 15 kN at 3100 mm

Load combinations

Load combination 1	Support A	Dead × 1.40
		Imposed × 1.60
		Dead × 1.40
		Imposed × 1.60
	Support B	Dead × 1.40
		Imposed × 1.60

Analysis results

Maximum moment;	M_{max} = **126.1** kNm;	M_{min} = **0** kNm
Maximum shear;	V_{max} = **52.5** kN;	V_{min} = **-52.5** kN
Deflection;	δ_{max} = **19.1** mm;	δ_{min} = **0** mm
Maximum reaction at support A;	R_{A_max} = **52.5** kN;	R_{A_min} = **52.5** kN
Unfactored dead load reaction at support A;	R_{A_Dead} = **27.6** kN	
Unfactored imposed load reaction at support A;	$R_{A_Imposed}$ = **8.6** kN	
Maximum reaction at support B;	R_{B_max} = **52.5** kN;	R_{B_min} = **52.5** kN
Unfactored dead load reaction at support B;	R_{B_Dead} = **27.6** kN	
Unfactored imposed load reaction at support B;	$R_{B_Imposed}$ = **8.6** kN	

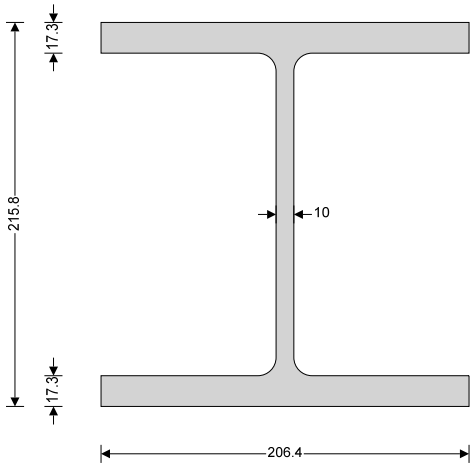
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	Sheet No:	
	Prepared by:	ARS
PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

Section details

Section type;	**UC 203x203x71 (BS4-1)**
Steel grade;	**S355**
From table 9: Design strength p_y	
Thickness of element;	max(T, t) = **17.3** mm
Design strength;	p_y = **345** N/mm^2
Modulus of elasticity;	E = **205000** N/mm^2



Lateral restraint

Span 1 has lateral restraint at supports only

Effective length factors

Effective length factor in major axis;	K_x = **1.00**
Effective length factor in minor axis;	K_y = **1.00**
Effective length factor for lateral-torsional buckling;	$K_{LT.A}$ = **1.20**; + 2 × D
	$K_{LT.B}$ = **1.00**;

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.89**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

d / t = 18.0 × ε <= 80 × ε; Class 1 plastic

Outstand flanges - Table 11

Width of section; b = B / 2 = **103.2** mm

b / T = 6.7 × ε <= 9 × ε; Class 1 plastic

Section is class 1 plastic

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB

Shear capacity - Section 4.2.3

Design shear force; $F_v = \max(\text{abs}(V_{max}), \text{abs}(V_{min})) =$ **52.5** kN

$d / t < 70 \times \varepsilon$

Web does not need to be checked for shear buckling

Shear area; $A_v = t \times D =$ **2158** mm^2

Design shear resistance; $P_v = 0.6 \times p_y \times A_v =$ **446.7** kN

PASS - Design shear resistance exceeds design shear force

Moment capacity - Section 4.2.5

Design bending moment; $M = \max(\text{abs}(M_{s1_max}), \text{abs}(M_{s1_min})) =$ **126.1** kNm

Moment capacity low shear - cl.4.2.5.2; $M_c = \min(p_y \times S_{xx}, 1.2 \times p_y \times Z_{xx}) =$ **275.6** kNm

Effective length for lateral-torsional buckling - Section 4.3.5

Effective length for lateral torsional buckling; $L_E = ((1.2 + 1.2) \times L_{s1} + 2 \times D) / 2 =$ **7656** mm

Slenderness ratio; $\lambda = L_E / r_{yy} =$ **144.529**

Equivalent slenderness - Section 4.3.6.7

Buckling parameter; $u =$ **0.853**

Torsional index; $x =$ **11.926**

Slenderness factor; $v = 1 / [1 + 0.05 \times (\lambda / x)^2]^{0.25} =$ **0.588**

Ratio - cl.4.3.6.9; $\beta_W =$ **1.000**

Equivalent slenderness - cl.4.3.6.7; $\lambda_{LT} = u \times v \times \lambda \times \sqrt{[\beta_W]} =$ **72.501**

Limiting slenderness - Annex B.2.2; $\lambda_{L0} = 0.4 \times (\pi^2 \times E / p_y)^{0.5} =$ **30.632**

$\lambda_{LT} > \lambda_{L0}$ - Allowance should be made for lateral-torsional buckling

Bending strength - Section 4.3.6.5

Robertson constant; $\alpha_{LT} =$ **7.0**

Perry factor; $\eta_{LT} = \max(\alpha_{LT} \times (\lambda_{LT} - \lambda_{L0}) / 1000, 0) =$ **0.293**

Euler stress; $p_E = \pi^2 \times E / \lambda_{LT}^2 =$ **384.9** N/mm^2

$\phi_{LT} = (p_y + (\eta_{LT} + 1) \times p_E) / 2 =$ **421.4** N/mm^2

Bending strength - Annex B.2.1; $p_b = p_E \times p_y / (\phi_{LT} + (\phi_{LT}^2 - p_E \times p_y)^{0.5}) =$ **209.8** N/mm^2

Equivalent uniform moment factor - Section 4.3.6.6

Equivalent uniform moment factor for LTB; $m_{LT} =$ **1.000**

Buckling resistance moment - Section 4.3.6.4

Buckling resistance moment; $M_b = p_b \times S_{xx} =$ **167.6** kNm

$M_b / m_{LT} =$ **167.6** kNm

PASS - Buckling resistance moment exceeds design bending moment

Check vertical deflection - Section 2.5.2

Consider deflection due to dead and imposed loads

Limiting deflection; $\delta_{lim} = \min(20 \text{ mm}, L_{s1} / 250) =$ **20** mm

Maximum deflection span 1; $\delta = \max(\text{abs}(\delta_{max}), \text{abs}(\delta_{min})) =$ **19.072** mm

PASS - Maximum deflection does not exceed deflection limit

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

STEEL COLUMN C1 – C5

HEIGHT = 3.0m

ASSUMING **203 X 203 X 46 UC** WITH **300 X 300 X 15mm** THICK M.S. PLATE AS BASE PLATE

LOADING: (WORST CASE)

Load from the Beam 3 = 281.00 Kn.

281.00 Kn

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

STEEL COLUMN C1 – C5 DESIGN:

STEEL MEMBER DESIGN (BS5950)

In accordance with BS5950-1:2000 incorporating Corrigendum No.1

TEDDS calculation version 3.0.07

Section details

Section type; **UC 203x203x46 (BS4-1)**

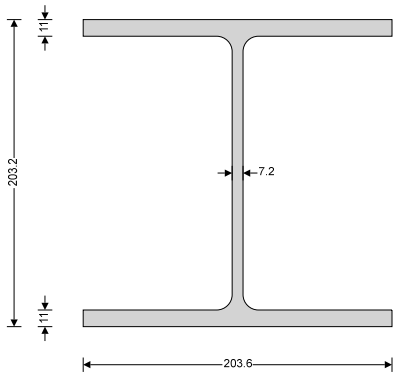
Steel grade; **S355**

From table 9: Design strength p_y

Thickness of element; max(T, t) = **11.0** mm

Design strength; p_y = **355** N/mm^2

Modulus of elasticity; E = **205000** N/mm^2



Lateral restraint

Distance between major axis restraints; L_x = **3000** mm

Distance between minor axis restraints; L_y = **3000** mm

Effective length factors

Effective length factor in major axis; K_x = **1.00**

Effective length factor in minor axis; K_y = **1.00**

Effective length factor for lateral-torsional buckling; K_{LT} = **1.20**; + 2 × D

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.88**

Internal compression parts - Table 11

Depth of section; d = **160.8** mm

Stress ratios; r1 = min(F_c / (d × t × p_{yw}), 1) = **0.684**

r2 = F_c / (A × p_{yw}) = **0.135**

d / t = 25.4 × ε <= max(80 × ε / (1 + r1), 40 × ε);

Class 1 plastic

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

Outstand flanges - Table 11

Width of section; $b = B / 2 =$ **101.8** mm

$b / T = 10.5 \times \varepsilon <= 15 \times \varepsilon$; Class 3 semi-compact

Section is class 3 semi-compact

Compression members - Section 4.7

Design compression force; $F_c =$ **281** kN

Effective length for major (x-x) axis buckling - Section 4.7.3

Effective length for buckling; $L_{Ex} = L_x \times K_x =$ **3000** mm

Slenderness ratio - cl.4.7.2; $\lambda_x = L_{Ex} / r_{xx} =$ **34.017**

Compressive strength - Section 4.7.5

Limiting slenderness; $\lambda_0 = 0.2 \times (\pi^2 \times E / p_y)^{0.5} =$ **15.099**

Strut curve - Table 23; b

Robertson constant; $\alpha_x =$ **3.5**

Perry factor; $\eta_x = \alpha_x \times (\lambda_x - \lambda_0) / 1000 =$ **0.066**

Euler stress; $p_{Ex} = \pi^2 \times E / \lambda_x^2 =$ **1748.4** N/mm^2

$\phi_x = (p_y + (\eta_x + 1) \times p_{Ex}) / 2 =$ **1109.6** N/mm^2

Compressive strength - Annex C.1; $p_{cx} = p_{Ex} \times p_y / (\phi_x + (\phi_x^2 - p_{Ex} \times p_y)^{0.5}) =$ **328.2** N/mm^2

Compression resistance - Section 4.7.4

Compression resistance - cl.4.7.4; $P_{cx} = A \times p_{cx} =$ **1927.8** kN

PASS - Compression resistance exceeds design compression force

Effective length for minor (y-y) axis buckling - Section 4.7.3

Effective length for buckling; $L_{Ey} = L_y \times K_y =$ **3000** mm

Slenderness ratio - cl.4.7.2; $\lambda_y = L_{Ey} / r_{yy} =$ **58.431**

Compressive strength - Section 4.7.5

Limiting slenderness; $\lambda_0 = 0.2 \times (\pi^2 \times E / p_y)^{0.5} =$ **15.099**

Strut curve - Table 23; c

Robertson constant; $\alpha_y =$ **5.5**

Perry factor; $\eta_y = \alpha_y \times (\lambda_y - \lambda_0) / 1000 =$ **0.238**

Euler stress; $p_{Ey} = \pi^2 \times E / \lambda_y^2 =$ **592.6** N/mm^2

$\phi_y = (p_y + (\eta_y + 1) \times p_{Ey}) / 2 =$ **544.4** N/mm^2

Compressive strength - Annex C.1; $p_{cy} = p_{Ey} \times p_y / (\phi_y + (\phi_y^2 - p_{Ey} \times p_y)^{0.5}) =$ **251.1** N/mm^2

Compression resistance - Section 4.7.4

Compression resistance - cl.4.7.4; $P_{cy} = A \times p_{cy} =$ **1474.9** kN

PASS - Compression resistance exceeds design compression force

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

STEEL COLUMN C6 – C9

HEIGHT = 3.0m

ASSUMING **152 X 152 X 37 UC** WITH **300 X 300 X 15mm** THICK M.S. PLATE AS BASE PLATE

LOADING: (WORST CASE)

Load from the Beam 7 = 51.00 Kn.

Load from the Beam 8 = 53.00 Kn.

104.00 Kn

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STEEL COLUMN C6- C9 DESIGN:

STEEL MEMBER DESIGN (BS5950)

In accordance with BS5950-1:2000 incorporating Corrigendum No.1

TEDDS calculation version 3.0.07

Section details

Section type; **UC 152x152x37 (BS4-1)**

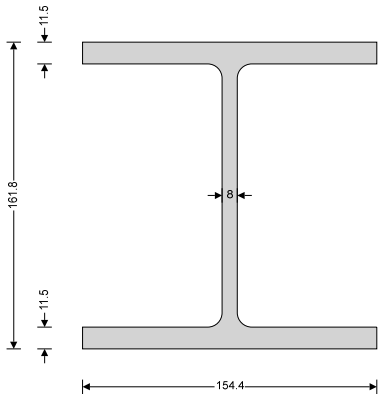
Steel grade; **S355**

From table 9: Design strength p_y

Thickness of element; max(T, t) = **11.5** mm

Design strength; p_y = **355** N/mm^2

Modulus of elasticity; E = **205000** N/mm^2



Lateral restraint

Distance between major axis restraints; L_x = **3000** mm

Distance between minor axis restraints; L_y = **3000** mm

Effective length factors

Effective length factor in major axis; K_x = **1.00**

Effective length factor in minor axis; K_y = **1.00**

Effective length factor for lateral-torsional buckling; K_{LT} = **1.20**; + 2 × D

Classification of cross sections - Section 3.5

$\varepsilon = \sqrt{[275\ N/mm^2 / p_y]}$ = **0.88**

Internal compression parts - Table 11

Depth of section; d = **123.6** mm

Stress ratios; r1 = min(F_c / (d × t × p_{yw}), 1) = **0.296**

r2 = F_c / (A × p_{yw}) = **0.062**

d / t = 17.6 × ε <= max(80 × ε / (1 + r1), 40 × ε);

Class 1 plastic

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB

Outstand flanges - Table 11

Width of section; $b = B / 2 =$ **77.2** mm

$b / T = 7.6 \times \varepsilon <= 9 \times \varepsilon$; Class 1 plastic

Section is class 1 plastic

Compression members - Section 4.7

Design compression force; $F_c =$ **104** kN

Effective length for major (x-x) axis buckling - Section 4.7.3

Effective length for buckling; $L_{Ex} = L_x \times K_x =$ **3000** mm

Slenderness ratio - cl.4.7.2; $\lambda_x = L_{Ex} / r_{xx} =$ **43.797**

Compressive strength - Section 4.7.5

Limiting slenderness; $\lambda_0 = 0.2 \times (\pi^2 \times E / p_y)^{0.5} =$ **15.099**

Strut curve - Table 23; b

Robertson constant; $\alpha_x =$ **3.5**

Perry factor; $\eta_x = \alpha_x \times (\lambda_x - \lambda_0) / 1000 =$ **0.100**

Euler stress; $p_{Ex} = \pi^2 \times E / \lambda_x^2 =$ **1054.8** N/mm^2

$\phi_x = (p_y + (\eta_x + 1) \times p_{Ex}) / 2 =$ **757.9** N/mm^2

Compressive strength - Annex C.1; $p_{cx} = p_{Ex} \times p_y / (\phi_x + (\phi_x^2 - p_{Ex} \times p_y)^{0.5}) =$ **310.8** N/mm^2

Compression resistance - Section 4.7.4

Compression resistance - cl.4.7.4; $P_{cx} = A \times p_{cx} =$ **1464** kN

PASS - Compression resistance exceeds design compression force

Effective length for minor (y-y) axis buckling - Section 4.7.3

Effective length for buckling; $L_{Ey} = L_y \times K_y =$ **3000** mm

Slenderness ratio - cl.4.7.2; $\lambda_y = L_{Ey} / r_{yy} =$ **77.483**

Compressive strength - Section 4.7.5

Limiting slenderness; $\lambda_0 = 0.2 \times (\pi^2 \times E / p_y)^{0.5} =$ **15.099**

Strut curve - Table 23; c

Robertson constant; $\alpha_y =$ **5.5**

Perry factor; $\eta_y = \alpha_y \times (\lambda_y - \lambda_0) / 1000 =$ **0.343**

Euler stress; $p_{Ey} = \pi^2 \times E / \lambda_y^2 =$ **337** N/mm^2

$\phi_y = (p_y + (\eta_y + 1) \times p_{Ey}) / 2 =$ **403.8** N/mm^2

Compressive strength - Annex C.1; $p_{cy} = p_{Ey} \times p_y / (\phi_y + (\phi_y^2 - p_{Ey} \times p_y)^{0.5}) =$ **195.4** N/mm^2

Compression resistance - Section 4.7.4

Compression resistance - cl.4.7.4; $P_{cy} = A \times p_{cy} =$ **920.6** kN

PASS - Compression resistance exceeds design compression force

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB		Date: JUL 2024

PAD FOOTING F1 & F2

Assuming **1.5 m X 1.5 m X 1.0 m** deep Mass Concrete footing.

LOADING: (Worst Case)

Load from the Beam 2 = 178.00 Kn.

Load from the Beam 3 = 52.00 Kn.

230.00 Kn

Total unfactored load = 230.00 Kn X 0.7 = 161.00 Kn.

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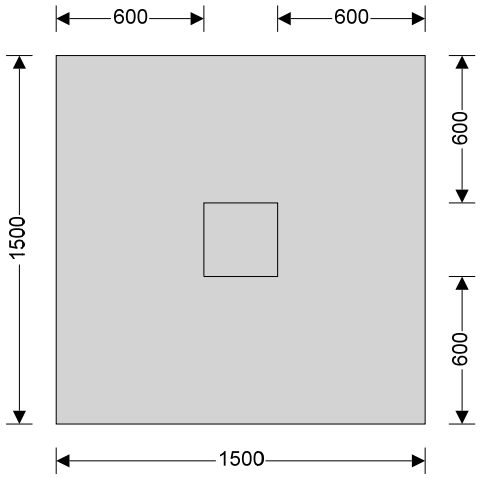
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PAD FOOTING F1 & F2 DESIGN:

PAD FOOTING ANALYSIS AND DESIGN (BS8110-1:1997)

Tedds calculation version 2.0.07



Pad footing details

Length of pad footing;	L = **1500** mm
Width of pad footing;	B = **1500** mm
Area of pad footing;	A = L × B = **2.250** m^2
Depth of pad footing;	h = **1000** mm
Depth of soil over pad footing;	h_{soil} = **100** mm
Density of concrete;	ρ_{conc} = **23.6** kN/m^3

Column details

Column base length;	l_A = **300** mm
Column base width;	b_A = **300** mm
Column eccentricity in x;	e_{PxA} = **0** mm
Column eccentricity in y;	e_{PyA} = **0** mm

Soil details

Density of soil;	ρ_{soil} = **20.0** kN/m^3
Design shear strength;	ϕ' = **25.0** deg
Design base friction;	δ = **19.3** deg
Allowable bearing pressure;	$P_{bearing}$ = **100** kN/m^2

Axial loading on column

Dead axial load on column;	P_{GA} = **161.0** kN
Imposed axial load on column;	P_{QA} = **0.0** kN
Wind axial load on column;	P_{WA} = **0.0** kN
Total axial load on column;	P_A = **161.0** kN

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Foundation loads

Dead surcharge load; F_{Gsur} = **0.000** kN/m^2

Imposed surcharge load; F_{Qsur} = **0.000** kN/m^2

Pad footing self weight; $F_{swt} = h \times \rho_{conc}$ = **23.600** kN/m^2

Soil self weight; $F_{soil} = h_{soil} \times \rho_{soil}$ = **2.000** kN/m^2

Total foundation load; $F = A \times (F_{Gsur} + F_{Qsur} + F_{swt} + F_{soil})$ = **57.6** kN

Calculate pad base reaction

Total base reaction; $T = F + P_A$ = **218.6** kN

Eccentricity of base reaction in x; $e_{Tx} = (P_A \times e_{PxA} + M_{xA} + H_{xA} \times h) / T$ = **0** mm

Eccentricity of base reaction in y; $e_{Ty} = (P_A \times e_{PyA} + M_{yA} + H_{yA} \times h) / T$ = **0** mm

Check pad base reaction eccentricity

$abs(e_{Tx}) / L + abs(e_{Ty}) / B$ = **0.000**

Base reaction acts within middle third of base

Calculate pad base pressures

$q_1 = T / A - 6 \times T \times e_{Tx} / (L \times A) - 6 \times T \times e_{Ty} / (B \times A)$ = **97.156** kN/m^2

$q_2 = T / A - 6 \times T \times e_{Tx} / (L \times A) + 6 \times T \times e_{Ty} / (B \times A)$ = **97.156** kN/m^2

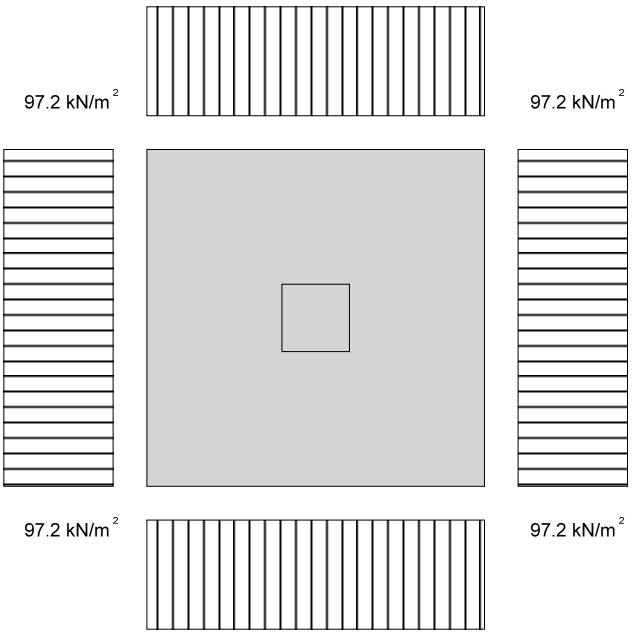
$q_3 = T / A + 6 \times T \times e_{Tx} / (L \times A) - 6 \times T \times e_{Ty} / (B \times A)$ = **97.156** kN/m^2

$q_4 = T / A + 6 \times T \times e_{Tx} / (L \times A) + 6 \times T \times e_{Ty} / (B \times A)$ = **97.156** kN/m^2

Minimum base pressure; $q_{min} = min(q_1, q_2, q_3, q_4)$ = **97.156** kN/m^2

Maximum base pressure; $q_{max} = max(q_1, q_2, q_3, q_4)$ = **97.156** kN/m^2

PASS - Maximum base pressure is less than allowable bearing pressure



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Material details

Characteristic strength of concrete; f_{cu} = **30** N/mm^2

Calculate pad lengths in x direction

Left hand length; $L_L = L / 2 + e_{PxA}$ = **750** mm

Right hand length; $L_R = L / 2 - e_{PxA}$ = **750** mm

Calculate rate of change of base pressure in x direction

Length of base reaction; $L_x = L$ = **1500** mm

Rate of change of base pressure; $C_x = [(q_1 + q_2) - (q_3 + q_4)] / (2 \times L_x)$ = **0.000** kN/m^2/m

Calculate pad lengths in y direction

Top length; $L_T = B / 2 - e_{PyA}$ = **750** mm

Bottom length; $L_B = B / 2 + e_{PyA}$ = **750** mm

Calculate rate of change of base pressure in y direction

Length of base reaction; $L_y = B$ = **1500** mm

Rate of change of base pressure; $C_y = [(q_1 + q_3) - (q_2 + q_4)] / (2 \times L_y)$ = **0.000** kN/m^2/m

Calculate minimum depth of unreinforced pad footing

Average pressure to left of pad footing; $q_L = (q_1 + q_2) / 2 - C_x \times (L_L - l_A / 2) / 2$ = **97.156** kN/m^2

Minimum depth to left of pad footing; $h_{Lmin} = (L_L - l_A/2) \times \max(0.15 \times [(q_L/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **600** mm

Average pressure to right of pad footing; $q_R = (q_3 + q_4) / 2 - C_x \times (L_R - l_A / 2) / 2$ = **97.156** kN/m^2

Minimum depth to right of pad footing; $h_{Rmin} = (L_R - l_A/2) \times \max(0.15 \times [(q_R/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **600** mm

Average pressure to top of pad footing; $q_T = (q_2 + q_4) / 2 - C_y \times (L_T - b_A / 2) / 2$ = **97.156** kN/m^2

Minimum depth to top of pad footing; $h_{Tmin} = (L_T - b_A/2) \times \max(0.15 \times [(q_T/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **600** mm

Average pressure to bottom of pad footing; $q_B = (q_1 + q_3) / 2 - C_y \times (L_B - b_A / 2) / 2$ = **97.156** kN/m^2

Minimum depth to bottom of pad footing; $h_{Bmin} = (L_B - b_A/2) \times \max(0.15 \times [(q_B/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **600** mm

Minimum depth of unreinforced pad footing; $h_{min} = \max(h_{Lmin}, h_{Rmin}, h_{Tmin}, h_{Bmin}, 300\text{ mm})$ = **600** mm

PASS - Unreinforced pad footing depth is greater than minimum

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PAD FOOTING F3 & F5

Assuming **1.3m X 1.3 m X 1.0 m** deep Mass Concrete footing.

LOADING: (Worst Case)

Load from the Beam 3 = 101.00 Kn.

Load from the Beam 4 = 52.00 Kn.

153.00 Kn

Total unfactored load = 153.00 Kn X 0.7 = 107.10 Kn.

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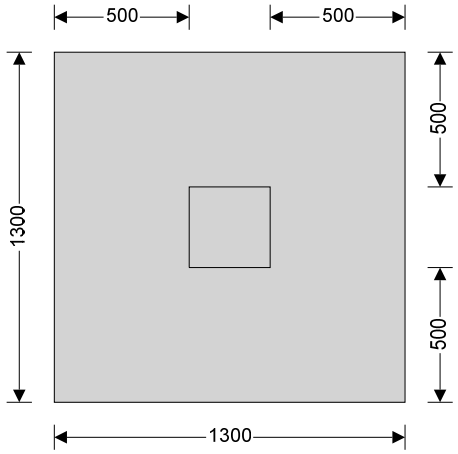
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PAD FOOTING F3 & F5 DESIGN:

PAD FOOTING ANALYSIS AND DESIGN (BS8110-1:1997)

Tedds calculation version 2.0.07



Pad footing details

Length of pad footing;	L = **1300** mm
Width of pad footing;	B = **1300** mm
Area of pad footing;	A = L × B = **1.690** m^2
Depth of pad footing;	h = **1000** mm
Depth of soil over pad footing;	h_{soil} = **200** mm
Density of concrete;	ρ_{conc} = **23.6** kN/m^3

Column details

Column base length;	l_A = **300** mm
Column base width;	b_A = **300** mm
Column eccentricity in x;	e_{PxA} = **0** mm
Column eccentricity in y;	e_{PyA} = **0** mm

Soil details

Density of soil;	ρ_{soil} = **20.0** kN/m^3
Design shear strength;	ϕ' = **25.0** deg
Design base friction;	δ = **19.3** deg
Allowable bearing pressure;	$P_{bearing}$ = **100** kN/m^2

Axial loading on column

Dead axial load on column;	P_{GA} = **107.1** kN
Imposed axial load on column;	P_{QA} = **0.0** kN
Wind axial load on column;	P_{WA} = **0.0** kN
Total axial load on column;	P_A = **107.1** kN

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Foundation loads

Dead surcharge load; F_{Gsur} = **0.000** kN/m^2

Imposed surcharge load; F_{Qsur} = **0.000** kN/m^2

Pad footing self weight; $F_{swt} = h \times \rho_{conc}$ = **23.600** kN/m^2

Soil self weight; $F_{soil} = h_{soil} \times \rho_{soil}$ = **4.000** kN/m^2

Total foundation load; $F = A \times (F_{Gsur} + F_{Qsur} + F_{swt} + F_{soil})$ = **46.6** kN

Calculate pad base reaction

Total base reaction; $T = F + P_A$ = **153.7** kN

Eccentricity of base reaction in x; $e_{Tx} = (P_A \times e_{PxA} + M_{xA} + H_{xA} \times h) / T$ = **0** mm

Eccentricity of base reaction in y; $e_{Ty} = (P_A \times e_{PyA} + M_{yA} + H_{yA} \times h) / T$ = **0** mm

Check pad base reaction eccentricity

$abs(e_{Tx}) / L + abs(e_{Ty}) / B$ = **0.000**

Base reaction acts within middle third of base

Calculate pad base pressures

$q_1 = T / A - 6 \times T \times e_{Tx} / (L \times A) - 6 \times T \times e_{Ty} / (B \times A)$ = **90.973** kN/m^2

$q_2 = T / A - 6 \times T \times e_{Tx} / (L \times A) + 6 \times T \times e_{Ty} / (B \times A)$ = **90.973** kN/m^2

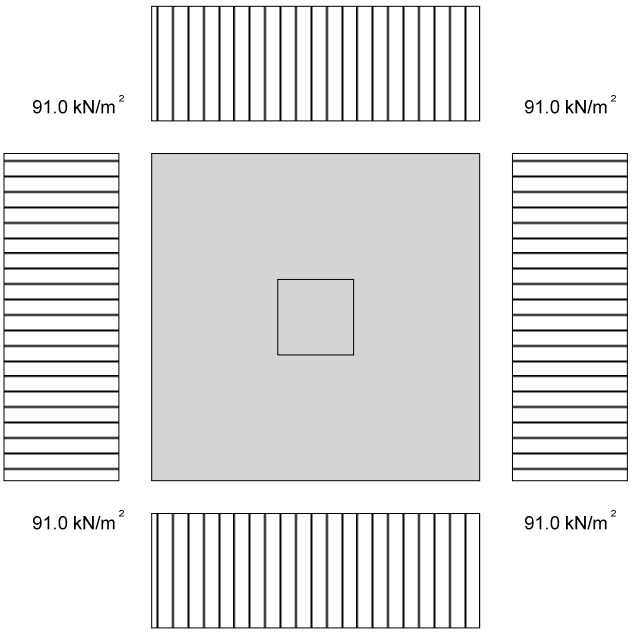
$q_3 = T / A + 6 \times T \times e_{Tx} / (L \times A) - 6 \times T \times e_{Ty} / (B \times A)$ = **90.973** kN/m^2

$q_4 = T / A + 6 \times T \times e_{Tx} / (L \times A) + 6 \times T \times e_{Ty} / (B \times A)$ = **90.973** kN/m^2

Minimum base pressure; $q_{min} = min(q_1, q_2, q_3, q_4)$ = **90.973** kN/m^2

Maximum base pressure; $q_{max} = max(q_1, q_2, q_3, q_4)$ = **90.973** kN/m^2

PASS - Maximum base pressure is less than allowable bearing pressure



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Material details

Characteristic strength of concrete; f_{cu} = **30** N/mm^2

Calculate pad lengths in x direction

Left hand length; $L_L = L / 2 + e_{PxA}$ = **650** mm

Right hand length; $L_R = L / 2 - e_{PxA}$ = **650** mm

Calculate rate of change of base pressure in x direction

Length of base reaction; $L_x = L$ = **1300** mm

Rate of change of base pressure; $C_x = [(q_1 + q_2) - (q_3 + q_4)] / (2 \times L_x)$ = **0.000** kN/m^2/m

Calculate pad lengths in y direction

Top length; $L_T = B / 2 - e_{PyA}$ = **650** mm

Bottom length; $L_B = B / 2 + e_{PyA}$ = **650** mm

Calculate rate of change of base pressure in y direction

Length of base reaction; $L_y = B$ = **1300** mm

Rate of change of base pressure; $C_y = [(q_1 + q_3) - (q_2 + q_4)] / (2 \times L_y)$ = **0.000** kN/m^2/m

Calculate minimum depth of unreinforced pad footing

Average pressure to left of pad footing; $q_L = (q_1 + q_2) / 2 - C_x \times (L_L - l_A / 2) / 2$ = **90.973** kN/m^2

Minimum depth to left of pad footing; $h_{Lmin} = (L_L - l_A/2) \times \max(0.15 \times [(q_L/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **500** mm

Average pressure to right of pad footing; $q_R = (q_3 + q_4) / 2 - C_x \times (L_R - l_A / 2) / 2$ = **90.973** kN/m^2

Minimum depth to right of pad footing; $h_{Rmin} = (L_R - l_A/2) \times \max(0.15 \times [(q_R/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **500** mm

Average pressure to top of pad footing; $q_T = (q_2 + q_4) / 2 - C_y \times (L_T - b_A / 2) / 2$ = **90.973** kN/m^2

Minimum depth to top of pad footing; $h_{Tmin} = (L_T - b_A/2) \times \max(0.15 \times [(q_T/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **500** mm

Average pressure to bottom of pad footing; $q_B = (q_1 + q_3) / 2 - C_y \times (L_B - b_A / 2) / 2$ = **90.973** kN/m^2

Minimum depth to bottom of pad footing; $h_{Bmin} = (L_B - b_A/2) \times \max(0.15 \times [(q_B/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **500** mm

Minimum depth of unreinforced pad footing; $h_{min} = \max(h_{Lmin}, h_{Rmin}, h_{Tmin}, h_{Bmin}, 300\text{ mm})$ = **500** mm

PASS - Unreinforced pad footing depth is greater than minimum

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PAD FOOTING F4

Assuming **1.7 m X 1.7 m X 1.0 m** deep Mass Concrete footing.

LOADING: (Worst Case)

Load from the Column C4 = 281.00 Kn.

281.00 Kn

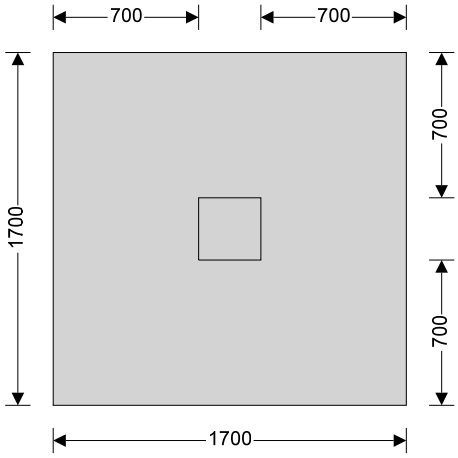
Total unfactored load = 281.00 Kn X 0.7 = 196.70 Kn.

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PAD FOOTING F4 DESIGN:

PAD FOOTING ANALYSIS AND DESIGN (BS8110-1:1997)

Tedds calculation version 2.0.07



Pad footing details

Length of pad footing;	L = **1700** mm
Width of pad footing;	B = **1700** mm
Area of pad footing;	A = L × B = **2.890** m^2
Depth of pad footing;	h = **1000** mm
Depth of soil over pad footing;	h_{soil} = **200** mm
Density of concrete;	ρ_{conc} = **23.6** kN/m^3

Column details

Column base length;	l_A = **300** mm
Column base width;	b_A = **300** mm
Column eccentricity in x;	e_{PxA} = **0** mm
Column eccentricity in y;	e_{PyA} = **0** mm

Soil details

Density of soil;	ρ_{soil} = **20.0** kN/m^3
Design shear strength;	ϕ' = **25.0** deg
Design base friction;	δ = **19.3** deg
Allowable bearing pressure;	$P_{bearing}$ = **100** kN/m^2

Axial loading on column

Dead axial load on column;	P_{GA} = **196.7** kN
Imposed axial load on column;	P_{QA} = **0.0** kN
Wind axial load on column;	P_{WA} = **0.0** kN
Total axial load on column;	P_A = **196.7** kN

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Foundation loads

Dead surcharge load; F_{Gsur} = **0.000** kN/m^2

Imposed surcharge load; F_{Qsur} = **0.000** kN/m^2

Pad footing self weight; $F_{swt} = h \times \rho_{conc}$ = **23.600** kN/m^2

Soil self weight; $F_{soil} = h_{soil} \times \rho_{soil}$ = **4.000** kN/m^2

Total foundation load; $F = A \times (F_{Gsur} + F_{Qsur} + F_{swt} + F_{soil})$ = **79.8** kN

Calculate pad base reaction

Total base reaction; $T = F + P_A$ = **276.5** kN

Eccentricity of base reaction in x; $e_{Tx} = (P_A \times e_{PxA} + M_{xA} + H_{xA} \times h) / T$ = **0** mm

Eccentricity of base reaction in y; $e_{Ty} = (P_A \times e_{PyA} + M_{yA} + H_{yA} \times h) / T$ = **0** mm

Check pad base reaction eccentricity

$abs(e_{Tx}) / L + abs(e_{Ty}) / B$ = **0.000**

Base reaction acts within middle third of base

Calculate pad base pressures

$q_1 = T / A - 6 \times T \times e_{Tx} / (L \times A) - 6 \times T \times e_{Ty} / (B \times A)$ = **95.662** kN/m^2

$q_2 = T / A - 6 \times T \times e_{Tx} / (L \times A) + 6 \times T \times e_{Ty} / (B \times A)$ = **95.662** kN/m^2

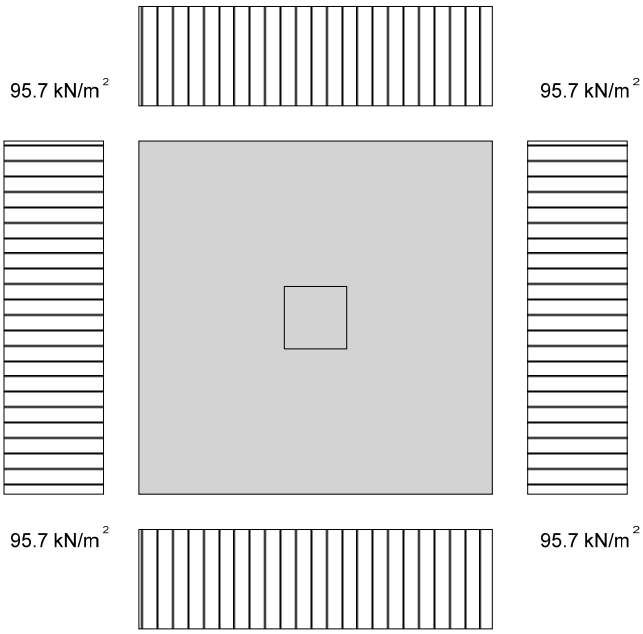
$q_3 = T / A + 6 \times T \times e_{Tx} / (L \times A) - 6 \times T \times e_{Ty} / (B \times A)$ = **95.662** kN/m^2

$q_4 = T / A + 6 \times T \times e_{Tx} / (L \times A) + 6 \times T \times e_{Ty} / (B \times A)$ = **95.662** kN/m^2

Minimum base pressure; $q_{min} = min(q_1, q_2, q_3, q_4)$ = **95.662** kN/m^2

Maximum base pressure; $q_{max} = max(q_1, q_2, q_3, q_4)$ = **95.662** kN/m^2

PASS - Maximum base pressure is less than allowable bearing pressure



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Material details

Characteristic strength of concrete; f_{cu} = **30** N/mm^2

Calculate pad lengths in x direction

Left hand length; $L_L = L / 2 + e_{PxA}$ = **850** mm

Right hand length; $L_R = L / 2 - e_{PxA}$ = **850** mm

Calculate rate of change of base pressure in x direction

Length of base reaction; $L_x = L$ = **1700** mm

Rate of change of base pressure; $C_x = [(q_1 + q_2) - (q_3 + q_4)] / (2 \times L_x)$ = **0.000** kN/m^2/m

Calculate pad lengths in y direction

Top length; $L_T = B / 2 - e_{PyA}$ = **850** mm

Bottom length; $L_B = B / 2 + e_{PyA}$ = **850** mm

Calculate rate of change of base pressure in y direction

Length of base reaction; $L_y = B$ = **1700** mm

Rate of change of base pressure; $C_y = [(q_1 + q_3) - (q_2 + q_4)] / (2 \times L_y)$ = **0.000** kN/m^2/m

Calculate minimum depth of unreinforced pad footing

Average pressure to left of pad footing; $q_L = (q_1 + q_2) / 2 - C_x \times (L_L - l_A / 2) / 2$ = **95.662** kN/m^2

Minimum depth to left of pad footing; $h_{Lmin} = (L_L - l_A/2) \times \max(0.15 \times [(q_L/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **700** mm

Average pressure to right of pad footing; $q_R = (q_3 + q_4) / 2 - C_x \times (L_R - l_A / 2) / 2$ = **95.662** kN/m^2

Minimum depth to right of pad footing; $h_{Rmin} = (L_R - l_A/2) \times \max(0.15 \times [(q_R/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **700** mm

Average pressure to top of pad footing; $q_T = (q_2 + q_4) / 2 - C_y \times (L_T - b_A / 2) / 2$ = **95.662** kN/m^2

Minimum depth to top of pad footing; $h_{Tmin} = (L_T - b_A/2) \times \max(0.15 \times [(q_T/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **700** mm

Average pressure to bottom of pad footing; $q_B = (q_1 + q_3) / 2 - C_y \times (L_B - b_A / 2) / 2$ = **95.662** kN/m^2

Minimum depth to bottom of pad footing; $h_{Bmin} = (L_B - b_A/2) \times \max(0.15 \times [(q_B/1\text{kN/m}^2)^2 / (f_{cu}/1\text{N/mm}^2)]^{1/4}, 1)$ = **700** mm

Minimum depth of unreinforced pad footing; $h_{min} = \max(h_{Lmin}, h_{Rmin}, h_{Tmin}, h_{Bmin}, 300\text{ mm})$ = **700** mm

PASS - Unreinforced pad footing depth is greater than minimum

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FOUNDATION DEPTH NEAR TREE:

FOUNDATIONS NEAR TREES

In accordance with Clause 4.2.12 of NHBC Part 4: Foundations - Chapter 4.2

Tedds calculation version 2.0.03

Site Details

Distance north of London;	N = **0** mi
Distance west of London;	W = **0** mi
Reduction depth due to climate variations - cl.4.2.4;	Z_c = **0.00** m

Soil Details

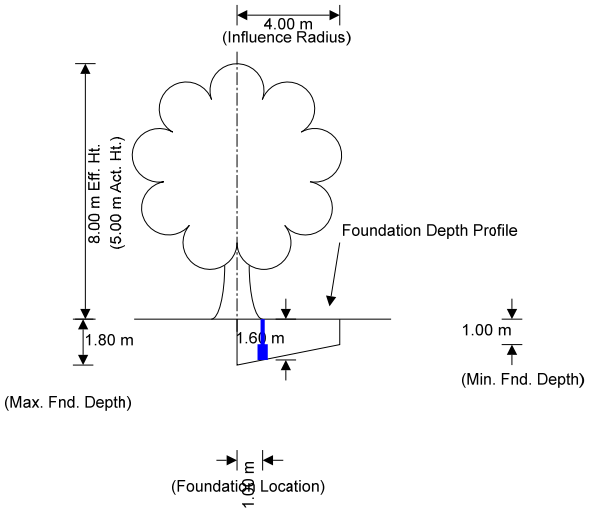
Plasticity index from lab tests;	I_p = **40** %
Percentage of particles < 425 μm;	p_{425} = **100** %
Modified plasticity index - cl. 4.2.4;	$I'_p = I_p \times p_{425} / 100$ % = **40** %
Volume change potential - Table 1;	High

Details for Tree - 1

Species of tree;	Broad leaf - Fig

tree is to remain at the site, with no further planting allowed.

Water demand of tree - Table 3;	Low
Mature height of tree - Table 3;	H_{m1} = **8.00** m
Influence radius - Table 3b;	$r_{inf1} = 0.5 \times H_{m1}$ = **4.00** m
Measured height of tree;	H_{act1} = **5.00** m
Distance from centre of tree to face of foundations;	D_1 = **1.00** m
Effective height of tree - Table 3a;	H_{eff1} = **8.00** m



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Minimum foundation depth - Table 4B; Z_{min} = **1.00** m

Look up value for foundation depth -; Chart 1 Soils with HIGH volume change potential;

$Z_{LookUp1}$ = **1.60** m

Required foundation depth; $Z_{req1} = Z_{LookUp1} - Z_c$ = **1.60** m

Summary Table

Tree	Name	Distance (m)	Measured Height (m)	Effective Height (m)	Tree to be removed	Required Foundation Depth (m)
1	Fig	1.0	5.0	8.0	No	1.60

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CONCENTRIC STRIP FOOTING

Assuming **0.6m Wide x1.6 m Deep** mass Concrete Foundation

LOADING:

I. Dead Load

Pitch Roof	$= \left\lfloor \frac{1.28kn/\mathrm{m}^2 \text{ x } (4.0)\mathrm{m}}{2} \right\rfloor$	= 2.56 Kn/m
Floor (Block & Beam)	$= \left\lfloor \frac{2.15kn/\mathrm{m}^2 \text{ x } (4.2)\mathrm{m}}{2} \right\rfloor$	= 4.52 Kn/m
Brick work	= 20 Kn/m^3 x 0.100 x 6.0	= 12.00 Kn/m
Block work	= 19 Kn/m^3 x 0.100 x 6.0	= 11.40 Kn/m
		30.48 Kn/m

II. Imposed Load

Pitch Roof	$= \left\lfloor \frac{0.92kn/\mathrm{m}^2 \text{ x } (4.0)\mathrm{m}}{2} \right\rfloor$	= 1.84 Kn/m
Floor (Block & Beam)	$= \left\lfloor \frac{3.0kn/\mathrm{m}^2 \text{ x } (4.2)\mathrm{m}}{2} \right\rfloor$	= 6.30 Kn/m
		8.14 Kn/m

Total unfactored Dead load = 30.48 Kn X 0.7 = 21.34 Kn.

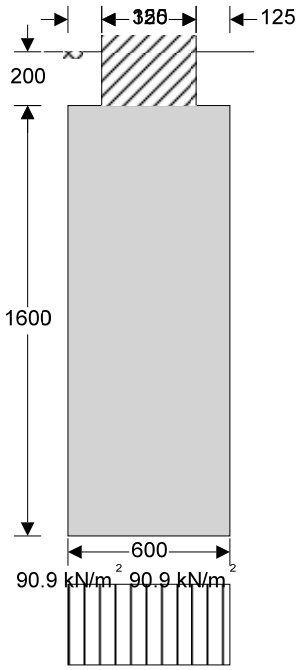
Total unfactored Live load = 8.14 Kn X 0.7 = 5.70 Kn.

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CENTRIC STRIP FOOTING DESIGN:

STRIP FOOTING ANALYSIS AND DESIGN (BS8110-1:1997)

Tedds calculation version 2.0.07



Strip footing details

Width of strip footing;	B = **600** mm
Depth of strip footing;	h = **1600** mm
Depth of soil over strip footing;	h_{soil} = **200** mm
Density of concrete;	ρ_{conc} = **23.6** kN/m^3

Load details

Load width;	b = **350** mm
Load eccentricity;	e_P = **0** mm

Soil details

Dense, moderately graded, sub-angular, gravel

Mobilisation factor;	m= ;**1.5**;
Density of soil;	ρ_{soil} = **20.0** kN/m^3
Design shear strength;	ϕ' = **25.0** deg
Design base friction;	δ = **19.3** deg
Allowable bearing pressure;	$P_{bearing}$ = **100** kN/m^2

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

Axial loading on strip footing

Dead axial load; P_G = **21.3** kN/m

Imposed axial load; P_Q = **8.1** kN/m

Wind axial load; P_W = **0.0** kN/m

Total axial load; P = **29.5** kN/m

Foundation loads

Dead surcharge load; F_{Gsur} = **0.000** kN/m^2

Imposed surcharge load; F_{Qsur} = **0.000** kN/m^2

Strip footing self weight; $F_{swt} = h \times \rho_{conc}$ = **37.760** kN/m^2

Soil self weight; $F_{soil} = h_{soil} \times \rho_{soil}$ = **4.000** kN/m^2

Total foundation load; $F = B \times (F_{Gsur} + F_{Qsur} + F_{swt} + F_{soil})$ = **25.1** kN/m

Calculate base reaction

Total base reaction; T = F + P = **54.5** kN/m

Eccentricity of base reaction in x; $e_T = (P \times e_P + M + H \times h) / T$ = **0** mm

Base reaction acts within middle third of base

Calculate base pressures

$q_1 = (T / B) \times (1 - 6 \times e_T / B)$ = **90.893** kN/m^2

$q_2 = (T / B) \times (1 + 6 \times e_T / B)$ = **90.893** kN/m^2

Minimum base pressure; $q_{min} = \min(q_1, q_2)$ = **90.893** kN/m^2

Maximum base pressure; $q_{max} = \max(q_1, q_2)$ = **90.893** kN/m^2

PASS - Maximum base pressure is less than allowable bearing pressure

Material details

Characteristic strength of concrete; f_{cu} = **30** N/mm^2

Calculate base lengths

Left hand length; $B_L = B / 2 + e_P$ = **300** mm

Right hand length; $B_R = B / 2 - e_P$ = **300** mm

Calculate rate of change of base pressure

Length of base reaction; $B_x = B$ = **600** mm

Rate of change of base pressure; $C_x = (q_1 - q_2) / B_x$ = **0.000** kN/m^2/m

Calculate minimum depth of unreinforced strip footing

Average pressure to left of strip footing; $q_L = q_1 - C_x \times (B_L - b / 2) / 2$ = **90.893** kN/m^2

Minimum depth to left of strip footing; $h_{Lmin} = (B_L - b/2) \times \max(0.15 \times [(q_L / 1\ kN/m^2)^2 / (f_{cu} / 1\ N/mm^2)]^{1/4}, 1)$ = **125** mm

Average pressure to right of strip footing; $q_R = q_2 + C_x \times (B_R - b / 2) / 2$ = **90.893** kN/m^2

Minimum depth to right of strip footing; $h_{Rmin} = (B_R - b/2) \times \max(0.15 \times [(q_R / 1kN/m^2)^2 / (f_{cu} / 1N/mm^2)]^{1/4}, 1)$ = **125** mm

Minimum depth of unreinforced strip footing; $h_{min} = \max(h_{Lmin}, h_{Rmin}, 300\ mm)$ = **300** mm

PASS - Unreinforced strip footing depth is greater than minimum

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

RAFTERS

SPAN OF THE JOIST = 4000 mm

Assuming **50 X 200 mm C24** grade timbers @ **400 mm** C/c.

LOADING:

I. Dead Load

Pitched roof = $\left\lfloor \frac{1.05kn/\mathrm{m}^2}{\cos(35°)} \right\rfloor$ = 1.28 Kn/m

---

1.28 Kn/m

II. Imposed Load

Pitched roof = $\left\lfloor \frac{0.75kn/\mathrm{m}^2}{\cos(35°)} \right\rfloor$ = 0.92 Kn/m

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0.92 Kn/m

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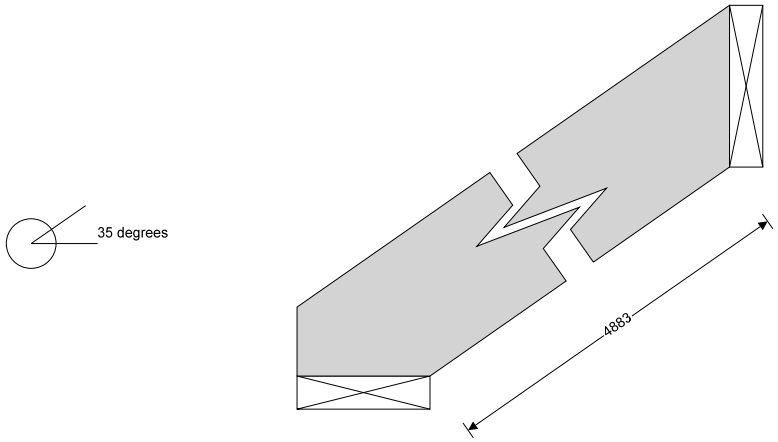
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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

RAFTERS DESIGN:

TIMBER RAFTER DESIGN (BS5268-2:2002)

TEDDS calculation version 1.0.03



Rafter details

Breadth of timber sections;	b = **50** mm
Depth of timber sections;	h = **200** mm
Rafter spacing;	s = **400** mm
Rafter slope;	α = **35.0** deg
Clear span of rafter on horizontal;	L_{clh} = **4000** mm
Clear span of rafter on slope;	$L_{cl} = L_{clh} / \cos(\alpha)$ = **4883** mm
Rafter span;	**Single span**
Timber strength class;	**C24**

Section properties

Cross sectional area of rafter;	$A = b \times h$ = **10000** mm^2
Section modulus;	$Z = b \times h^2 / 6$ = **333333** mm^3
Second moment of area;	$I = b \times h^3 / 12$ = **33333333** mm^4
Radius of gyration;	$r = \sqrt{(I / A)}$ = **57.7** mm

Loading details

Rafter self weight;	$F_j = b \times h \times \rho_{char} \times g_{acc}$ = **0.03** kN/m
Dead load on slope;	F_d = **1.05** kN/m^2
Imposed load on plan;	F_u = **0.92** kN/m^2
Imposed point load;	F_p = **0.90** kN

Modification factors

Section depth factor;	$K_7 = (300 \text{ mm} / h)^{0.11}$ = **1.05**
Load sharing factor;	K_8 = **1.10**

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

Consider long term load condition

Load duration factor; K_3 = **1.00**

Total UDL perpendicular to rafter; $F = F_d \times \cos(\alpha) \times s + F_j \times \cos(\alpha)$ = **0.372** kN/m

Notional bearing length; $L_b = F \times L_{cl} / [2 \times (b \times \sigma_{cp1} \times K_8 - F)]$ = **7** mm

Effective span; $L_{eff} = L_{cl} + L_b$ = **4890** mm

Check bending stress

Bending stress parallel to grain; σ_m = **7.500** N/mm^2

Permissible bending stress; $\sigma_{m_adm} = \sigma_m \times K_3 \times K_7 \times K_8$ = **8.626** N/mm^2

Applied bending stress; $\sigma_{m_max} = F \times L_{eff}^2 / (8 \times Z)$ = **3.337** N/mm^2

PASS - Applied bending stress within permissible limits

Check compressive stress parallel to grain

Compression stress parallel to grain; σ_c = **7.900** N/mm^2

Minimum modulus of elasticity; E_{min} = **7200** N/mm^2

Compression member factor; K_{12} = **0.49**

Permissible compressive stress; $\sigma_{c_adm} = \sigma_c \times K_3 \times K_8 \times K_{12}$ = **4.279** N/mm^2

Applied compressive stress; $\sigma_{c_max} = F \times L_{eff} \times (\cot(\alpha) + 3 \times \tan(\alpha)) / (2 \times A)$ = **0.321** N/mm^2

PASS - Applied compressive stress within permissible limits

Check combined bending and compressive stress parallel to grain

Euler stress; $\sigma_e = \pi^2 \times E_{min} / \lambda^2$ = **9.906** N/mm^2

Euler coefficient; $K_{eu} = 1 - (1.5 \times \sigma_{c_max} \times K_{12} / \sigma_e)$ = **0.976**

Combined axial compression and bending check; $\sigma_{m_max} / (\sigma_{m_adm} \times K_{eu}) + \sigma_{c_max} / \sigma_{c_adm}$ = **0.471**; **< 1**

PASS - Combined compressive and bending stresses are within permissible limits

Check shear stress

Shear stress parallel to grain; τ = **0.710** N/mm^2

Permissible shear stress; $\tau_{adm} = \tau \times K_3 \times K_8$ = **0.781** N/mm^2

Applied shear stress; $\tau_{max} = 3 \times F \times L_{eff} / (4 \times A)$ = **0.136** N/mm^2

PASS - Applied shear stress within permissible limits

Check deflection

Permissible deflection; $\delta_{adm} = 0.003 \times L_{eff}$ = **14.670** mm

Bending deflection; $\delta_b = 5 \times F \times L_{eff}^4 / (384 \times E_{mean} \times I)$ = **7.697** mm

Shear deflection; $\delta_s = 12 \times F \times L_{eff}^2 / (5 \times E_{mean} \times A)$ = **0.198** mm

Total deflection; $\delta_{max} = \delta_b + \delta_s$ = **7.894** mm

PASS - Total deflection within permissible limits

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Consider medium term load condition

Load duration factor; K_3 = **1.25**

Total UDL perpendicular to rafter; $F = [F_u \times \cos(\alpha)^2 + F_d \times \cos(\alpha)] \times s + F_j \times \cos(\alpha)$ = **0.619** kN/m

Notional bearing length; $L_b = F \times L_{cl} / [2 \times (b \times \sigma_{cp1} \times K_8 - F)]$ = **12** mm

Effective span; $L_{eff} = L_{cl} + L_b$ = **4895** mm

Check bending stress

Bending stress parallel to grain; σ_m = **7.500** N/mm^2

Permissible bending stress; $\sigma_{m_adm} = \sigma_m \times K_3 \times K_7 \times K_8$ = **10.783** N/mm^2

Applied bending stress; $\sigma_{m_max} = F \times L_{eff}^2 / (8 \times Z)$ = **5.562** N/mm^2

PASS - Applied bending stress within permissible limits

Check compressive stress parallel to grain

Compression stress parallel to grain; σ_c = **7.900** N/mm^2

Minimum modulus of elasticity; E_{min} = **7200** N/mm^2

Compression member factor; K_{12} = **0.44**

Permissible compressive stress; $\sigma_{c_adm} = \sigma_c \times K_3 \times K_8 \times K_{12}$ = **4.809** N/mm^2

Applied compressive stress; $\sigma_{c_max} = F \times L_{eff} \times (\cot(\alpha) + 3 \times \tan(\alpha)) / (2 \times A)$ = **0.535** N/mm^2

PASS - Applied compressive stress within permissible limits

Check combined bending and compressive stress parallel to grain

Euler stress; $\sigma_e = \pi^2 \times E_{min} / \lambda^2$ = **9.887** N/mm^2

Euler coefficient; $K_{eu} = 1 - (1.5 \times \sigma_{c_max} \times K_{12} / \sigma_e)$ = **0.964**

Combined axial compression and bending check; $\sigma_{m_max} / (\sigma_{m_adm} \times K_{eu}) + \sigma_{c_max} / \sigma_{c_adm}$ = **0.646**; **< 1**

PASS - Combined compressive and bending stresses are within permissible limits

Check shear stress

Shear stress parallel to grain; τ = **0.710** N/mm^2

Permissible shear stress; $\tau_{adm} = \tau \times K_3 \times K_8$ = **0.976** N/mm^2

Applied shear stress; $\tau_{max} = 3 \times F \times L_{eff} / (4 \times A)$ = **0.227** N/mm^2

PASS - Applied shear stress within permissible limits

Check deflection

Permissible deflection; $\delta_{adm} = 0.003 \times L_{eff}$ = **14.684** mm

Bending deflection; $\delta_b = 5 \times F \times L_{eff}^4 / (384 \times E_{mean} \times I)$ = **12.852** mm

Shear deflection; $\delta_s = 12 \times F \times L_{eff}^2 / (5 \times E_{mean} \times A)$ = **0.330** mm

Total deflection; $\delta_{max} = \delta_b + \delta_s$ = **13.181** mm

PASS - Total deflection within permissible limits

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Consider short term load condition

Load duration factor; K_3 = **1.50**

Total UDL perpendicular to rafter; $F = F_d \times \cos(\alpha) \times s + F_j \times \cos(\alpha)$ = **0.372** kN/m

Notional bearing length; $L_b = [F \times L_{cl} + F_p \times \cos(\alpha)] / [2 \times (b \times \sigma_{cp1} \times K_8 - F)]$ = **10** mm

Effective span; $L_{eff} = L_{cl} + L_b$ = **4893** mm

Check bending stress

Bending stress parallel to grain; σ_m = **7.500** N/mm^2

Permissible bending stress; $\sigma_{m_adm} = \sigma_m \times K_3 \times K_7 \times K_8$ = **12.939** N/mm^2

Applied bending stress; $\sigma_{m_max} = F \times L_{eff}^2/(8 \times Z) + F_p \times \cos(\alpha) \times L_{eff}/(4 \times Z)$ = **6.046** N/mm^2

PASS - Applied bending stress within permissible limits

Check compressive stress parallel to grain

Compression stress parallel to grain; σ_c = **7.900** N/mm^2

Minimum modulus of elasticity; E_{min} = **7200** N/mm^2

Compression member factor; K_{12} = **0.40**

Permissible compressive stress; $\sigma_{c_adm} = \sigma_c \times K_3 \times K_8 \times K_{12}$ = **5.209** N/mm^2

Applied compressive stress; $\sigma_{c_max} = F \times L_{eff} \times (\cot(\alpha) + 3 \times \tan(\alpha))/(2 \times A) + F_p \times \sin(\alpha)/A$ = **0.373** N/mm^2

PASS - Applied compressive stress within permissible limits

Check combined bending and compressive stress parallel to grain

Euler stress; $\sigma_e = \pi^2 \times E_{min} / \lambda^2$ = **9.895** N/mm^2

Euler coefficient; $K_{eu} = 1 - (1.5 \times \sigma_{c_max} \times K_{12} / \sigma_e)$ = **0.977**

Combined axial compression and bending check; $\sigma_{m_max} / (\sigma_{m_adm} \times K_{eu}) + \sigma_{c_max} / \sigma_{c_adm}$ = **0.550**; **< 1**

PASS - Combined compressive and bending stresses are within permissible limits

Check shear stress

Shear stress parallel to grain; τ = **0.710** N/mm^2

Permissible shear stress; $\tau_{adm} = \tau \times K_3 \times K_8$ = **1.172** N/mm^2

Applied shear stress; $\tau_{max} = 3 \times F \times L_{eff} / (4 \times A) + 3 \times F_p \times \cos(\alpha) / (2 \times A)$ = **0.247** N/mm^2

PASS - Applied shear stress within permissible limits

Check deflection

Permissible deflection; $\delta_{adm} = 0.003 \times L_{eff}$ = **14.678** mm

Bending deflection; $\delta_b = L_{eff}^3 \times (5 \times F \times L_{eff} / 384 + F_p \times \cos(\alpha)/ 48) / (E_{mean} \times I)$ = **12.712** mm

Shear deflection; $\delta_s = 12 \times L_{eff} \times (F \times L_{eff} + 2 \times F_p \times \cos(\alpha))/(5 \times E_{mean} \times A)$ = **0.358** mm

Total deflection; $\delta_{max} = \delta_b + \delta_s$ = **13.070** mm

PASS - Total deflection within permissible limits

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB		Date: JUL 2024

FLAT ROOF JOIST

SPAN OF THE JOIST = 1500 mm

Assuming **50 X 200 mm C24** grade timber @ **400 mm** C/c.

LOADING:

I. Dead Load

Flat Roof = $\left\lfloor \frac{1.0kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$ = 0.40 Kn/m

---

0.40 Kn/m

II. Imposed Load

Floor = $\left\lfloor \frac{0.75kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2} \right\rfloor$ = 0.30 Kn/m

---

0.30 Kn/m

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FLAT ROOF JOIST DESIGN:

TIMBER JOIST DESIGN (BS5268-2:2002)

Tedds calculation version 1.1.04

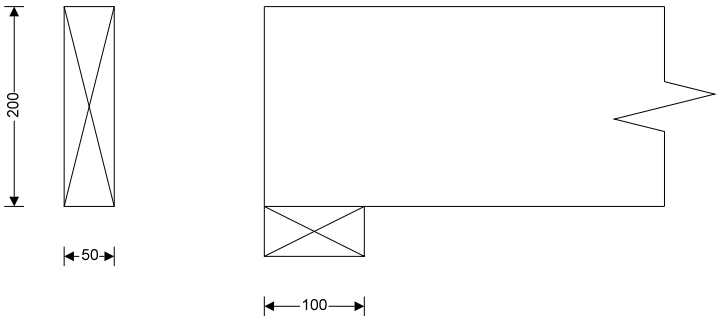
Joist details

Joist breadth;	b = **50** mm
Joist depth;	h = **200** mm
Joist spacing;	s = **400** mm
Timber strength class;	**C24**
Service class of timber;	**1**



Span details

Number of spans;	N_{span} = **1**
Length of bearing;	L_b = **100** mm
Effective length of span;	L_{s1} = **1500** mm



Section properties

Second moment of area;	$I = b \times h^3 / 12$ = **33333333** mm^4
Section modulus;	$Z = b \times h^2 / 6$ = **333333** mm^3

Loading details

Joist self weight;	$F_{swt} = b \times h \times \rho_{char} \times g_{acc}$ = **0.03** kN/m
Dead load;	F_{d_udl} = **0.90** kN/m^2
Imposed UDL(Medium term);	F_{i_udl} = **0.75** kN/m^2
Imposed point load (Short term);	F_{i_pt} = **0.90** kN

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Modification factors

Service class for bending parallel to grain	K_{2m} = **1.00**
Service class for compression	K_{2c} = **1.00**
Service class for shear parallel to grain	K_{2s} = **1.00**
Service class for modulus of elasticity	K_{2e} = **1.00**
Section depth factor;	K_7 = **1.05**
Load sharing factor;	K_8 = **1.10**

Consider medium term loads

Load duration factor;	K_3 = **1.25**
Maximum bending moment;	M = **0.195** kNm
Maximum shear force;	V = **0.521** kN
Maximum support reaction;	R = **0.521** kN
Maximum deflection;	δ = **0.162** mm

Check bending stress

Bending stress;	σ_m = **7.500** N/mm^2
Permissible bending stress;	$\sigma_{m_adm} = \sigma_m \times K_{2m} \times K_3 \times K_7 \times K_8$ = **10.783** N/mm^2
Applied bending stress;	σ_{m_max} = M / Z = **0.586** N/mm^2

PASS - Applied bending stress within permissible limits

Check shear stress

Shear stress;	τ = **0.710** N/mm^2
Permissible shear stress;	$\tau_{adm} = \tau \times K_{2s} \times K_3 \times K_8$ = **0.976** N/mm^2
Applied shear stress;	τ_{max} = 3 × V / (2 × b × h) = **0.078** N/mm^2

PASS - Applied shear stress within permissible limits

Check bearing stress

Compression perpendicular to grain (no wane);	σ_{cp1} = **2.400** N/mm^2
Permissible bearing stress;	$\sigma_{c_adm} = \sigma_{cp1} \times K_{2c} \times K_3 \times K_8$ = **3.300** N/mm^2
Applied bearing stress;	σ_{c_max} = R / (b × L_b) = **0.104** N/mm^2

PASS - Applied bearing stress within permissible limits

Check deflection

Permissible deflection;	δ_{adm} = min(L_{s1} × 0.003, 14 mm) = **4.500** mm
Bending deflection (based on E_{mean});	$\delta_{bending}$ = **0.127** mm
Shear deflection;	δ_{shear} = **0.035** mm
Total deflection;	$\delta = \delta_{bending} + \delta_{shear}$ = **0.162** mm

PASS - Actual deflection within permissible limits

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Consider short term loads

Load duration factor;	K_3 = **1.50**
Maximum bending moment;	M = **0.448** kNm
Maximum shear force;	V = **1.196** kN
Maximum support reaction;	R = **1.196** kN
Maximum deflection;	δ = **0.328** mm

Check bending stress

Bending stress;	σ_m = **7.500** N/mm^2
Permissible bending stress;	$\sigma_{m_adm} = \sigma_m \times K_{2m} \times K_3 \times K_7 \times K_8$ = **12.939** N/mm^2
Applied bending stress;	σ_{m_max} = M / Z = **1.345** N/mm^2

PASS - Applied bending stress within permissible limits

Check shear stress

Shear stress;	τ = **0.710** N/mm^2
Permissible shear stress;	$\tau_{adm} = \tau \times K_{2s} \times K_3 \times K_8$ = **1.172** N/mm^2
Applied shear stress;	$\tau_{max} = 3 \times V / (2 \times b \times h)$ = **0.179** N/mm^2

PASS - Applied shear stress within permissible limits

Check bearing stress

Compression perpendicular to grain (no wane);	σ_{cp1} = **2.400** N/mm^2
Permissible bearing stress;	$\sigma_{c_adm} = \sigma_{cp1} \times K_{2c} \times K_3 \times K_8$ = **3.960** N/mm^2
Applied bearing stress;	$\sigma_{c_max} = R / (b \times L_b)$ = **0.239** N/mm^2

PASS - Applied bearing stress within permissible limits

Check deflection

Permissible deflection;	$\delta_{adm} = \min(L_{s1} \times 0.003, 14\text{ mm})$ = **4.500** mm
Bending deflection (based on E_{mean});	$\delta_{bending}$ = **0.248** mm
Shear deflection;	δ_{shear} = **0.080** mm
Total deflection;	$\delta = \delta_{bending} + \delta_{shear}$ = **0.328** mm

PASS - Actual deflection within permissible limits

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

FLOOR JOIST

SPAN OF THE JOIST = 2700 mm

Assuming **50 X 200 mm C24** grade timber @ **400 mm** C/c.

LOADING:

I. Dead Load

Floor = $\left[\frac{0.60kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2}\right]$ = 0.22 Kn/m

---

0.22 Kn/m

II. Imposed Load

Floor = $\left[\frac{1.5kn/\mathrm{m}^2 \text{ x } (0.4+0.4)\mathrm{m}}{2}\right]$ = 0.60 Kn/m

---

0.60 Kn/m

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FLOOR JOIST DESIGN:

TIMBER JOIST DESIGN (BS5268-2:2002)

Tedds calculation version 1.1.04

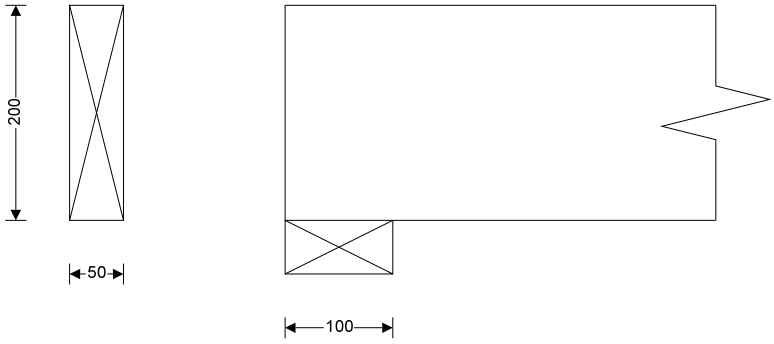
Joist details

Joist breadth;	b = **50** mm
Joist depth;	h = **200** mm
Joist spacing;	s = **400** mm
Timber strength class;	**C24**
Service class of timber;	**1**



Span details

Number of spans;	N_{span} = **1**
Length of bearing;	L_b = **100** mm
Effective length of span;	L_{s1} = **2700** mm



Section properties

Second moment of area;	$I = b \times h^3 / 12$ = **33333333** mm^4
Section modulus;	$Z = b \times h^2 / 6$ = **333333** mm^3

Loading details

Joist self weight;	$F_{swt} = b \times h \times \rho_{char} \times g_{acc}$ = **0.03** kN/m
Dead load;	F_{d_udl} = **0.60** kN/m^2
Imposed UDL(Long term);	F_{i_udl} = **1.50** kN/m^2
Imposed point load (Medium term);	F_{i_pt} = **1.40** kN

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Modification factors

Service class for bending parallel to grain	K_{2m} = **1.00**
Service class for compression	K_{2c} = **1.00**
Service class for shear parallel to grain	K_{2s} = **1.00**
Service class for modulus of elasticity	K_{2e} = **1.00**
Section depth factor;	K_7 = **1.05**
Load sharing factor;	K_8 = **1.10**

Consider long term loads

Load duration factor;	K_3 = **1.00**
Maximum bending moment;	M = **0.797** kNm
Maximum shear force;	V = **1.180** kN
Maximum support reaction;	R = **1.180** kN
Maximum deflection;	δ = **1.822** mm

Check bending stress

Bending stress;	σ_m = **7.500** N/mm^2
Permissible bending stress;	$\sigma_{m_adm} = \sigma_m \times K_{2m} \times K_3 \times K_7 \times K_8$ = **8.626** N/mm^2
Applied bending stress;	σ_{m_max} = M / Z = **2.390** N/mm^2

PASS - Applied bending stress within permissible limits

Check shear stress

Shear stress;	τ = **0.710** N/mm^2
Permissible shear stress;	$\tau_{adm} = \tau \times K_{2s} \times K_3 \times K_8$ = **0.781** N/mm^2
Applied shear stress;	$\tau_{max} = 3 \times V / (2 \times b \times h)$ = **0.177** N/mm^2

PASS - Applied shear stress within permissible limits

Check bearing stress

Compression perpendicular to grain (no wane);	σ_{cp1} = **2.400** N/mm^2
Permissible bearing stress;	$\sigma_{c_adm} = \sigma_{cp1} \times K_{2c} \times K_3 \times K_8$ = **2.640** N/mm^2
Applied bearing stress;	$\sigma_{c_max} = R / (b \times L_b)$ = **0.236** N/mm^2

PASS - Applied bearing stress within permissible limits

Check deflection

Permissible deflection;	$\delta_{adm} = \min(L_{s1} \times 0.003, 14\text{ mm})$ = **8.100** mm
Bending deflection (based on E_{mean});	$\delta_{bending}$ = **1.681** mm
Shear deflection;	δ_{shear} = **0.142** mm
Total deflection;	$\delta = \delta_{bending} + \delta_{shear}$ = **1.822** mm

PASS - Actual deflection within permissible limits

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Consider medium term loads

Load duration factor;	K_3 = **1.25**
Maximum bending moment;	M = **1.195** kNm
Maximum shear force;	V = **1.770** kN
Maximum support reaction;	R = **1.770** kN
Maximum deflection;	δ = **2.334** mm

Check bending stress

Bending stress;	σ_m = **7.500** N/mm^2
Permissible bending stress;	$\sigma_{m_adm} = \sigma_m \times K_{2m} \times K_3 \times K_7 \times K_8$ = **10.783** N/mm^2
Applied bending stress;	σ_{m_max} = M / Z = **3.585** N/mm^2

PASS - Applied bending stress within permissible limits

Check shear stress

Shear stress;	τ = **0.710** N/mm^2
Permissible shear stress;	$\tau_{adm} = \tau \times K_{2s} \times K_3 \times K_8$ = **0.976** N/mm^2
Applied shear stress;	τ_{max} = 3 × V / (2 × b × h) = **0.266** N/mm^2

PASS - Applied shear stress within permissible limits

Check bearing stress

Compression perpendicular to grain (no wane);	σ_{cp1} = **2.400** N/mm^2
Permissible bearing stress;	$\sigma_{c_adm} = \sigma_{cp1} \times K_{2c} \times K_3 \times K_8$ = **3.300** N/mm^2
Applied bearing stress;	σ_{c_max} = R / (b × L_b) = **0.354** N/mm^2

PASS - Applied bearing stress within permissible limits

Check deflection

Permissible deflection;	δ_{adm} = min(L_{s1} × 0.003, 14 mm) = **8.100** mm
Bending deflection (based on E_{mean});	$\delta_{bending}$ = **2.122** mm
Shear deflection;	δ_{shear} = **0.212** mm
Total deflection;	$\delta = \delta_{bending} + \delta_{shear}$ = **2.334** mm

PASS - Actual deflection within permissible limits

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ELEMENT: **MISCELLANEOUS DESIGN**

DESIGN OF SPREAER PLATE:

A. FOR RIDGE BEAM 1,2 & BEAM 6

Assuming Spreader Plate **700x100x40mm** thick Mild Steel plate

Max. reaction from beam = 71.80 Kn X 0.7 = 50.26 Kn (unfactored)

Allowable compressive stress = 0.79 N/mm^2

$$\text{Length of the spreader} = \frac{50.26 \times 10^3}{0.79 \times 100}$$

Length of the spreader = 637mm

Therefore provided **700 x 100 x 40 mm thick M.S. plate** as spreader is safe

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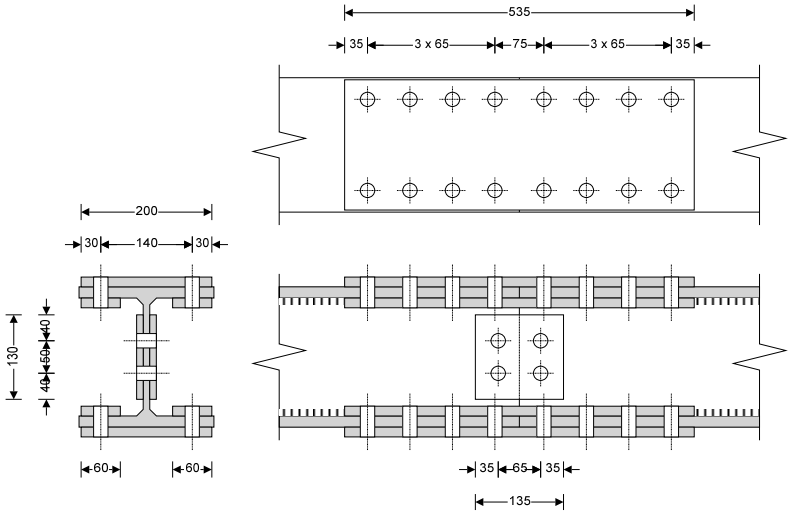
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SPLICE CONNECTION FOR RIDGE BEAM 1 (203 X 203 X 71 UC) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **84** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **78** kN

Steel beam details

Beam section classification;	**UC 203x203x71**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **15** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **130** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **40** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **423** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **423** kN
Force in the flange;	$F_f = \max(T, C)$ = **423** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **345** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **1227** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **3090** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **1192** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **3828** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **285.4** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **495.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **1093.1** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **2.7**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **78** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2600** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1720** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2009** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **470** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **554** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **90** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **496** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **470** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1538087** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **2.5** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **16.8** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **50.7** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **39.0** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **64.0** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **165.0** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **158.4** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x71**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 15 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 15 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**130 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**

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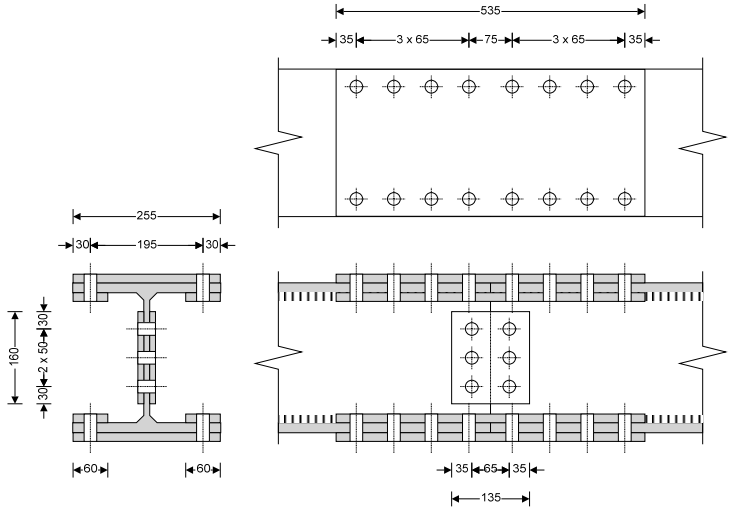
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SPLICE CONNECTION FOR RIDGE BEAM 2 (254 X 254 X 89 UC) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **194** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **72** kN

Steel beam details

Beam section classification;	**UC 254x254x89**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **15** mm
Width of outer flange plates;	b_{fp} = **255** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **195** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **160** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **3**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **3**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **30** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **798** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **798** kN
Force in the flange;	$F_f = \max(T, C)$ = **798** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **345** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **2314** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **4040** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **2249** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **4736** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **285.4** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **495.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **1093.1** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **5.0**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **72** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **3200** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1880** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2473** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **514** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **682** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **130** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **666** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **514** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **2286713** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **2.3** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **20.3** kNm

PASS - Effective web plate area is adequate in bending

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PROJECT: 1 LAMBOURNE DRIVE COBHAM KT11 3LB	Date:	JUL 2024

Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **100** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **23.4** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **24.0** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **33.5** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **44.3** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **42** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **170.0** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **230.4** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **158.4** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 254x254x89**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 255 mm x 15 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 15 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**160 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**6 No. total - 3 No. rows of 1 No. bolts on each side of the joint**

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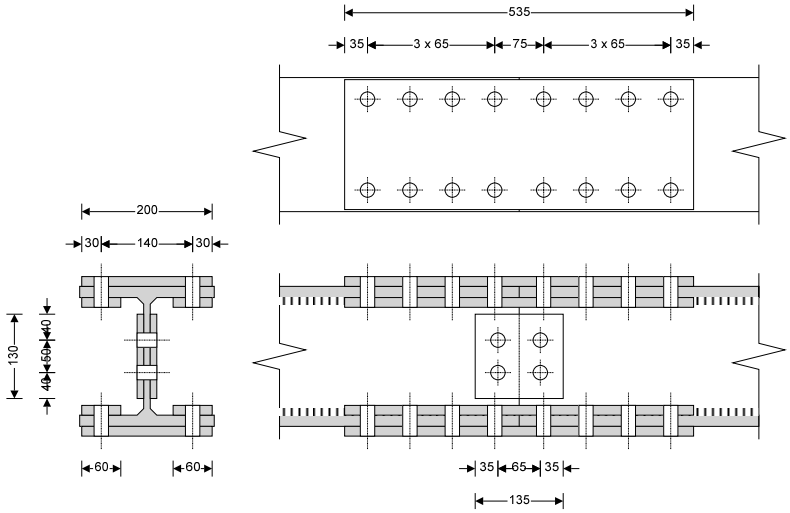
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SPLICE CONNECTION FOR BEAM 1 (203 X 203 X 71) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **151** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **114** kN

Steel beam details

Beam section classification;	**UC 203x203x71**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **15** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **130** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **40** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **761** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **761** kN
Force in the flange;	$F_f = \max(T, C)$ = **761** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **345** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **2205** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **3090** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **2143** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **3828** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **285.4** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **495.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **1093.1** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **4.8**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **114** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2600** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1720** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2009** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **470** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **554** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **90** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **496** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **470** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1538087** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **3.7** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **16.8** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **74.1** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **57.0** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **93.5** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **165.0** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **158.4** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x71**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 15 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 15 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**130 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**

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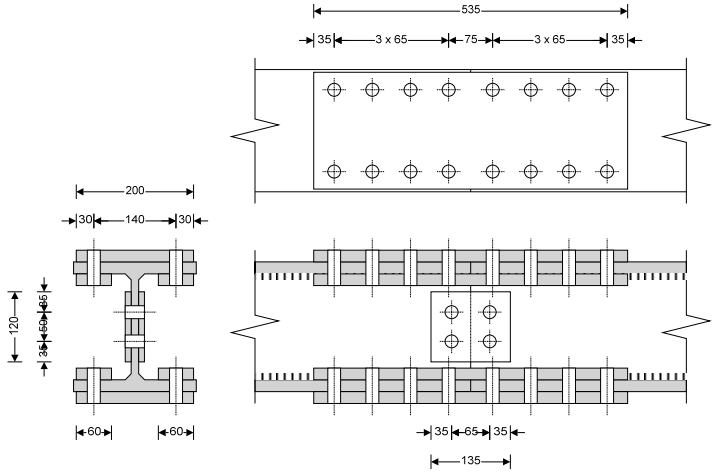
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SPLICE CONNECTION FOR BEAM 2 (203 X 203 X 86) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **240** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **178** kN

Steel beam details

Beam section classification;	**UC 203x203x86**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **20** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **120** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **35** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **1175** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **1175** kN
Force in the flange;	$F_f = \max(T, C)$ = **1175** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **345** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **3406** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **3723** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **345** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **3406** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **5104** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **338.3** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **660.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **1457.5** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **7.4**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **178** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2400** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1520** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **1855** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **404** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **497** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **85** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **461 kN**

Shear capacity of web plates; pv = min(pv_net, pv_gross, pv_block) = **404 kN**

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1147253** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **5.8** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **13.2** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **115.7** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **89.0** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **146.0** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **209.6** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **158.4** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x86**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 20 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 20 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**120 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**

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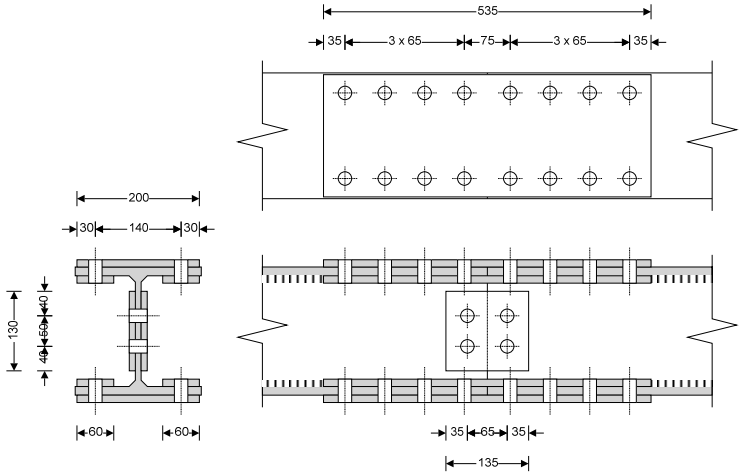
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SPLICE CONNECTION FOR BEAM 3 (203 X 203 X 60) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **110** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **159** kN

Steel beam details

Beam section classification;	**UC 203x203x60**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **12** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **130** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **40** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **563** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **563** kN
Force in the flange;	$F_f = \max(T, C)$ = **563** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **355** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **1586** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **2527** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **1586** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **3062** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **234.3** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **396.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **874.5** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **3.6**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **159** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2600** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1720** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2009** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **470** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **554** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **90** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **496** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **470** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1538087** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **5.2** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **16.8** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **103.3** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **79.5** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **130.4** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **155.1** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **155.1** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x60**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 12 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 12 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**130 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**

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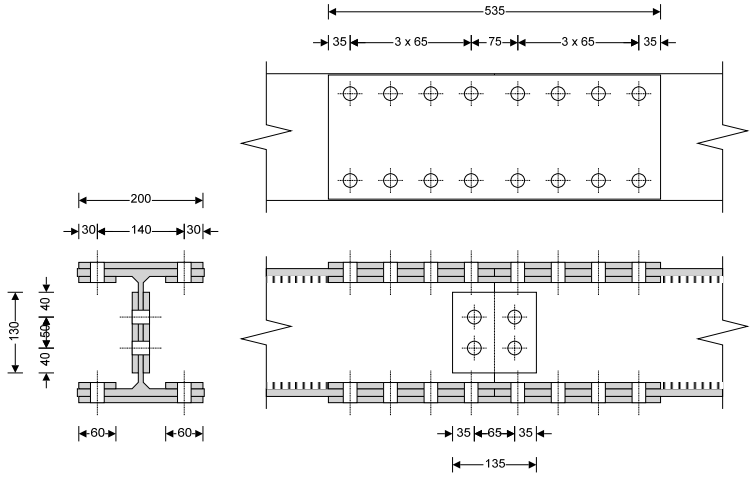
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SPLICE CONNECTION FOR BEAM 4,5,7 (203 X 203 X 52) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **69** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **52** kN

Steel beam details

Beam section classification;	**UC 203x203x52**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **10** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **130** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **40** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **356** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **356** kN
Force in the flange;	$F_f = \max(T, C)$ = **356** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **355** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **1003** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **2204** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **1003** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **2552** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **206.3** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **330.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **728.8** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **2.2**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **52** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2600** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1720** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2009** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **470** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **554** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **90** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **496** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **470** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1538087** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **1.7** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **16.8** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **33.8** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **26.0** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **42.6** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **130.4** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **130.4** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x52**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 10 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 10 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**130 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**

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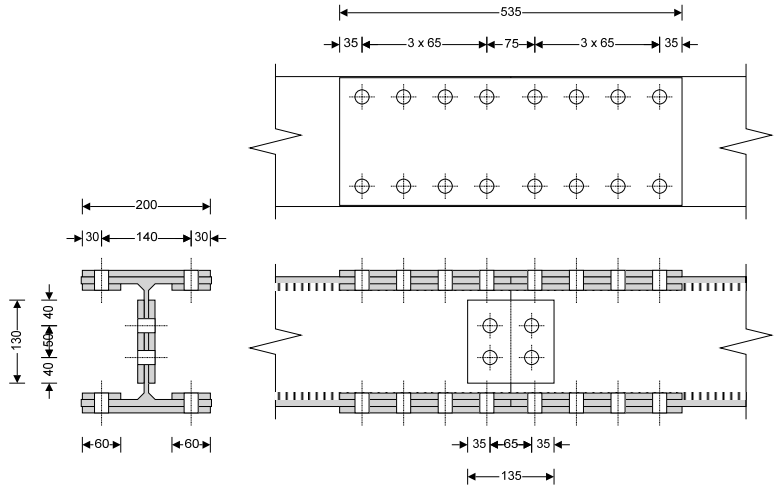
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SPLICE CONNECTION FOR BEAM 6 (203 X 203 X 46) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **50** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **34** kN

Steel beam details

Beam section classification;	**UC 203x203x46**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **10** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **130** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **40** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **260** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **260** kN
Force in the flange;	$F_f = \max(T, C)$ = **260** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **355** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **733** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **1931** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **733** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **2552** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **181.5** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **330.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **728.8** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **1.6**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **34** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2600** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1720** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2009** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **470** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **554** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **90** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **496** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **470** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1538087** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **1.1** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **16.8** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **22.1** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **17.0** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **27.9** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **118.8** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **118.8** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x46**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 10 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 10 mm to the inside of each flange
Flange bolting;	**16 No. total per flange - 4 No. rows of 2 No. bolts on each side of the joint**
Web plates;	**130 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**

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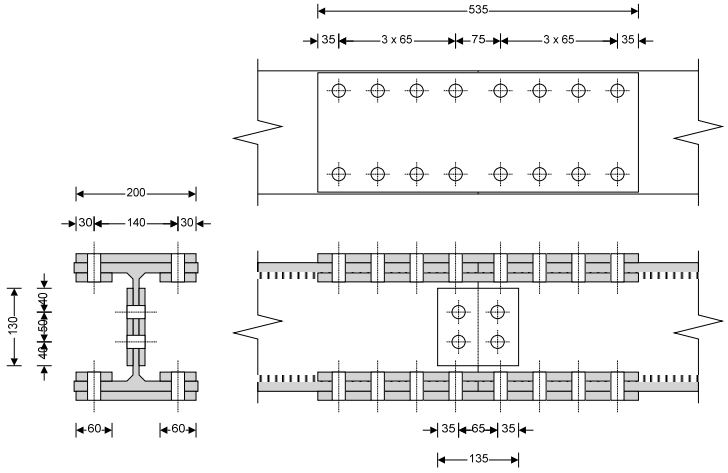
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SPLICE CONNECTION FOR BEAM 8 (203 X 203 X 71) @ L/3

BOLTED COVER PLATE SPLICE CONNECTION TO BS5950-1:2000

TEDDS calculation version 1.0.08



Connection loads

Design moment;	M = **126** kNm
Axial force in the member (compression +ve);	N = **0** kN
Shear force in the member;	V = **53** kN

Steel beam details

Beam section classification;	**UC 203x203x71**
Grade of steel section;	**S355**
Section bearing strength;	p_{bs_s} = **550** N/mm^2

General connection details

Grade of steel plate;	**S355**
Plate bearing strength;	p_{bs_p} = **550** N/mm^2
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Hole diameter;	D_h = **22** mm
Bolt slip factor;	μ = **0.50**
Hole type factor;	K_s = 1.0

Flange plate details – plates bolted to both sides of each flange

Thickness of flange plates;	t_{fp} = **15** mm
Width of outer flange plates;	b_{fp} = **200** mm
Width of inner flange plates;	b_{fp_i} = **60** mm
Length of flange plates;	l_{fp} = **535** mm

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Flange bolting details

Rows of flange bolts on each side of joint;	n_{fb_r} = **4**
Bolts per row;	n_{fb_p} = **2**
Total number of flange bolts each side of joint;	$n_{fb} = n_{fb_r} \times n_{fb_p}$ = **8**
Spacing between rows of bolts;	S_f = **65** mm
Spacing between rows of bolts across joint;	S_{fc} = **75** mm
Spacing at end of flange plates;	S_{fe} = **35** mm
Lateral spacing between central bolts;	S_{flc} = **140** mm
Lateral spacing at edge of flange plates;	S_{fle} = **30** mm

Web plate details - plates bolted to both sides of the web

Thickness of web plates;	t_{wp} = **10** mm
Width of web plates;	b_{wp} = **135** mm
Length of web plates;	l_{wp} = **130** mm

Web bolting details

Rows of web bolts;	n_{wb_r} = **2**
Bolts per row each side of joint;	n_{wb_p} = **1**
Total number of web bolts each side of joint;	$n_{wb} = n_{wb_r} \times n_{wb_p}$ = **2**
Spacing between rows of bolts;	S_w = **50** mm
Spacing at end of web plates;	S_{we} = **40** mm
Lateral spacing between central bolts;	S_{wlc} = **65** mm
Lateral spacing at edge of web plates;	S_{wle} = **35** mm

Step 1 - Distribution of forces in member flanges

Forces in member tension flange;	$T = [M / (D_b - T_b)] – N / 2$ = **635** kN
Forces in member compression flange;	$C = [M / (D_b - T_b)] + N / 2$ = **635** kN
Force in the flange;	$F_f = \max(T, C)$ = **635** kN

Step 2 - Calculate distribution of forces in member flanges

Check area of flange

Design strength of section;	p_{ys} = **345** N/mm^2
Minimum required effective flange area;	F_f / p_{ys} = **1840** mm^2
Effective net area coefficient;	K_e = **1.1**
Effective flange area;	$A_{ef} = \min(K_e \times [B_b – (n_{fb_p} \times D_h)] \times T_b, B_b \times T_b)$ = **3090** mm^2

PASS - Effective flange area is adequate

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Check area of flange plates

Design strength of plates; p_{yp} = **355** N/mm^2

Minimum required effective flange plate area; F_f / p_{yp} = **1788** mm^2

Effective flange plate area; $A_{ep} = \min(K_e \times [b_{fp} + 2 \times (b_{fp_i} - (n_{fb_p} \times D_h))] \times t_{fp}, (b_{fp} + 2 \times b_{fp_i}) \times t_{fp})$

A_{ep} = **3828** mm^2

PASS - Effective flange plate area is adequate

Step 3 - Design of flange bolts

Slip resistance of the bolt per interface; $S_{fb} = 1.1 \times K_s \times \mu \times P_p$ = **79.2** kN

Bearing capacity of the bolt in the flange; $P_{bg_s} = 1.5 \times d \times T_b \times p_{bs_s}$ = **285.4** kN

Bearing capacity of the bolt in the plate; $P_{bg_p} = 3 \times d \times t_{fp} \times p_{bs_p}$ = **495.0** kN

Average flange bolt end distance; $S_{fe_ave} = S_{fe} + (n_{fb_r} - 1) \times S_f / 2$ = **133** mm

Bearing capacity limit of the bolt in the plate; $P_{bg_p_lim} = S_{fe_ave} \times t_{fp} \times p_{bs_p}$ = **1093.1** kN

Bolt capacity; $P_s = \min(2 \times S_{fb}, P_{bg_s}, P_{bg_p}, P_{bg_p_lim})$ = **158.4** kN

Number of bolts required; $n_{fb_req} = F_f / P_s$ = **4.0**

Number of bolts provided; n_{fb} = **8**

PASS - Flange plate bolting is adequate

Step 4 - Design of web plates and bolts

Check web plate in shear

Shear force in web plates; V = **53** kN

Gross shear area; $A_v = n_{wp} \times l_{wp} \times t_{wp}$ = **2600** mm^2

Net shear area (allowing for bolt holes); $A_{v_net} = n_{wp} \times (l_{wp} - n_{wb_r} \times D_h) \times t_{wp}$ = **1720** mm^2

Net shear area limit; $0.85 \times A_v / K_e$ = **2009** mm^2

Net shear capacity of web plates; $p_{v_net} = 0.7 \times K_e \times A_{v_net} \times p_{yp}$ = **470** kN

Gross shear capacity of web plates; $p_{v_gross} = 0.6 \times A_v \times p_{yp}$ = **554** kN

Length of block shear face; $L_v = S_{we} + (n_{wb_r} - 1) \times S_w$ = **90** mm

Length of block tension face; $L_t = S_{wle} + (n_{wb_p} - 1) \times S_{wl}$ = **35** mm

Block shear coefficient; $k = \text{if}(n_{wb_p} > 1, 2.5, 0.5)$ = **0.5**

Block shear capacity of web plates; $p_{v_block} = 0.6 \times p_{yp} \times t_{wp} \times n_{wp} \times [L_v + K_e \times (L_t - k \times D_h)]$ = **496** kN

Shear capacity of web plates; $p_v = \min(p_{v_net}, p_{v_gross}, p_{v_block})$ = **470** kN

PASS - Effective web plate area is adequate in shear

Check web plate in bending

Second moment of area of web plate; $I = (t_{wp} \times l_{wp}^3 / 12) - (n_{wb_r} \times t_{wp} \times D_h^3 / 12) - (t_{wp} \times D_h \times K \times S_w^2)$

I = **1538087** mm^4

Distance from joint to centroid of bolt group; $a = [((n_{wb_p} - 1) \times S_{wl}) + S_{wlc}] / 2$ = **33** mm

Moment in web plate; $M_{wp} = V \times a$ = **1.7** kNm

Moment capacity of web plates; $M_{cap} = p_{yp} \times n_{wp} \times I / (l_{wp} / 2)$ = **16.8** kNm

PASS - Effective web plate area is adequate in bending

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Check web plate bolts

Elastic section modulus of bolt group;	$Z_b = n_{wb_r} \times (n_{wb_r} + 1) \times S_w / 6 =$ **50** mm
Force on bolt due to moment;	$F_m = V \times a / Z_b =$ **34.4** kN
Force on bolt due to direct shear;	$F_v = V / n_{wb_r} =$ **26.5** kN
Resultant bolt load;	$F_r = \sqrt{(F_m^2 + F_v^2)} =$ **43.5** kN
Angle of the resultant bolt load;	$\theta = \text{atan}(F_m / F_v) =$ **52.4** deg
Minimum edge distance;	$e_r = \min(S_{we} / \cos(\theta), S_{wle} / \cos(90 - \theta)) =$ **44** mm
Edge distance factor for web plate bearing;	$K_{edge} = \min(e_r / (3 \times d), 1) =$ **0.7**
Slip resistance of the bolt per interface;	$S_{fb} = 1.1 \times K_s \times \mu \times P_p =$ **79.2** kN
Bearing capacity of the bolt in the web;	$P_{bg_s} = 1.5 \times d \times t_b \times p_{bs_s} =$ **165.0** kN
Bearing capacity of the bolt in the plate;	$P_{bg_p} = 1.5 \times K_{edge} \times d \times t_{wp} \times n_{wp} \times p_{bs_p} =$ **242.9** kN
Bolt capacity;	$P_s = \min(n_{wp} \times S_{fb}, P_{bg_s}, P_{bg_p}) =$ **158.4** kN

PASS - Web plate bolting is adequate

Connection summary

Beam classification;	**UC 203x203x71**
Bolt classification;	**M20 (Torqued General Grade HSFG)**
Flange plates;	**535 mm x 200 mm x 15 mm to the outside of each flange**
	2 No. 535 mm x 60 mm x 15 mm to the inside of each flange
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Web plates;	**130 mm x 135 mm x 10 mm on each side of the web**
Web bolting;	**4 No. total - 2 No. rows of 1 No. bolts on each side of the joint**